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Inventor(s)

Vugt-Van Lutz; Ingrid Maria et al.

RECOMBINANT YEAST CELL

Abstract

A recombinant yeast cell that functionally expresses: —a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity; —a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity; and —a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the “GT promoter”), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more, and a process for the production of ethanol using such recombinant yeast cell.

Inventors: Vugt-Van Lutz; Ingrid Maria (Echt, NL), Schmitz; Jozef Petrus Johannes (Echt, NL), Jansen; Mickel Leonardus August (Echt, NL), de Bruijn; Hans Marinus Charles Johannes (Echt, NL), Rossell-Aragort; Sergio Luis (Echt, NL), Van Rij; Evert Tjeerd (Echt, NL), Bureman; Phillip E. (Echt, NL)

Applicant: DANISCO US INC. (Palo Alto, CA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2022/068983, filed Jul. 7, 2022, which claims priority to U.S. Application No. 63/220,905, filed Jul. 12, 2021, and European Application No. 21185147.2, filed Jul. 12, 2021, all of which are hereby incorporated by reference in their entireties.

INCORPORATION BY REFERENCE OF THE SEQUENCE LISTING

[0002] The present application is being filed with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled IFF34060-US-PCT_SEQ_Listing.xml, created on Jul. 9, 2024, which is 161,064 in size. The information in the electronic format of the Sequence Listing is incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The invention relates to a recombinant yeast cell and to a process for the production of ethanol wherein said recombinant yeast cell is used.

BACKGROUND OF THE INVENTION

[0004] Microbial fermentation processes are applied to industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks. Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD^{sup.+} can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to re-oxidize NADH formed in biosynthetic reactions.

[0005] Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology. Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification. A major challenge relating to the stoichiometry of yeast-based production of ethanol, is that substantial amounts of NADH-dependent side-products such as glycerol are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al., “Anaerobic and aerobic batch cultivations of *Saccharomyces cerevisiae* mutants impaired in glycerol Synthesis”, (2000), Yeast, vol. 16, pages 463-474). Under conditions that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.

[0006] Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In

this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed by decarboxylation of pyruvate to ethanol via NAD.sup.+ -dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD.sup.+ to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD.sup.+ -dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.

[0007] In the literature, several different approaches have been reported that could help to reduce the amount of the byproduct glycerol and divert carbon to ethanol resulting in an increased yield of ethanol per gram of fermented carbohydrate.

[0008] WO2015/028583 describes a yeast cell that is genetically modified comprising: a) one or more nucleic acid sequence encoding a glycerol dehydrogenase (E.C. 1.1.1.6); b) one or more nucleic acid sequence encoding a dihydroxyacetone kinase (E.C. 2.7.1.28 or E.C. 2.7.1.29) and c) one or more nucleic acid sequence encoding a glycerol transporter. In addition, the cell may comprise one or more nucleic acid sequences encoding a NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase. WO2015/028583 further describes a process comprising the preparation of a fermentation product from acetate and from a fermentable carbohydrate—in particular a carbohydrate selected from the group of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose and mannose—which preparation is carried out under anaerobic conditions using the above yeast cell.

[0009] WO2015/028583 explains that as acetic acid is often considered to be the most toxic compound present in hydrolysates, there is a desire to further decrease the acetate (acetic acid) concentration in hydrolysates. It is mentioned that one way of increasing the anaerobic acetate conversion potential of the yeast is by introducing a glycerol conversion pathway that for example converts externally added glycerol forcing the yeast cell to convert more acetic acid in order to maintain the redox balance.

[0010] In the examples and Table 11, WO2015/028583 illustrates that especially transformant T5, including a glycerol transporter originating from *Zygosaccharomyces rouxii*, resulted in the conversion of more glycerol, relative to the reference strain. Also more acetic acid was consumed. The ethanol titer, however, was not the highest in case of this T5, because not all sugars were consumed. Hence, although good results are obtained with the yeast cell and process described in WO2015/028583, there is still room for further improvement.

[0011] It would be an advancement in the art to provide a yeast and process for the production of ethanol wherein the yeast comprises a glycerol conversion pathway and/or a glycerol transporter, similar to the yeast in WO2015/028583, but wherein the speed of the sugar conversion and/or the total amount of sugar consumed is improved.

SUMMARY OF THE INVENTION

[0012] The inventors have now surprising found that the process and yeast cell of WO2015/028583 can be even further improved by promoting the glycerol transporter with a specific promoter.

[0013] Accordingly the invention provides a recombinant yeast cell functionally expressing: [0014] a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity (preferably within enzyme class E.C. 1.1.1.6); [0015] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (preferably within enzyme class E.C. 2.7.1.28 or E.C. 2.7.1.29); and [0016] a nucleic acid sequence encoding a protein having glycerol transporter activity,

wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the “GT promoter”), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more.

[0017] In addition, the invention provides a process for the production of ethanol, comprising converting a carbon source, such as a carbohydrate or another organic carbon source, using the above recombinant yeast cell, suitably thereby forming ethanol.

[0018] Advantageously, use of the above recombinant yeast cell and/or the above process results in an improved speed of the sugar conversion and/or a higher total amount of sugar consumed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The invention is illustrated by the following figures:

[0020] FIG. 1: Ethanol and CO₂ gas production during the full 66 hours of the fermentation of corn mash with respectively reference strain RX16, new strain NX17 and new strain NX18 as described in Example 5 and illustrated in Table 18.

[0021] FIG. 2: Ethanol and CO₂ gas production during the first 10 hours of the fermentation of corn mash with respectively reference strain RX16, new strain NX17 and new strain NX18 as described in Example 5 and illustrated in Table 19.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

[0022] This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. An overview is provided by Table 1 below.

TABLE-US-00001 TABLE 1 Overview of sequence listings: SEQ ID NO: Enzyme Name Origin
Type SEQ ID NO: 1 Glycerol transporter *S. pombe* protein SEQ ID NO: 2 Glycerol transporter *Plasmodium falciparum*; protein SEQ ID NO: 3 Glycerol transporter (T3) *Danio rerio* protein (aquaglyceroporin AQP9) SEQ ID NO: 4 Glycerol transporter *Xenopus tropicalis*; protein SEQ ID NO: 5 Glycerol transporter (T5) *Z. rouxii* protein (ZYRO0E0121) SEQ ID NO: 6 Glycerol transporter (T3) *Danio rerio* DNA (aquaglyceroporin AQP9) SEQ ID NO: 7 Glycerol transporter (T5) *Z. rouxii* DNA (ZYRO0E0121) SEQ ID NO: 8 Promoter Motif Artificial sequence DNA SEQ ID NO: 9 Promoter Motif Artificial sequence DNA SEQ ID NO: 10 Promoter ANB1 *Saccharomyces cerevisiae* DNA SEQ ID NO: 11 Promoter DAN1 *Saccharomyces cerevisiae* DNA SEQ ID NO: 12 Promoter HEM13 *Saccharomyces cerevisiae* DNA SEQ ID NO: 13 Glycerol dehydrogenase *Klebsiella pneumoniae* protein (Kpne_gldA) SEQ ID NO: 14 Glycerol dehydrogenase (Eaer_gldA) *Enterococcus aerogenes* protein SEQ ID NO: 15 Glycerol dehydrogenase (Ya_gldA) *Yersinia aldovae* protein SEQ ID NO: 16 Glycerol dehydrogenase (Ec_gldA) *Escherichia coli* protein SEQ ID NO: 17 Glycerol dehydrogenase (Ec_gldA) Artificial sequence; based on DNA gldA gene from *Escherichia coli* - codon optimized SEQ ID NO: 18 Dihydroxyacetone kinase (DAK1) *Saccharomyces cerevisiae* protein SEQ ID NO: 19 Dihydroxyacetone kinase (DAK2) *Saccharomyces cerevisiae* protein SEQ ID NO: 20 Dihydroxyacetone kinase *Klebsiella pneumoniae* protein (Kpne_dhaK) SEQ ID NO: 21 Dihydroxyacetone kinase *Yarrowia lipolytica* protein (Ylip_DAK1) SEQ ID NO: 22 Dihydroxyacetone kinase *Schizosaccharomyces pombe* protein (Spom_DAK1) SEQ ID NO: 23 Dihydroxyacetone kinase (DAK1) *Saccharomyces cerevisiae* DNA SEQ ID NO: 24 Dihydroxyacetone kinase (DAK2) *Saccharomyces cerevisiae* DNA SEQ ID NO: 25 Ribulose-1,5-biphosphate *Thiobacillus denitrificans* protein carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898 SEQ ID NO: 26 Ribulose-1,5-biphosphate Artificial sequence; based on DNA carboxylase oxygenase (RuBisCO) cbbM gene from *Thiobacillus* as previously described in *denitrificans* - codon optimized WO2014/129898 SEQ ID NO: 27 Phosphoribulokinase (PRK) as *Spinacea oleracea* protein previously described in WO2014/129898 SEQ ID NO: 28

Phosphoribulokinase (PRK) as Artificial sequence; based on DNA previously described in prk gene from *Spinacea* WO2014/129898 *oleracea*, codon optimized SEQ ID NO: 29 (long) chaperone groEL as Artificial sequence; Translated protein previously described in protein sequence based on WO2014/129898 GroEL of *Escherichia coli* SEQ ID NO: 30 (long) chaperone groEL as Artificial sequence; based on DNA previously described in GroEL gene from *Escherichia* WO2014/129898 *coli*, codon optimized SEQ ID NO: 31 (short) chaperone groES as Artificial sequence; Translated protein previously described in protein sequence based on WO2014/129898 GroES of *Escherichia coli* SEQ ID NO: 32 (short) chaperone groES as Artificial sequence; based on DNA previously described in GroES gene from *Escherichia* WO2014/129898 *coli*, codon optimized SEQ ID NO: 33 Phosphoketolase *Bifidobacterium animalis* protein SEQ ID NO: 34 Phosphoketolase *Bifidobacterium adolescentis* protein SEQ ID NO: 35 Phosphoketolase *Bifidobacterium lactis* protein SEQ ID NO: 36 Phosphoketolase *Leuconostoc mesenteroides* protein SEQ ID NO: 37 Phosphotransacetylase *Bacillus subtilis* protein SEQ ID NO: 38 Phosphotransacetylase *Lactobacillus plantarum* protein SEQ ID NO: 39 Phosphotransacetylase *Bifidobacterium adoloscentis* protein SEQ ID NO: 40 Phosphotransacetylase *Methosarcina thermophila* protein SEQ ID NO: 41 Acetate kinase *Bifidobacterium adolescentis* protein SEQ ID NO: 42 Acetate kinase *Aspergillus nidulans* protein SEQ ID NO: 43 Bifunctional NAD.sup.+ dependent *Escherichia coli* protein acetylating acetaldehyde/alcohol dehydrogenase (adhE) SEQ ID NO: 44 Ethanolamine utilizing protein (eutE) *Escherichia coli* protein SEQ ID NO: 45 acetaldehyde dehydrogenase (acdH) *Lactobacillus plantarum* protein SEQ ID NO: 46 acetaldehyde dehydrogenase (acdH) *Listeria innocua* protein SEQ ID NO: 47 acetaldehyde/alcohol *Staphylococcus aureus* protein dehydrogenase (adhE) SEQ ID NO: 48 acetylating acetaldehyde *Escherichia coli* protein dehydrogenase (mhpF) SEQ ID NO: 49 Acetyl-coenzyme A synthetase 1 *Saccharomyces cerevisiae* protein (ACS1) SEQ ID NO: 50 Acetyl-coenzyme A synthetase 2 *Saccharomyces cerevisiae* protein (ACS2) SEQ ID NO: 51 Alcohol dehydrogenase 1 (ADH1) *Saccharomyces cerevisiae* protein SEQ ID NO: 52 Alcohol dehydrogenase 2 (ADH2) *Saccharomyces cerevisiae* protein SEQ ID NO: 53 Alcohol dehydrogenase 3 (ADH3) *Saccharomyces cerevisiae* protein SEQ ID NO: 54 Alcohol dehydrogenase 4 (ADH4) *Saccharomyces cerevisiae* protein SEQ ID NO: 55 Alcohol dehydrogenase 5 (ADH5) *Saccharomyces cerevisiae* protein SEQ ID NO: 56 Glucoamylase (GA) *Punctularia strigosozonata* protein SEQ ID NO: 57 Glucoamylase (GA) *Punctularia strigosozonata* protein SEQ ID NO: 58 Glucoamylase (GA) Artificial sequence, based on protein *Punctularia strigosozonata* SEQ ID NO: 59 Fragment A Artificial sequence DNA SEQ ID NO: 60 Fragment B Artificial sequence DNA SEQ ID NO: 61 Fragment C Artificial sequence DNA SEQ ID NO: 62 Fragment D Artificial sequence DNA SEQ ID NO: 63 Fragment E Artificial sequence DNA SEQ ID NO: 64 Fragment F Artificial sequence DNA SEQ ID NO: 65 Fragment G Artificial sequence DNA SEQ ID NO: 66 Connector 25 Artificial sequence DNA SEQ ID NO: 67 Connector 2A Artificial sequence DNA SEQ ID NO: 68 Connector 2B Artificial sequence DNA SEQ ID NO: 69 Connector 2C Artificial sequence DNA SEQ ID NO: 70 Sc_INT7.03_FLANK5 Artificial sequence DNA SEQ ID NO: 71 Sc_INT7.03_FLANK3 Artificial sequence DNA SEQ ID NO: 72 INT28_FLANK5 Artificial sequence DNA SEQ ID NO: 73 INT28_FLANK3 Artificial sequence DNA SEQ ID NO: 74 NR1 nitrate reductase *Zea mays* Protein [Uniprot accession no: Q9XGW5] SEQ ID NO: 75 NIA Nitrate reductase *Spinacia oleracea* Protein [Uniprot accession no: P23312] SEQ ID NO: 76 nirD Nitrite reductase (small subunit) *Escherichia coli* Protein SEQ ID NO: 77 nirB Nitrite reductase (large subunit) *Escherichia coli* Protein SEQ ID NO: 78 niiA Nitrite reductase *Emericella nidulans* Protein (*Aspergillus nidulans*) SEQ ID NO: 79 YNT1 Nitrate/Nitrite transporter *Pichia angusta* Protein (*Hansenula polymorpha*)

[0023] In the context of this patent application, each of the above protein/amino acid sequences is preferably encoded by a DNA/nucleic acid sequence that is codon-pair optimized for expression in a yeast, more preferably for expression in a *Saccharomyces cerevisiae* yeast.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0024] Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

[0025] Throughout the present specification and the accompanying claims, the words “comprise” and “include” and variations such as “comprises”, “comprising”, “includes” and “including” are to be interpreted inclusively. That is, these words are intended to convey the possible inclusion of other elements or integers not specifically recited, where the context allows.

[0026] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to one or at least one) of the grammatical object of the article. By way of example, “an element” may mean one element or more than one element. When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “gene”, this means “at least one” of that gene, e.g. “at least one gene”, unless specified otherwise.

[0027] When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular aspect of the invention; in particular when referring to such as compound, it includes the natural isomer(s).

[0028] Unless explicitly indicated otherwise, the various embodiments of the invention described herein can be cross-combined.

[0029] The term “carbon source” refers to a source of carbon, preferably a compound or molecule comprising carbon. Preferably the carbon source is a carbohydrate. A carbohydrate is understood herein to be an organic compound made of carbon, oxygen and hydrogen. Suitably the carbon source may be selected from the group consisting of mono-, di- and/or polysaccharides, acids and acid salts. More preferably the carbon source is a compound selected from the group consisting of glucose, arabinose, xylose, galactose, mannose, rhamnose, fructose, glycerol, and acetic acid or a salt thereof.

[0030] The terms “dry matter” and “dry solids”, abbreviated respectively as “DM” and “DS”, are used interchangeably herein and refer to material remaining after removal of water. Dry matter content can be determined by any method known to the person skilled in the art therefore.

[0031] The term “ferment”, and variations thereof such as “fermenting”, “fermentation” and/or “fermentative”, is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. An anaerobic fermentation is herein defined to be a fermentation carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell. Conditions in which essentially no oxygen is consumed suitably corresponds to an oxygen consumption of less than 5 mmol/l.Math.h.sup.-1, in particular to an oxygen consumption of less than 2.5 mmol/l.Math.h.sup.-1, or less than 1 mmol/l.Math.h.sup.-1. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This suitably corresponds to a dissolved oxygen concentration in a culture broth of less than 5% of air saturation, more suitably to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.

[0032] The term “fermentation process” refers to a process for the preparation or production of a fermentation product.

[0033] The term “cell” refers to a eukaryotic or prokaryotic organism, preferably occurring as a single cell. In the present invention the cell is a recombinant yeast cell. That is, the recombinant cell is selected from the group of genera consisting of yeast.

[0034] The terms “yeast” and “yeast cell” are used herein interchangeably and refer to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The budding yeasts (“true yeasts”) are classified in the order Saccharomycetales. The yeast cell according to the invention is preferably a yeast cell derived from the genus of *Saccharomyces*. More preferably the yeast cell is a yeast cell of the species

Saccharomyces cerevisiae.

[0035] The term “recombinant”, for example referring to a “recombinant yeast”, a “recombinant cell”, “recombinant micro-organism” and/or “recombinant strain” as used herein, refers to a yeast, cell, micro-organism or strain, respectively, containing nucleic acid which is the result of one or more genetic modifications. Simply put the yeast, cell, micro-organism or strain contains a different combination of nucleic acid from (either of) its parent(s). To construe a recombinant yeast, cell, micro-organism or strain, recombinant DNA technique(s) and/or another mutagenic technique(s) can be used. For example a recombinant yeast and/or a recombinant yeast cell may comprise nucleic acid not present in the corresponding wild-type yeast and/or cell, which nucleic acid has been introduced into that yeast and/or yeast cell using recombinant DNA techniques (i.e. a transgenic yeast and/or cell), or which nucleic acid not present in said wild-type yeast and/or cell is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type yeast and/or yeast cell (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant” may suitably relate to a yeast, cell, micro-organism or strain from which nucleic acid sequences have been removed, for example using recombinant DNA techniques.

[0036] By a recombinant yeast comprising or having a certain activity is herein understood that the recombinant yeast may comprise one or more nucleic acid sequences encoding for a protein having such activity. Hence allowing the recombinant yeast to functionally express such a protein or enzyme.

[0037] The term “functionally expressing” means that there is a functioning transcription of the relevant nucleic acid sequence, allowing the nucleic acid sequence to actually be transcribed, for example resulting in the synthesis of a protein.

[0038] The term “transgenic” as used herein, for example referring to a “transgenic yeast” and/or a “transgenic cell”, refers to a yeast and/or cell, respectively, containing nucleic acid not naturally occurring in that yeast and/or cell and which has been introduced into that yeast and/or cell using for example recombinant DNA techniques, such as a recombinant yeast and/or cell.

[0039] The term “mutated” as used herein regarding proteins or polypeptides means that, as compared to the wild-type or naturally occurring protein or polypeptide sequence, at least one amino acid has been replaced with a different amino acid, inserted into, or deleted from the amino acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., *Molecular Cloning—A Laboratory Manual*, 2nd ed., Vol. 1-3 (1989), published by Cold Spring Harbor Publishing).

[0040] The term “mutated” as used herein regarding genes means that, as compared to the wild-type or naturally occurring nucleic acid sequence, at least one nucleotide in the nucleic acid sequence of a gene or a regulatory sequence thereof, has been replaced with a different nucleotide, inserted into, or deleted from the nucleic acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis, resulting for example in the transcription of a protein sequence with a qualitatively or quantitatively altered function or the knock-out of that gene. In the context of this invention an “altered gene” has the same meaning as a mutated gene.

[0041] The term “gen” or “gene”, as used herein, refers to a nucleic acid sequence that can be transcribed into mRNAs that are then translated into protein. A gene encoding for a certain protein refers to the one or more nucleic acid sequence(s) encoding for such a protein.

[0042] The term “nucleic acid” or “nucleotide” as used herein, refers to a monomer unit in a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-

stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e.g., peptide nucleic acids). For example, a certain enzyme that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to the reference nucleotide sequence encoding the enzyme. A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.

[0043] The terms “nucleotide sequence” and “nucleic acid sequence” are used interchangeably herein. An example of a nucleic acid sequence is a DNA sequence.

[0044] The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues, for example illustrated by an amino acid sequence. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

[0045] The term “enzyme” refers herein to a protein having a catalytic function. Where a protein catalyzes a certain biological reaction, the terms “protein” and “enzyme” may be used interchangeable herein. When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at <http://www.chem.qmul.ac.uk/iubmb/enzyme/>. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.

[0046] If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to a accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found via www.ncbi.nlm.nih.gov/, (as available on 1 Oct. 2020) unless specified otherwise.

[0047] Every nucleic acid sequence herein that encodes a polypeptide also includes any conservatively modified variants thereof. This includes that, by reference to the genetic code, it describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described

without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.

[0048] The term “functional homologue” (or in short “homologue”) of a polypeptide and/or amino acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polypeptide and/or amino acid sequence comprising said specific sequence with the proviso that one or more amino acids are mutated, substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion.

[0049] The term “functional homologue” (or in short “homologue”) of a polynucleotide and/or nucleic acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polynucleotide and/or nucleic acid sequence comprising said specific sequence with the proviso that one or more nucleic acids are mutated, substituted, deleted, added, and/or inserted, and which polynucleotide encodes for a polypeptide sequence that has (qualitatively) the same enzymatic functionality for substrate conversion. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

[0050] Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, “identity” also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as determined by the match between strings of such sequences.

[0051] Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by “percent identity” or “percent similarity”, which is high or low respectively. Although disputed, to indicate “percent identity” or “percent similarity”, “level of homology” or “percent homology” are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal et al., “*An overview of sequence comparison: Time warps, string edits, and macromolecules*”, (1983), Society for Industrial and Applied Mathematics (SIAM), Vol 25, No. 2, pages 201-237 and D. and the handbook edited by Sankoff and J. B. Kruskal, (ed.), “*Time warps, string edits and macromolecules: the theory and practice of sequence comparison*”, (1983), pp. 1-44, published by Addison-Wesley Publishing Company, Massachusetts USA).

[0052] The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman et al “*A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins*” (1970) J. Mol. Biol. Vol. 48, pages 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOSS package is used (version 2.8.0 or higher, see Rice et al, “*EMBOSS: The European Molecular Biology Open Software Suite*” (2000), Trends in Genetics vol. 16, (6) pages 276-277, <http://emboss.bioinformatics.nl/>). For protein sequences, EBLOSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

[0053] The homology or identity is the percentage of identical matches between the two full

sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as "IDENTITY".

[0054] The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labelled in the output of the program as "longest-identity".

[0055] A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or more substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in the sequence listing).

[0056] Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called "conservative" amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. In an embodiment, conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine. Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. In an embodiment, conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to Ser; Arg to Lys; Asn to Gln or His; Asp to Glu; Cys to Ser or Ala; Gln to Asn; Glu to Asp; Gly to Pro; His to Asn or Gln; Ile to Leu or Val; Leu to Ile or Val; Lys to Arg; Gln or Glu; Met to Leu or Ile; Phe to Met, Leu or Tyr; Ser to Thr; Thr to Ser; Trp to Tyr; Tyr to Trp or Phe; and, Val to Ile or Leu.

[0057] Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity. Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing

solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

[0058] “Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0059] “Overexpression” refers to expression of a gene, respectively a nucleic acid sequence, by a recombinant cell in excess to its expression in a corresponding wild-type cell. Such overexpression can for example be arranged for by: increasing the frequency of transcription of one or more nucleic acid sequences, for example by operational linking of the nucleic acid sequence to a promoter functional within the recombinant cell; and/or by increasing the number of copies of a certain nucleic acid sequence.

[0060] The terms “upregulate”, “upregulated” and “upregulation” refer to a process by which a cell increases the quantity of a cellular component, such as RNA or protein. Such an upregulation may be in response to or caused by a genetic modification.

[0061] By the term “pathway” or “metabolic pathway” is herein understood a series of chemical reactions in a cell that build and breakdown molecules.

[0062] Nucleic acid sequences (i.e. polynucleotides) or proteins (i.e. polypeptides) may be native or heterologous to the genome of the host cell.

[0063] “Native”, “homologous” or “endogenous” with respect to a host cell, means that the nucleic acid sequence does naturally occur in the genome of the host cell or that the protein is naturally produced by that cell. The terms “native”, “homologous” and “endogenous” are used interchangeable herein.

[0064] As used herein, “heterologous” may refer to a nucleic acid sequence or a protein. For example, “heterologous”, with respect to the host cell, may refer to a polynucleotide that does not naturally occur in that way in the genome of the host cell or that a polypeptide or protein is not naturally produced in that manner by that cell. A heterologous nucleic acid sequence is a nucleic acid that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a native structural gene is from a species different from that from which the structural gene is derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. That is, heterologous protein expression involves expression of a protein that is not naturally expressed in that way in the host cell. The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding a certain protein or enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., *Methods in Yeast Genetics*, (1986), published by Cold Spring Harbor Laboratory, is a well-recognized work describing the various methods available to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.

[0065] As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene or other (part of) nucleic acid sequence. Suitably, a promoter is located in the 5'-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be

constitutive, inducible or repressible. In an embodiment there is no (external) inducer needed.

[0066] The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.

[0067] The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid sequence that comprises in the 5' to 3' direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region.

[0068] “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism's genome and is usually circular in nature.

[0069] An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism's genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.

[0070] By “host cell” is herein understood a cell, such as a yeast cell, that is to be transformed with one or more nucleic acid sequences encoding for one or more heterologous proteins, to construe a transformed cell, also referred to as a recombinant cell. For example, the transformed cell may contain a vector and may support the replication and/or expression of the vector.

[0071] “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome. “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide (i.e. an exogenous nucleic acid sequence) into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.

[0072] By “constitutive expression” and “constitutively expressing” is herein understood that there is a continuous transcription of a nucleic acid sequence. That is, the nucleic acid sequence is transcribed in an ongoing manner. Constitutively expressed genes are always “on”.

[0073] By “anaerobic constitutive expression” is herein understood that nucleic acid sequence is constitutively expressed in an organism under anaerobic conditions. That is, under anaerobic conditions the nucleic acid sequence is transcribed in an ongoing manner, i.e. under such anaerobic conditions the genes are always “on”.

[0074] By “disruption” is herein understood any disruption of activity, including, but not limited to, deletion, mutation and reduction of the affinity of the disrupted gene and expression of RNA complementary to such disrupted gene. It includes all nucleic acid modifications such as nucleotide deletions or substitutions, gene knock-outs, and other actions which affect the translation or

transcription of the corresponding polypeptide and/or which affect the enzymatic (specific) activity, its substrate specificity, and/or or stability. It also includes modifications that may be targeted on the coding sequence or on the promotor of the gene. A gene disruptant is a cell that has one or more disruptions of the respective gene. Native to yeast herein is understood as that the gene is present in the yeast cell before the disruption.

[0075] The term “encoding” has the same meaning as “coding for”. Thus, by way of example, “one or more genes encoding a protein having activity X” has the same meaning as “one or more genes coding for a protein having activity X”.

[0076] As far as genes or nucleic acid sequences encoding a protein or an enzyme are concerned, the phrase “a nucleic acid sequence encoding a X”, respectively “one or more nucleic acid sequences encoding a X”, wherein X denotes a certain protein or (enzymatic) activity, has the same meaning as “a nucleic acid sequence encoding a protein having X activity”, respectively “one or more nucleic acid sequences encoding a protein having X activity”. Thus, by way of example, “one or more nucleic acid sequences encoding a transketolase” has the same meaning as “one or more nucleic acid sequences encoding a protein having transketolase activity”. As indicated above, the article “a” refers to “one or more”.

[0077] By a “redox sink” is herein understood a metabolic pathway that, overall, consumes or oxidizes NADH into NAD⁺ and/or prevents or reduces the consumption or reduction of NAD⁺ into NADH. A non-native metabolic pathway is a metabolic pathway that does not occur in the corresponding wild-type cell. Hence, a non-native metabolic pathway forming a redox sink is preferably a non-native metabolic pathway that, as compared to a corresponding wild-type yeast cell, increases NADH consumption and/or decreases NAD⁺ consumption. By increasing NADH consumption and/or decreasing NAD⁺ consumption advantageously an (additional) non-native redox sink can be created within the cell.

[0078] The abbreviation “NADH” refers to reduced, hydrogenated form of nicotinamide adenine dinucleotide. The abbreviation “NAD⁺” refers to the oxidized form of nicotinamide adenine dinucleotide. Nicotinamide adenine dinucleotide may act as a so-called cofactor, assisting in biochemical reactions and/or transformations in a cell. [074] “NADH dependent” or “NAD⁺ dependent” is herein equivalent to NADH specific and “NADH dependency” or “NAD⁺ dependency” is herein equivalent to NADH specificity.

[0079] By a “NADH dependent” or “NAD⁺ dependent” enzyme is herein understood an enzyme that is exclusively depended on NADH/NAD⁺ as a co-factor or that is predominantly dependent on NADH/NAD⁺ as a cofactor, i.e. as contrasted to other types of co-factor. By an “exclusive NADH/NAD⁺ dependent” enzyme is herein understood an enzyme that has an absolute requirement for NADH/NAD⁺ over NADPH/NADP⁺. That is, it is only active when NADH/NAD⁺ is applied as cofactor. By a “predominantly NADH/NDA⁺-dependent” enzyme is herein understood an enzyme that has a higher specificity and/or a higher catalytic efficiency for NADH/NAD⁺ as a cofactor than for NADPH/NADP⁺ as a cofactor.

[0080] The enzyme's specificity characteristics can be described by the formula:

$$[00001] 1 < K_m \text{NADP}^+ / K_m \text{NAD}^+ < \infty (\text{infinity})$$

wherein K.sub.m is the so-called Michaelis constant.

[0081] For a predominantly NADH-dependent enzyme, preferably

K.sub.mNADP.sup.+ / K.sub.mNAD.sup.+ is between 1 and 1000, between 1 and 500, between 1 and 200, between 1 and 100, between 1 and 50, between 1 and 10, between 5 and 100, between 5 and 50, between 5 and 20 or between 5 and 10.

[0082] The K.sub.m's for the enzymes herein can be determined as enzyme specific, for NAD.sup.+ and NADP.sup.+ respectively, using know analysis techniques, calculations and protocols. These are described for instance in Lodish et al., Molecular Cell Biology 6.sup.th Edition, Ed. Freeman, pages 80 and 81, e.g. FIG. 3-22. For an predominantly NADH-dependent enzyme, preferably the ratio of the catalytic efficiency for NADPH/NADP⁺ as a cofactor

(k.sub.cat/K.sub.m).sup.NADP+ to NADH/NAD+ as cofactor (k.sub.cat/K.sub.m).sup.NAD+, i.e. the catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP+:(k.sub.cat/K.sub.m).sup.NAD+, is more than 1:1, more preferably equal to or more than 2:1, still more preferably equal to or more than 5:1, even more preferably equal to or more than 10:1, yet even more preferably equal to or more than 20:1, even still more preferably equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the predominantly NADH-dependent enzyme may have a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP+:(k.sub.cat/K.sub.m).sup.NAD+ of equal to or less than 1,000,000,000:1 (i.e. 1.Math.10.sup.9:1).

The Yeast Cell

[0083] The recombinant yeast cell is preferably a yeast cell, or derived from a yeast cell, from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae.

[0084] Examples of suitable yeast cells include *Saccharomyces*, such as *Saccharomyces cerevisiae*, *Saccharomyces eubayanus*, *Saccharomyces jurei*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum* and *Saccharomyces bayanus*.

[0085] Examples of suitable yeast cells further include *Schizosaccharomyces*, such as *Schizosaccharomyces pombe*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces octosporus* and *Schizosaccharomyces cryophilus*.

[0086] Other exemplary yeasts include *Torulaspora* such as *Torulaspora delbrueckii*; *Kluyveromyces* such as *Kluyveromyces marxianus*; *Pichia* such as *Pichia stipitis*, *Pichia pastoris* or *Pichia angusta*; *Zygosaccharomyces* such as *Zygosaccharomyces bailii*; *Brettanomyces* such as *Brettanomyces inter medius*; *Brettanomyces bruxellensis*, *Brettanomyces anomalus*, *Brettanomyces custersianus*, *Brettanomyces naardenensis*, *Brettanomyces nanus*, *Dekkera bruxellensis* and *Dekkera anomala*; *Metschnikowia*, *Issatchenkia*, such as *Issatchenkia orientalis*, *Kloeckera* such as *Kloeckera apiculata*; and *Aureobasidium* such as *Aureobasidium pullulans*.

[0087] The yeast cell is preferably a yeast cell of the genus *Schizosaccharomyces*, herein also referred to as a *Schizosaccharomyces* yeast cell, or a yeast cell of the genus *Saccharomyces*, herein also referred to as a *Saccharomyces* yeast cell. More preferably the yeast cell is a yeast cell derived from a yeast cell of the species *Saccharomyces cerevisiae*, herein also referred to as a *Saccharomyces cerevisiae* yeast cell. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the species *Saccharomyces cerevisiae*.

[0088] Preferably the yeast cell is an industrial yeast cell. The living environments of yeast cells in industrial processes are significantly different from that in the laboratory. Industrial yeast cells must be able to perform well under multiple environmental conditions which may vary during the process. Such variations include changes in nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, etc., which together have potential impact on the cellular growth and ethanol production of the yeast cell. An industrial yeast cell can be understood to refer to a yeast cell that, when compared to a laboratory counterpart, has a more robust performance. That is, when compared to a laboratory counterpart, the industrial yeast cell shows less variation in performance when one or more environmental conditions selected from the group of nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, are varied during fermentation. Preferably, the yeast cell is constructed on the basis of an industrial yeast cell as a host, wherein the construction is conducted as described hereinafter. Examples of industrial yeast cells are Ethanol Red® (Fermentis) Fermiol® (DSM) and Thermosacc® (Lallemand).

[0089] The recombinant yeast cell described herein may be derived from any host cell capable of producing a fermentation product. Preferably the host cell is a yeast cell, more preferably an industrial yeast cell as described herein above. Preferably the yeast cell described herein is derived from a host cell having the ability to produce ethanol.

[0090] The yeast cell described herein may be derived from the host cell through any technique known by one skilled in the art to be suitable therefore. Such techniques may include any one or more of mutagenesis, recombinant DNA technology (including, but not limited to, CRISPR-CAS techniques), selective and/or adaptive evolution, mating, cell fusion, and/or cytoduction between yeast strains. Suitably the one or more desired genes are incorporated in the yeast cell by a combination of one or more of the above techniques.

[0091] The recombinant yeast cells according to the invention are preferably inhibitor tolerant, i.e. they can withstand common inhibitors at the level that they typically have with common pretreatment and hydrolysis conditions, so that the recombinant yeast cells can find broad application, i.e. it has high applicability for different feedstock, different pretreatment methods and different hydrolysis conditions. In an embodiment the recombinant yeast cell is inhibitor tolerant. Inhibitor tolerance is resistance to inhibiting compounds. The presence and level of inhibitory compounds in lignocellulose may vary widely with variation of feedstock, pretreatment method hydrolysis process. Examples of categories of inhibitors are carboxylic acids, furans and/or phenolic compounds. Examples of carboxylic acids are lactic acid, acetic acid or formic acid. Examples of furans are furfural and hydroxy-methylfurfural. Examples of phenolic compounds are vanillin, syringic acid, ferulic acid and coumaric acid. The typical amounts of inhibitors are for carboxylic acids: several grams per liter, up to 20 grams per liter or more, depending on the feedstock, the pretreatment and the hydrolysis conditions. For furans: several hundreds of milligrams per liter up to several grams per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions. For phenolics: several tens of milligrams per liter, up to a gram per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions.

[0092] In an embodiment, the recombinant yeast cell is a cell that is naturally capable of alcoholic fermentation, preferably, anaerobic alcoholic fermentation. A recombinant yeast cell preferably has a high tolerance to ethanol, a high tolerance to low pH (i.e. capable of growth at a pH lower than about 5, about 4, about 3, or about 2.5) and towards organic and/or a high tolerance to elevated temperatures.

Glycerol Transporter

[0093] The recombinant yeast comprises a nucleic acid sequence encoding a protein having glycerol transporter activity. By glycerol transporter activity is herein understood the activity of transporting glycerol across the membrane of the recombinant yeast cell.

[0094] The glycerol transporter can suitably allow the recombinant yeast cell to transport glycerol, that is externally available in the medium (e.g. from the backset in corn mash) or secreted after internal cellular synthesis, into the cell. Subsequently the recombinant yeast cell can convert the glycerol to ethanol with help of for example a suitable glycerol dehydrogenase and/or a suitable dihydroxyacetone kinase.

[0095] The protein having glycerol transporter activity is herein also referred to as “glycerol transporter enzyme”, “glycerol transporter protein” or simply “glycerol transporter”. The protein having glycerol transporter activity is also abbreviated herein as “GT”. Preferences for the glycerol transporter protein and the nucleic sequences encoding for such are as described in WO2015/028583, incorporated herein by reference.

[0096] As explained in detail below, preferably the recombinant yeast cell comprises glycerol-proton symporter activity. That is, preferably the protein having glycerol transporter activity is a protein having glycerol-proton symporter activity and preferably the nucleic acid sequence encoding a protein having glycerol transporter activity is a nucleic acid sequence encoding a protein having glycerol-proton symporter activity. Preferably the recombinant yeast cell functionally expresses such nucleic acid sequence encoding for a protein having glycerol-proton symporter activity. Still more preferably the recombinant yeast cell comprises a heterologous glucose-tolerant gene encoding a protein with glycerol-proton symporter activity, suitably allowing the recombinant yeast cell to functionally express such a protein.

[0097] Nowadays many glycerol transporters (such as channels, facilitators and symporters) have been identified, characterized biochemically and the corresponding genes have been cloned (Neves et al., “*Yeast orthologues associated with glycerol transport and metabolism*”, (2004), FEMS Yeast Res. Vol. 5, pages 51-62 and Neves et al “*New insights on glycerol transport in Saccharomyces cerevisiae*”, (2004), FEBS Letters 565 (2004) 160-162), both incorporated herein by reference). [0098] As explained in WO2015/028583, in case of *S. cerevisiae*, four different genes have been implicated with glycerol transport (see Table 4 of WO2015/028583, incorporated herein by reference): FPS1, GUP1, GUP2 and STL1.

[0099] In WO2015/028583, five alternative proteins were selected, heterologous to *S. cerevisiae*. These glycerol transporters, either being a facilitator, a channel, a uniporter or a symporter, were shown, upon overexpression in strains having anaerobic glycerol and acetic acid conversion pathways, to result in improved glycerol uptake activity in yeast cells.

[0100] Preferably the recombinant yeast cell in the present invention functionally expresses one or more nucleic acid sequence(s) and/or corresponding proteins as listed in Table 2 below, or a functional homologue of any of these having a nucleic acid sequence, respectively amino acid sequence, with at least 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 98 or 99% nucleic acid sequence identity, respectively amino acid sequence identity, therewith. Specific examples of suitable protein(s) having glycerol transporter activity and their sequence identity with the protein first listed are summarized in Table 3(a) to 3(e).

TABLE-US-00002 TABLE 2 Preferred glycerol transporter proteins and genes encoding for such:
Species Gene Name Protein Sequence DNA Sequence *Schizosaccharomyces* SPAC977.17 SEQ ID NO: 1 — *pombe Plasmodium falciparum* CAC88373 SEQ ID NO: 2 — *Danio rerio* AQP9 SEQ ID NO: 3 — NP_001171215 *Xenopus tropicalis* NP_001087946 SEQ ID NO: 4 SEQ ID NO: 6 *Zygosaccharomyces rouxii* ZYRO0E01210p SEQ ID NO: 5 SEQ ID NO: 7

TABLE-US-00003 TABLE 3 Description Identity (%) Accession number a) SPAC977.17 from *Schizosaccharomyces pombe* and proteins with a similar amino acid sequence identity. MIP water channel (predicted) [*Schizosaccharomyces* 100 NP_592788.1 *pombe* 972h-] > sp|Q9P7T9.1|YI7H hypothetical protein TDEL_0C00200 [*Torulaspora* 54 XP_003680120.1 *delbrueckii*] > emb|CCE90909.1 KLTH0B00440p [*Lachancea thermotolerans*] > emb|CAR21378.1 55 XP_002551817.1 YFL054C-like protein [*Saccharomyces cerevisiae* × 55 EHN02649.1 *Saccharomyces kudriavzevii* VIN7] BN860_19284g1_1 [*Zygosaccharomyces bailii* CLIB 213] 53 CDF87997.1 aquaglyceroporin, putative [*Aspergillus fumigatus* A1163] 38 EDP47128.1 b) CAC88373 from *Plasmodium falciparum* and proteins with a similar amino acid sequence identity. aquaglyceroporin [*Plasmodium falciparum* 3D7] > pdb|3C02 100 XP_001348009.1 putative aquaglyceroporin [*Plasmodium vivax*] 66 ADD39010.1 glycerol uptake facilitator protein [*Pseudomonas* 38 WP_004881860.1 *viridiflava*] > gb|EKN47076.1 glycerol transporter [*Pseudomonas syringae* group 38 WP_007248745.1 genomosp. 3] > gb|EGH58018.1 glycerol uptake facilitator GlpF [*Caloramator australicus*] > emb|CCC57748.1 33 WP_008907471.1 PREDICTED: aquaporin-10 [*Gorilla gorilla gorilla*] 33 XP_004026862.1 c) AQP9 (NP_001171215) from *Danio rerio* and proteins with a similar amino acid sequence identity. aquaporin-9 [*Danio rerio*] > gb|ACB10576.1| aquaporin-9b 100 NP_001171215.1 [*Danio rerio*] aquaglyceroporin [*Osmerus mordax*] 73 ABG24574.1 PREDICTED: aquaporin-9 [*Gorilla gorilla gorilla*] 51 XP_004056310.1 aquaporin 3 (Gill blood group) [*Xenopus laevis*] > emb|CAA10517.1| 52 NP_001081876.1 aquaporin-3 [*Xenopus laevis*] d) NP_001087946 from *Xenopus tropicalis* and proteins with a similar amino acid sequence identity. aquaporin 3 (Gill blood group) [*Xenopus laevis*] > gb|AAH77577.1 100 NP_001087946.1 aquaporin 3 (Gill blood group) [*Xenopus (Silurana)* 94 NP_001016845.1 *tropicalis*] > emb|CAJ82459.1 PREDICTED: aquaporin-3 [*Gorilla gorilla gorilla*] 78 XP_004047976.1 PREDICTED: aquaporin-3-like [*Takifugu rubripes*] 65 XP_003975282.1 aquaporin [*Sparus aurata*] 50 AAR13054.1 e) ZYRO0E01210p from *Zygosaccharomyces rouxii* and proteins with a similar amino acid sequence identity.

ZYRO0E01210p [*Zygosaccharomyces rouxii*] > emb|CAR30744.1| 100 XP_002498999.1
ZYRO0E01210p [*Zygosaccharomyces rouxii*] BN860_18536g1_1 [*Zygosaccharomyces bailii*
CLIB 213] 82 CDF87965.1 hypothetical protein TDEL_0B07220 [*Torulaspora* 66
XP_003680062.1 *delbrueckii*] > emb|CCE90851.1| hypothetical protein TDEL_0B07220
[*Torulaspora delbrueckii*] Stl1p [*Saccharomyces cerevisiae* S288c] > sp|P39932.2 66 NP_010825.3
sugar transporter STL1 [*Candida albicans* WO-1] 64 EEQ46634.1 monosaccharide transporter
[*Cryptococcus gattii* WM276] > gb|ADV21423.1 45 XP_003193210.1

[0101] As indicated above, the recombinant yeast preferably comprises glycerol-proton symporter activity. That is, the recombinant yeast preferably comprises one or more nucleic acid sequences encoding for a heterologous protein having glycerol-proton symporter activity. A preferred example of such glycerol-proton symporter proteins are STL1 proteins. STL1 proteins belong to the category of “Sugar Transporter-Like proteins” and can be subject to glucose-induced inactivation. STL1 proteins are glycerol proton symporters of the plasma membrane, they can be strongly but transiently induced when cells are subjected to osmotic shock. Preferably the glycerol transporter protein is a STL1 protein and preferably the nucleic acid sequence encoding for the protein having glycerol transporter activity is a nucleic acid sequence encoding for a STL1 protein. Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the protein having glycerol transporter activity is a STL1 protein, most preferably a STL1 protein derived from *Zygosaccharomyces rouxii*.

[0102] More preferably the recombinant yeast comprises one or more glucose-tolerant nucleic acid sequence(s) encoding one or more heterologous protein(s) with glycerol-proton symporter activity.

[0103] Preferably the protein having glycerol transporter activity comprises or consists of: [0104] an amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5; or [0105] a functional homologue of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5; or [0106] a functional homologue of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4 or SEQ ID NO: 5.

[0107] More preferably the recombinant yeast comprises, respectively functionally expresses, a nucleic acid sequence encoding for a protein comprising an amino acid sequence represented by SEQ ID NO: 1, 2, 3, 4 or 5, most preferably represented by SEQ ID NO: 5.

[0108] The proteins having an amino acid sequence of SEQ ID NO: 3 or SEQ ID NO: 5 and functional homologues thereof are most preferred.

[0109] Preferable the nucleic acid sequence encoding the protein having glycerol transporter activity comprises or consists of: [0110] a nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7; or [0111] a functional homologue of SEQ ID NO: 6 or SEQ ID NO: 7, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7; or [0112] a functional homologue of SEQ ID NO: 6 or SEQ ID NO: 7, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200,

no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 6 or SEQ ID NO: 7.

[0113] More preferably the recombinant yeast comprises a glucose-tolerant STL gene, most preferably a STL1 protein derived from *Zygosaccharomyces rouxii*.

[0114] The nucleic acid sequence (e.g. the gene) encoding for the glycerol transporter protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

GT Promoter

[0115] The recombinant yeast cell functionally expresses, a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the “GT promoter”), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more. Herewith is suitably meant that the expression of the glycerol transporter (“GT”) is at least a factor 2 higher under anaerobic conditions than under aerobic conditions. The above can alternatively be phrased as the recombinant yeast cell functionally expressing one or more nucleic acid sequences encoding for a glycerol transporter, wherein the glycerol transporter is under control of a promoter (the “GT promoter”) which has a GT expression ratio anaerobic/aerobic of 2 or more.

[0116] The GT promoter can suitably be operably linked to the nucleic acid sequence encoding the protein having glycerol transporter activity. Preferably, the GT promoter is located in the 5'-region of a glycerol transporter gene, more preferably it is located proximal to the transcriptional start site of a glycerol transporter gene. As indicated above, the glycerol transporter gene is preferably a glycerol-proton symporter gene and more preferably an STL1 gene.

[0117] Preferably the GT promoter is ROX1 repressed. ROX1 is herein Heme-dependent repressor of hypoxic gene(s); that mediates aerobic transcriptional repression of hypoxia induced genes such as COX5b and CYC7; the repressor function is regulated through decreased promoter occupancy in response to oxidative stress; and contains an HMG domain that is responsible for DNA bending activity; involved in the hyperosmotic stress resistance. ROX1 is regulated by oxygen.

[0118] Without wishing to be limited by any kind of theory it is believed that the regulation of ROX1 may function as follows: According to Kwast et al., “*Genomic Analysis of Anaerobically induced genes in Saccharomyces cerevisiae: Functional roles of ROX1 and other factors in mediating the anoxic response*”, (2002), Journal of bacteriology vol 184, no 1 pages 250-265, herein incorporated by reference: “Although Rox1 functions in an O₂-independent manner, its expression is oxygen (heme) dependent, activated by the heme-dependent transcription factor Hap1 [19]. Thus, as oxygen levels fall to those that limit heme biosynthesis [20], ROX1 is no longer transcribed [21], its protein levels fall [22], and the genes it regulates are de-repressed”. Further details and suitable motifs are provided by Keng, T. (1992), “HAP1 and ROX1 form a regulatory pathway in the repression of HEM13 transcription in *Saccharomyces cerevisiae*”, Mol. Cell. Biol. 12: pages 2616-2623, and Ter Kinde and de Steensma, “*A microarray-assisted screen for potential Hap1 and Rox1 target genes in Saccharomyces cerevisiae*”, (2002), Yeast 19: pages 825-840, incorporated herein by reference.

[0119] Preferably, the GT promoter comprises a ROX1 binding motif. The GT promoter may suitably comprise one or more ROX1 binding motif(s).

[0120] Preferably the GT promoter can comprise in its nucleic acid sequence a copy, or one or more copies, of the motif NNNATTGTTNNN (illustrated by SEQ ID NO: 8), wherein “N” represents a nucleic acid chosen from the group consisting of Adenine (A), Guanine (G), Cytosine (C) and Thymine (T).

[0121] More preferably, the GT promoter comprises or consists of a nucleic acid sequence that is

identical to the nucleic acid sequence of the, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0122] Most preferably the recombinant yeast cell according to the invention is a recombinant yeast cell, wherein the GT promoter is the, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the GT promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1.

[0123] In addition or in the alternative, the GT promoter preferably comprises in its nucleic acid sequence one or more copies of the motifs: TCGTTYAG and/or AAAAATTGTTGA (illustrated by SEQ ID NO: 9), wherein "Y" represents C or T.

[0124] The GT promoter can also comprise or consist of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a DAN, TIR or PAU gene.

[0125] Preferably, the GT promoter comprises or consists of a nucleic acid sequence that is the same as that of the, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4 or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0126] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the GT promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.

[0127] More suitably, the GT promoter can comprise or consist of a sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, and YLL025W or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0128] More preferably the recombinant yeast cell according to the invention is a recombinant yeast cell, wherein the GT promoter is the, preferably native, promoter of a, preferably native, gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%,

at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0129] Most preferably the recombinant yeast cell is a recombinant yeast cell, wherein the GT promoter is the native promoter of ANB1, DAN1 or HEM13 of *Saccharomyces cerevisiae*.

[0130] The promoter is herein also simply abbreviated respectively as ANB1 promoter, DAN1 promoter and HEM13 promoter.

[0131] The nucleic acid sequence of the *S. cerevisiae* ANB1 promoter is illustrated in SEQ ID NO: 10. The nucleic acid sequence of the *S. cerevisiae* DAN1 promoter is illustrated in SEQ ID NO: 11. The nucleic acid sequence of the *S. cerevisiae* HEM13 promoter is illustrated in SEQ ID NO: 12.

[0132] Preferable the GT promoter comprises or consists of: [0133] a nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12; or [0134] a functional homologue of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12; or [0135] a functional homologue of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 10, SEQ ID NO: 11 or SEQ ID NO: 12.

[0136] The GT promoter can also be a synthetic oligonucleotide. That is, the GT promoter may be a product of artificial oligonucleotide synthesis. Artificial oligonucleotide synthesis is a method in synthetic biology that is used to create artificial oligonucleotides, such as genes, in the laboratory. Commercial gene synthesis services are now available from numerous companies worldwide, some of which have built their business model around this task. Current gene synthesis approaches are most often based on a combination of organic chemistry and molecular biological techniques and entire genes may be synthesized “de novo”, without the need for precursor template DNA.

[0137] The GT promoter preferably has a GT expression ratio anaerobic/aerobic of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. That is, the GT promoter preferably has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Preferably the expression of the glycerol transporter enzyme (“GT”) is thus at least a factor 2, at least a factor 3, at least a factor 4, at least a factor 5, at least a factor 6, at least a factor 7, at least a factor 8, at least a factor 9, at least a factor 10, at least a factor 20 or at least a factor 50, higher under anaerobic conditions than under aerobic conditions.

[0138] There is no upper limit, and the GT promoter can be a GT promoter that allows the promoted glycerol transporter gene to be expressed only at anaerobic conditions and not at aerobic conditions. That is, preferably the recombinant yeast cell is a recombinant yeast cell, wherein the GT promoter enables expression only during anaerobic conditions.

[0139] For practical reasons a GT expression ratio anaerobic/aerobic in the range from equal to or more than 2 to equal to or less than 10×10 (i.e. 10×10) or to or less than 10×4 (i.e. 10×4) can be considered.

[0140] As indicated above, “Expression” herein refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0141] The GT expression ratio can for example be determined by measuring the amount of glycerol transporter (GT) protein of cells grown under aerobic and anaerobic conditions. The amount of GT protein can be determined by proteomics.

[0142] It is also possible to determine the level or glycerol transporter (GT) expression ratio by measuring the glycerol transporter (GT) activity of cells grown under aerobic and anaerobic conditions, e.g. in a cell-free extract.

[0143] In addition or in the alternative to the above, the level or GT expression ratio can be determined by measuring the transcription level (e.g. as amount of mRNA) of the Glycerol transporter gene of cells grown under aerobic and anaerobic conditions. The skilled person knows how to determine translation levels using methods commonly known in the art, e.g. Q-PCR, real-time PCR, northern blot, RNA-seq.

[0144] The GT promoter advantageously enables higher expression of the glycerol transporter during anaerobic conditions than under aerobic conditions. In the process according to the invention, the recombinant yeast cell preferably expresses the glycerol transporter, where the amount of the glycerol transporter expressed under anaerobic conditions is a multiplication factor higher than the amount of glycerol transporter expressed under aerobic conditions and wherein this multiplication factor is preferably 2 or more, more preferably 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more.

Glycerol Dehydrogenase

[0145] The recombinant yeast cell also functionally expresses a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity.

[0146] The recombinant yeast cell may comprise a NAD.sup.+ dependent glycerol dehydrogenase (EC 1.1.1.6) and/or a NADP.sup.+ dependent glycerol dehydrogenase (EC 1.1.1.72). That is, the recombinant yeast cell may comprise a nucleic acid sequence encoding a protein having NAD.sup.+ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and/or a nucleic acid sequence encoding a protein having NADP.sup.+ dependent glycerol dehydrogenase activity (EC 1.1.1.72).

[0147] Preferably the protein having glycerol dehydrogenase activity is a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein having NAD.sup.+ dependent glycerol dehydrogenase activity (EC 1.1.1.6). Such protein may be from bacterial origin or for instance from fungal origin. An example is *gldA* from *E. coli*.

[0148] In an alternative or additional embodiment, a NADP.sup.+ dependent glycerol dehydrogenase can be present (EC 1.1.1.72).

[0149] A protein having glycerol dehydrogenase activity is herein also referred to as “glycerol dehydrogenase protein”, “glycerol dehydrogenase enzyme” or simply as “glycerol dehydrogenase”. In analogy thereto a protein having NAD⁺ dependent glycerol dehydrogenase activity is herein also referred to as “NAD⁺ dependent glycerol dehydrogenase protein”, “NAD⁺ dependent glycerol dehydrogenase enzyme” or simply as “NAD⁺ dependent glycerol dehydrogenase”. The glycerol dehydrogenase is abbreviated as GLD.

[0150] Preferences for a glycerol dehydrogenase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0151] NAD⁺ dependent glycerol dehydrogenase (EC 1.1.1.6) is an enzyme that catalyzes the chemical reaction:

glycerol+NAD.sup.+custom-characterglycerone+NADH+H.sup.+

[0152] Thus, the two substrates of this enzyme are glycerol and NAD.sup.+, whereas its three products are glycerone, NADH, and H.sup.+. Glyceron and dihydroxyacetone are herein synonyms.

[0153] This glycerol dehydrogenase enzyme belongs to the family of oxidoreductases, specifically those acting on the CH—OH group of donor with NAD.sup.+ or NADP.sup.+ as acceptor. The systematic name of this enzyme class is glycerol:NAD.sup.+ 2-oxidoreductase. Other names in common use include glycerin dehydrogenase, and NAD.sup.+ -dependent glycerol dehydrogenase. This enzyme participates in glycerolipid metabolism. A glycerol dehydrogenase protein may be

further defined by its amino acid sequence. Likewise a glycerol dehydrogenase protein may be further defined by a nucleotide sequence encoding the glycerol dehydrogenase protein. As explained in detail above under definitions, a certain glycerol dehydrogenase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glycerol dehydrogenase protein. [0154] Preferably the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity is a heterologous nucleic acid sequence. Preferably the protein having glycerol dehydrogenase activity is a heterologous protein having NAD⁺ dependent glycerol dehydrogenase activity.

[0155] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell preferably further comprises suitable co-factors to enhance the activity of the glycerol dehydrogenase. For example, the recombinant yeast cell may comprise zinc, zinc ions or zinc salts and/or one or more pathways to include such in the cell.

[0156] Suitable examples of heterologous proteins having glycerol dehydrogenase activity include the glycerol dehydrogenase proteins of respectively *Klebsiella pneumoniae*, *Enterococcus aerogenes*, *Yersinia aldovae*, and *Escherichia coli*. Their amino acid sequences of such proteins have been illustrated respectively by SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 and SEQ ID NO: 16.

[0157] A preferred glycerol dehydrogenase protein is the glycerol dehydrogenase protein encoded by the *gldA* gene from *E. coli*. SEQ ID NO: 16 shows the amino acid sequence of this preferred NAD⁺ dependent glycerol dehydrogenase protein, encoded by the *gldA* gene from *E. coli*. The nucleic acid sequence of the *gldA* gene of *E. coli* is illustrated by SEQ ID NO: 17.

[0158] The recombinant yeast cell therefore most preferably comprises a heterologous nucleotide sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell, as exemplified by the nucleic acid sequence shown in SEQ ID NO:17.

[0159] Preferably the protein having glycerol dehydrogenase activity thus comprises or consists of: [0160] an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16; or [0161] a functional homologue of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16; or [0162] a functional homologue of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15 or SEQ ID NO: 16.

The protein having an amino acid sequence of SEQ ID NO: 16 and functional homologues thereof are most preferred.

[0163] Preferable the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity comprises or consists of: [0164] a nucleic acid sequence of SEQ ID NO: 17; or [0165] a functional homologue of SEQ ID NO: 17, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 17; or [0166] a functional homologue of SEQ ID NO: 17, having one or more mutations,

substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 17, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 17.

[0167] The nucleic acid sequence (e.g. the gene) encoding for the glycerol dehydrogenase protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

[0168] Further examples of suitable glycerol dehydrogenases are listed in table 4(a) to 4(d). At the top of each table the gldA used in the examples and that is BLASTED is mentioned.

TABLE-US-00004 TABLE 4 Description Identity (%) Accession number (a): BLAST Query - gldA from *Escherichia coli* glycerol dehydrogenase, NAD [*Escherichia coli* str. K-12 100 NP_418380.4 substr. MG1655] glycerol dehydrogenase [*Escherichia coli* O127:H6 str. 99 YP_002331714.1 E2348/69] glycerol dehydrogenase [*Citrobacter youngae*] 94 WP_006686227.1 glycerol dehydrogenase [*Citrobacter freundii*] 92 WP_003840533.1 (b): BLAST Query - gldA from *Klebsiella pneumoniae* glycerol dehydrogenase [*Klebsiella pneumoniae* 342] 100 YP_002236495.1 glycerol dehydrogenase [*Citrobacter freundii*] 93 WP_003024745.1 Glycerol dehydrogenase (EC 1.1.1.6) [*Enterobacter* 92 YP_004590977.1 *aerogenes* EA1509E] glycerol dehydrogenase [*Escherichia coli*] 91 WP_016241524.1 glycerol dehydrogenase [*Enterococcus aerogenes*] 87 See examples herein strains with CAS15 glycerol dehydrogenase [*Yersinia aldovae*] 74 WP_004701845.1 glycerol dehydrogenase [*Enterobacteriaceae bacterium* 61 WP_017375113.1 LSJC7] glycerol dehydrogenase [*Citrobacter youngae*] 60 WP_006686227.1 (c): BLAST Query - gldA from *Enterococcus aerogenes* glycerol dehydrogenase [*Enterobacter aerogenes* KCTC 100 YP_004591726.1 2190] Glycerol dehydrogenase (EC 1.1.1.6) [*Enterobacter* 99 YP_007390021.1 *aerogenes* EA1509E] glycerol dehydrogenase [*Klebsiella pneumoniae*] 92 WP_004203683.1 glycerol dehydrogenase [*Escherichia coli*] 88 WP_001322519.1 See examples herein strains with CAS13 glycerol dehydrogenase [*Enterobacter cloacae* subsp. 87 YP_003615506.1 *cloacae* ATCC 13047] (d): BLAST Query - gldA from *Yersinia aldovae* glycerol dehydrogenase [*Yersinia aldovae*] 100 WP_004701845.1 glycerol dehydrogenase [*Yersinia intermedia*] 95 WP_005189747.1 glycerol dehydrogenase [*Serratia liquefaciens* ATCC 81 YP_008232202.1 27592] glycerol dehydrogenase [*Escherichia coli*] 76 WP_016241524.1 See examples herein strains with CAS13. hypothetical protein EAE_03845 [*Enterobacter* 75 YP_004590977.1 *aerogenes* KCTC 2190] glycerol dehydrogenase [*Aeromonas hydrophila*] 65 WP_017410769.1

Dihydroxyacetone Kinase

[0169] The recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity.

[0170] A protein having dihydroxyacetone kinase activity is herein also referred to as “dihydroxyacetone kinase protein”, “dihydroxyacetone kinase enzyme” or simply as “dihydroxyacetone kinase”. The dihydroxyacetone kinase is abbreviated herein as DAK.

[0171] Preferences for a dihydroxyacetone kinase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0172] The protein having dihydroxy kinase activity may suitably belong to the enzyme categories of E.C. 2.7.1.28 and/or E.C. 2.7.1.29. The recombinant yeast cell thus suitably functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 and/or E.C. 2.7.1.29).

[0173] A dihydroxyacetone kinase is preferably herein understood as an enzyme that catalyzes the chemical reaction (EC 2.7.1.29):

ATP+glycerone.Math.ADP+glycerone phosphate

and/or the chemical reaction (EC 2.7.1.28):

ATP+D-glyceraldehyde.Math.ADP+D-glyceraldehyde 3-phosphate.

[0174] Other names in common use for a dihydroxyacetone kinase include glycerone kinase, ATP:glycerone phosphotransferase and (phosphorylating) acetol kinase. It is further understood that glycerone and dihydroxyacetone are the same molecule. A dihydroxyacetone kinase protein may be further defined by its amino acid sequence. Likewise a dihydroxyacetone kinase protein may be further defined by a nucleotide sequence encoding the dihydroxyacetone kinase protein. As explained in detail above under definitions, a certain dihydroxyacetone kinase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the dihydroxyacetone kinase protein.

[0175] Preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a native protein having dihydroxyacetone kinase activity. More preferably, the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.

[0176] Yeast comprises two native isozymes of dihydroxyacetone kinase (DAK1 and DAK2). These native dihydroxyacetone kinase enzymes are preferred according to the invention. Preferably the host cell is a *Saccharomyces cerevisiae* cell and preferably the above native dihydroxyacetone kinase enzymes are the native dihydroxyacetone kinase enzymes of a *Saccharomyces cerevisiae* yeast cell. The amino acid sequences of the native dihydroxyacetone kinase proteins of *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 18 and SEQ ID NO: 19.

[0177] It is also possible for the recombinant yeast cell to functionally express a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity, where the nucleic acid sequence is a heterologous nucleic acid sequence. In an embodiment the recombinant yeast cell comprises a heterologous gene encoding a dihydroxyacetone kinase. Suitable heterologous genes include the genes encoding dihydroxyacetone kinases from *Saccharomyces kudriavzevii*, *Zygosaccharomyces bailii*, *Kluyveromyces lactis*, *Candida glabrata*, *Yarrowia lipolytica*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, *Escherichia coli*, *Yarrowia lipolytica*, *Schizosaccharomyces pombe*, *Botryotinia fuckeliana*, and *Exophiala dermatitidis*. Preferred heterologous proteins having dihydroxyacetone kinase activity include those derived from respectively *Klebsiella pneumoniae*, *Yarrowia lipolytica* and *Schizosaccharomyces pombe*, as illustrated respectively by SEQ ID NO: 20, SEQ ID NO: 21 and SEQ ID NO: 22.

[0178] The recombinant yeast cell may or may not comprise a genetic modification that causes overexpression of a dihydroxyacetone kinase, for example by overexpression of a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity. The nucleotide sequence encoding the dihydroxyacetone kinase may be native or heterologous to the cell. Nucleic acid sequences that may be used for overexpression of dihydroxyacetone kinase in the cells of the invention are for example the dihydroxyacetone kinase genes from *S. cerevisiae* (DAK1) and (DAK2) as e.g. described by Molin et al., "Dihydroxy-acetone kinases in *Saccharomyces cerevisiae* are involved in detoxification of dihydroxyacetone" (2003), J. Biol. Chem., vol. 278: pages 1415-1423, incorporated herein by reference.

[0179] The native nucleic acid sequences encoding dihydroxyacetone kinase proteins in *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 23 and SEQ ID NO: 24.

[0180] Preferably the recombinant yeast cell does comprise a genetic modification that increases the specific activity of any dihydroxyacetone kinase in the cell. For example, the recombinant yeast cell may comprise one or more native and/or heterologous nucleic acid sequence encoding one or more native and/or heterologous dihydroxyacetone kinase protein(s), such as DAK1 and/or DAK2,

that is/are overexpressed. A native dihydroxyacetone kinase, such as DAK1 and/or DAK2, may for example be overexpressed via one or more genetic modifications resulting in more copies of the gene encoding for the dihydroxy acetone kinase than present in the non-genetically modified cell, and/or a non-native promoter may be applied.

[0181] Preferably the recombinant yeast cell is a recombinant yeast cell, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is under control of a promoter. The promoter can for example be a promoter that is native to another gene in the host cell.

[0182] For overexpression of the nucleotide sequence encoding the dihydroxyacetone kinase, the nucleotide sequence (to be overexpressed) can also be placed in an expression construct wherein it is operably linked to suitable expression regulatory regions/sequences to ensure overexpression of the dihydroxyacetone kinase enzyme upon transformation of the expression construct into the host cell of the invention (see above). Suitable promoters for (over)expression of the nucleotide sequence coding for the enzyme having dihydroxyacetone kinase activity include promoters that are preferably insensitive to catabolite (glucose) repression and/or that are active under anaerobic conditions. A dihydroxyacetone kinase that is overexpressed, is preferably overexpressed by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. Preferably, the dihydroxyacetone kinase is overexpressed under anaerobic conditions by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. It is to be understood that these levels of overexpression may apply to the steady state level of the enzyme's activity (specific activity in the cell), the steady state level of the enzyme's protein as well as to the steady state level of the transcript coding for the enzyme in the cell. Overexpression of the nucleotide sequence in the host cell produces a specific dihydroxyacetone kinase activity of at least 0.002, 0.005, 0.01, 0.02 or 0.05 U min⁻¹ (mg protein)⁻¹, determined in cell extracts of the transformed host cells at 30° C. as described e.g. in the Examples of WO2013/081456.

[0183] A most preferred dihydroxyacetone kinase protein is the dihydroxyacetone kinase protein encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 18 shows the amino acid sequence of a suitable dihydroxyacetone kinase protein, encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 23 illustrates the nucleic acid sequence of the *Dak1* gene itself.

[0184] If the recombinant yeast cell comprises one or more overexpressed nucleic acid sequences encoding for a dihydroxyacetone kinase, the recombinant yeast cell therefore most preferably comprises one or more overexpressed nucleotide sequence encoding a dihydroxyacetone kinase derived from *Saccharomyces cerevisiae*, as exemplified by the nucleic acid sequence shown in SEQ ID NO: 23.

[0185] In a preferred embodiment the dihydroxy acetone kinase is encoded by an endogenous gene, e.g. a *DAK1* gene, which endogenous gene is preferably placed under control of a constitutive promoter.

[0186] Preferably the protein having dihydroxy acetone kinase activity thus comprises or consists of: [0187] an amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22; or [0188] a functional homologue of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22; or [0189] a functional homologue of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ

ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, SEQ ID NO: 21 or SEQ ID NO: 22.

The protein having an amino acid sequence of SEQ ID NO: 18 and functional homologues thereof are most preferred.

[0190] Preferable the nucleic acid sequence encoding the protein having dihydroxy acetone kinase activity comprises or consists of: [0191] a nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or [0192] a functional homologue of SEQ ID NO: 23 or SEQ ID NO: 24, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or [0193] a functional homologue of SEQ ID NO: 23 or SEQ ID NO: 24, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24.

[0194] The nucleic acid sequence (e.g. the gene) encoding for the dihydroxy acetone kinase protein may suitably be incorporated in the genome of the recombinant yeast cell.

[0195] Examples of suitable dihydroxyacetone kinases are listed in table 5(a) to 5(d). At the top of each table the DAK's used in the examples and that is BLASTED is mentioned.

TABLE-US-00005 TABLE 5 Accession Description Identity (%) number (a): BLAST Query - DAK1 from *Saccharomyces cerevisiae* Dak1p [*Saccharomyces cerevisiae* S288c] 100 NP_013641.1 dihydroxyacetone kinase [*Saccharomyces cerevisiae* 99 EDN64325.1 YJM789] DAK1-like protein [*Saccharomyces kudriavzevii* IFO 95 EJT44075.1 1802] ZYBA0S11-03576g1_1 [*Zygosaccharomyces bailii* CLIB 77 CDF91470.1 213] hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] 70 XP_451751.1 hypothetical protein [*Candida glabrata* CBS 138] 63 XP_449263.1 Dak2p [*Saccharomyces cerevisiae* S288c] 44 NP_116602.1 DAK1 [*Yarrowia lipolytica*] 41 See examples herein strains with CAS23 (b): BLAST Query - dhaK from *Klebsiella pneumoniae* dihydroxyacetone kinase subunit DhaK [*Klebsiella* 100 YP_002236493.1 *pneumoniae* 342] dihydroxyacetone kinase subunit K [*Klebsiella* 99 WP_004149886.1 *pneumoniae*] dihydroxyacetone kinase subunit K [*Enterobacter* 96 WP_020077889.1 *aerogenes*] dihydroxyacetone kinase subunit DhaK [*Escherichia coli* 88 YP_002407536.1 IAI39] dihydroxyacetone kinase, DhaK subunit [*Escherichia* 87 WP_001398949.1 *coli*] (c): BLAST Query - DAK1 from *Yarrowia lipolytica* YALI0F09273p [*Yarrowia lipolytica*] 100 XP_505199.1 dihydroxyacetone kinase [*Schizosaccharomyces pombe*] 46 AAC83220.1 dihydroxyacetone kinase Dak1 [*Schizosaccharomyces* 45 NP_593241.1 *pombe* 972h-] dihydroxyacetone kinase [*Saccharomyces cerevisiae* 44 EDV12567.1 RM11-1a] Dak2p [*Saccharomyces cerevisiae* JAY291] 44 EEU04233.1 BN860_19306g1_1 [*Zygosaccharomyces bailii* CLIB 213] 44 CDF87998.1 Dak1p [*Saccharomyces cerevisiae* CEN.PK113-7D] 42 EIW08612.1 See examples herein strains with CAS21 (d): BLAST Query - DAK1 from *Schizosaccharomyces pombe* dihydroxyacetone kinase Dak1 [*Schizosaccharomyces* 100 NP_593241.1 *pombe* 972h-] putative dihydroxyacetone kinase protein [*Botryotinia* 48 EMR88164.1 *fuckeliana* BcDW1] Dihydroxyacetone kinase 1 [*Fusarium oxysporum* f. sp. 48 ENH64704.1 *cubense* race 1] Dak1p [*Saccharomyces cerevisiae* CEN.PK113-7D] 46 EIW08612.1 Dak2p [*Saccharomyces cerevisiae* JAY291] 44 EEU04233.1 dihydroxyacetone kinase [*Exophiala dermatitidis* 42 EHY55064.1 NIH/UT8656]

Redox Sink

[0196] Preferably the recombinant yeast cell can further comprise one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.

[0197] For example, these one or more genetic modifications can be one or more genetic modifications for the functional expression of one or more, optionally heterologous, nucleic acid sequences encoding for one or more NAD⁺/NADH dependent proteins that function in a metabolic pathway to convert NADH to NAD⁺. Several examples of such metabolic pathways exist, as illustrated further below.

[0198] For example, the “one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink” can be chosen from the group consisting of: [0199] a) one or more genetic modifications comprising or consisting of: [0200] a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or [0201] a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or [0202] a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

and/or [0203] b) one or more genetic modifications comprising or consisting of: [0204] a nucleic acid sequence encoding for a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or [0205] a nucleic acid sequences encoding for a protein having phosphoribulokinase (PRK) activity; and [0206] optionally, a nucleic acid sequence encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity. and/or [0207] c) one or more genetic modifications comprising or consisting of: [0208] a nucleic acid sequence encoding a protein comprising NADH dependent acetylating acetaldehyde dehydrogenase activity.

[0209] For example, WO2014/081803 describes a recombinant microorganism expressing a heterologous phosphoketolase, phosphotransacetylase or acetate kinase and bifunctional acetaldehyde-alcohol dehydrogenase, incorporated herein by reference; and WO2015/148272 describes a recombinant *S. cerevisiae* strain expressing a heterologous phosphoketolase, phosphotransacetylase and acetylating acetaldehyde dehydrogenase, incorporated herein by reference. Further WO2018172328A1 describes a recombinant cell that may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity. The phosphoketolase (PKL) routes described in WO2014/081803, WO2015/148272 and WO2018172328A1, all incorporated herein by reference, provide preferred metabolic pathways to convert NADH to NAD⁺ and the NADH dependent phosphoketolase described therein is a preferred NADH dependent protein for application in the current invention.

Rubisco

[0210] As indicated above, the recombinant yeast cell may or may not functionally express one or more heterologous nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more molecular chaperones for Rubisco.

[0211] More preferably the recombinant yeast cell functionally expresses: [0212] a heterologous nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or [0213] a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or [0214] optionally one or more heterologous nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0215] The protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity is herein also referred to as “ribulose-1,5-biphosphate carboxylase oxygenase”, “ribulose-1,5-biphosphate carboxylase oxygenase protein”, “ribulose-1,5-biphosphate carboxylase oxygenase enzyme”, “Rubisco enzyme”, “Rubisco protein” or simply “Rubisco”. A ribulose-1,5-biphosphate

carboxylase oxygenase may be further defined by its amino acid sequence. Likewise a ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by a nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. As explained in detail above under definitions, a certain ribulose-1,5-biphosphate carboxylase oxygenase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. Preferences for the Rubisco protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0216] The Rubisco protein may suitably be selected from the group of eukaryotic and prokaryotic Rubisco proteins. The Rubisco protein is preferably from a non-phototrophic organism. For example, the Rubisco protein may be from a chemolithoautotrophic microorganism. Good results have been achieved with a bacterial Rubisco protein. Preferably, the Rubisco protein originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.

[0217] The Rubisco protein may be a single-subunit Rubisco protein or a Rubisco protein having more than one subunit. Preferably the Rubisco protein is a single-subunit Rubisco protein. Good results have been obtained with a Rubisco protein that is a so-called form-II Rubisco protein. Especially good results were achieved with a Rubisco protein encoded by a *cbbM* gene, also referred to as *CbbM*.

[0218] A preferred Rubisco protein is the Rubisco protein encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 25 shows the amino acid sequence of a suitable Rubisco protein, encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 26 illustrates the nucleic acid sequence of the *cbbM* gene from *Thiobacillus denitrificans*, codon optimized for *S. cerevisiae*.

[0219] Preferably the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity thus comprises or consists of: [0220] an amino acid sequence of SEQ ID NO: 25; or [0221] a functional homologue of SEQ ID NO: 25, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 25; or [0222] a functional homologue of SEQ ID NO: 25, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25.

[0223] Preferable the nucleic acid sequence encoding the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity comprises or consists of: [0224] a nucleic acid sequence of SEQ ID NO: 26; or [0225] a functional homologue of SEQ ID NO: 26, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 26; or [0226] a functional homologue of SEQ ID NO: 26, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26.

[0227] Examples of other suitable Rubisco polypeptides and their origin are given in Table 1 of WO2014/129898, incorporated herein by reference, and in Table 6 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:25.

[0228] The nucleic acid sequence (e.g. the gene) encoding for the ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898 and by the article of Guadalupe-Medina et al., “*Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast*”, published in Biotechnol, Biofuels, 2013, vol. 6, p. 125, both herein incorporated by reference.

TABLE-US-00006 TABLE 6 Natural Rubisco polypeptides suitable for expression MAX Source Accession no. ID (%) *Thiobacillus denitrificans* AAA99178.2 100 *Sideroxydans lithotrophicus* ES-1 YP_003522651.1 94 *Thiothrix nivea* DSM 5205 ZP_10101642.1 91 *Halothiobacillus neapolitanus* c2 YP_003262978.1 90 *Acidithiobacillus ferrooxidans* YP_002220242.1 88 ATCC 53993 *Rhodospirillum rubrum* T118 YP_522655.1 86 *Thiorhodococcus dreschii* AZ1 ZP_08824342.1 85 uncultured prokaryote AGE14067.1 82

[0229] As indicated above, the Rubisco protein is suitably functionally expressed in the recombinant yeast cell, at least during use in a fermentation process.

[0230] The nucleic acid sequence encoding for the Rubisco protein can be present in one, two or more copies with the recombinant yeast cell. Without wishing to be bound by any kind of theory it is believed that the robustness of the recombinant yeast cell is best served when the nucleic acid sequence (e.g. the gene) encoding for the Rubisco protein is present in the recombinant yeast cell in less than 12 copies, more preferably less than 8 copies. Preferably the recombinant yeast cell therefore comprises in the range from equal to or more than 1 copy, more preferably equal to or more than 2 copies, to equal to or less than 7 copies, more preferably equal to or less than 6 copies of a nucleic acid sequence (e.g. a gene) encoding for a Rubisco protein. The recombinant yeast cell may for example comprise one, two, three, four, five, six or seven copies of a nucleic acid sequence encoding for ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco).

[0231] To increase the likelihood that the Rubisco protein is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleic acid sequence encoding the Rubisco protein and other proteins as described herein (see below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleic acid sequence encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, “*The codon adaptation index—a measure of directional synonymous codon usage bias, and its potential applications*”, (1987), published in Nucleic Acids Research vol. 15, pages 1281-1295; also see: Jansen et al., “*Revisiting the codon adaptation index from a whole-genome perspective: analyzing the relationship between gene expression and codon occurrence in yeast using a variety of models*”, (2003), Nucleic Acids Res. Vol. 31(8), pages 2242-51). An adapted nucleic acid sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Preferably, the sequences have been codon optimized for expression in the fungal host cell in question, such as for example *Saccharomyces cerevisiae* cells.

[0232] Preferably the functionally expressed Rubisco protein has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent ^{14}C -bicarbonate incorporation by cell extracts of at least 1 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1, in particular an activity of at least 2 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1, more in particular an activity of at least 4 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1. The upper limit for the activity is not critical. In practice, the activity may be about 200 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1 or less, in

particular 25 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1, more in particular 15 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1 or less, e.g. about 10 nmol.Math.min.sup.-1.Math.(mg protein).sup.-1 or less. The conditions for an assay for determining this Rubisco activity are as found in the Examples (e.g. Example 4) of WO2014/129898, incorporated herein by reference.

Phosphoribulokinase

[0233] Preferably recombinant yeast cell is also functionally expressing a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity (EC2.7.1.19; PRK).

[0234] The protein having phosphoribulokinase (PRK) activity is herein also referred to as “phosphoribulokinase protein”, “phosphoribulokinase enzyme”, “phosphoribulokinase”, “PRK enzyme”, “PRK protein” or simply “PRK”. Preferences for the PRK protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0235] A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalyzing the chemical reaction:

ATP+D-ribulose 5-phosphate $\rightleftharpoons$ ADP+D-ribulose 1,5-bisphosphate

Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate; its two products are ADP and D-ribulose 1,5-bisphosphate.

[0236] The PRK protein belongs to the family of transferases, specifically those transferring phosphorus-containing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. The PRK enzyme participates in carbon fixation. A phosphoribulokinase (PRK) protein may be further defined by its amino acid sequence. Likewise a phosphoribulokinase (PRK) protein may be further defined by a nucleotide sequence encoding the phosphoribulokinase (PRK). As explained in detail above under definitions, a certain phosphoribulokinase (PRK) that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the phosphoribulokinase (PRK).

[0237] The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the PRK protein originates from a plant selected from Caryophyllales, in particular from Amaranthaceae, more in particular from *Spinacia*.

[0238] A preferred PRK protein is the PRK protein from *Spinacia*. SEQ ID NO: 27 shows the amino acid sequence of such PRK protein from *Spinacia*. SEQ ID NO: 28 illustrates the nucleic acid sequence of the prk gene from *Spinacia oleracea*—codon optimized for *S. cerevisiae*.

[0239] Preferably the protein having phosphoribulokinase (PRK) activity thus comprises or consists of: [0240] an amino acid sequence of SEQ ID NO: 27; or [0241] a functional homologue of SEQ ID NO: 27, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 27; or [0242] a functional homologue of SEQ ID NO: 27, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27.

[0243] Preferable the nucleic acid sequence encoding the protein having phosphoribulokinase (PRK) activity comprises or consists of: [0244] a nucleic acid sequence of SEQ ID NO: 28; or

[0245] a functional homologue of SEQ ID NO: 28, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 28; or [0246] a functional homologue of SEQ ID NO: 28, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28.

[0247] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoribulokinase (PRK) activity may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

[0248] Examples of suitable PRK polypeptides and their origin are given in Table 2 of WO2014/129898, incorporated herein by reference, and in Table 7 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:27.

TABLE-US-00007 TABLE 7 Natural PRK polypeptides suitable for expression with identity to PRK from *Spinacia* Source Accession no. MAX ID (%) *Spinacia oleracea* P09559.1 100 *Medicago truncatula* XP_003612664.1 88 *Arabidopsis thaliana* NP_174486.1 87 *Vitis vinifera* XP_002263724.1 86 *Closterium peracerosum* BAL03266.1 82 *Zea mays* NP_001148258.1 78

[0249] The nucleic acid sequences encoding for the PRK protein may be under the control of a promoter (the “PRK promoter”) that enables higher expression under anaerobic conditions than under aerobic conditions. Examples of such promoters are described in WO2017/216136A1 and WO2018/228836, both herein incorporated by reference. More preferably such promoter has a PRK expression ratio anaerobic/aerobic of 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Further preferences are as described in WO2018/228836, incorporated herein by reference.

Rubisco Chaperones

[0250] Optionally, the recombinant yeast cell further comprises one or more, preferably heterologous, nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0251] Suitably such molecular chaperones are also referred herein as “chaperone protein”, “chaperonin” or simply “chaperone”. Preferences for the chaperones and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0252] Preferably the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0253] Chaperonins are proteins that provide favorable conditions for the correct folding of other proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine triphosphate (ATP). A review article about chaperones that is useful herein is written by Yébenes et al., “*Chaperonins: two rings for folding*” (2011), Trends in Biochemical Sciences, Vol. 36, No. 8, pages 424-432, incorporated herein by reference.

[0254] The chaperone or chaperones may be prokaryotic chaperones or eukaryotic chaperones. In addition, the chaperones may be homologous or heterologous. For example, the recombinant yeast cell may comprises one or more nucleic acid sequence encoding one or more homologous or heterologous, prokaryotic or eukaryotic, molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the recombinant yeast cell, in particular with

at least one of Rubisco and PRK.

[0255] Suitably the chaperone or chaperones are derived from a bacterium, more preferably from *Escherichia*, in particular *E. coli*. Preferred chaperones are GroEL and GroES from *E. coli*. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60.

[0256] If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins. In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any recombinant yeast cell may serve as a chaperone for the Rubisco. This is described for example by Zeilstra-Ryalls et al., “*The universally conserved GroE (Hsp60) chaperonins*”, (1991), Annu Rev Microbiol. vol. 45, pages 301-325; and Horwich et al., “*Two Families of Chaperonin: Physiology and Mechanism*” (2007), Annu. Rev. Cell. Dev. Biol. Vol. 23, pages 115-145, both herewith incorporated by reference.

[0257] Good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.

[0258] As an alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroES, respectively the amino sequence of SEQ ID NO: 31.

[0259] SEQ ID NO:31 provides a preferred translated protein sequence, based on GroES of *Escherichia coli*. SEQ ID NO: 32 provides a synthetic nucleic acid sequence, based on GroES from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0260] Examples of suitable natural chaperones polypeptide homologous to GroES are given in Table 8.

TABLE-US-00008 TABLE 8 Natural chaperones homologous to GroES polypeptides suitable for expression >gi|115388105|ref|XP_001211558.1|:2-101 10 kDa heat shock protein, mitochondrial [*Aspergillus terreus* NIH2624] >gi|116196854|ref|XP_001224239.1|:1-102 conserved hypothetical protein [*Chaetomium globosum* CBS 148.51] >gi|119175741|ref|XP_001240050.1|:3-102 hypothetical protein CIMG_09671 [*Coccidioides immitis* RS] >gi|119471607|ref|XP_001258195.1|:12-111 chaperonin, putative [*Neosartorya fischeri* NRRL181] >gi|121699818|ref|XP_001268174.1|:8-106 chaperonin, putative [*Aspergillus clavatus* NRRL 1] >gi|126274604|ref|XP_001387607.1|:2-102 predicted protein [*Scheffersomyces stipitis* CBS 6054] >gi|146417701|ref|XP_001484818.1|:5-106 conserved hypothetical protein [*Meyerozyma guilliermondii* ATCC 6260] >gi|154303611|ref|XP_001552212.1|:1-102 10 kDa heat shock protein, mitochondrial [*Botryotinia fuckeliana* B05.10] >gi|156049571|ref|XP_001590752.1|:1-102 hypothetical protein SS1G_08492 [*Sclerotinia sclerotiorum* 1980] >gi|156840987|ref|XP_001643870.1|:1-103 hypothetical protein Kpol_495p10 [*Vanderwaltozyma polyspora* DSM 70924] >gi|169608295|ref|XP_001797567.1|:1-101 hypothetical protein SNOG_07218 [*Phaeosphaeria nodorum* SN15] >gi|171688384|ref|XP_001909132.1|:1-102 hypothetical protein [*Podospira anserina* S mat+] >gi|189189366|ref|XP_001931022.1|:71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis* Pt- 1C-BFP] >gi|19075598|ref|NP_588098.1|:1-102 mitochondrial heat shock protein Hsp10 (predicted) [*Schizosaccharomyces pombe* 972h-] >gi|212530240|ref|XP_002145277.1|:3-100 chaperonin, putative [*Talaromyces marneffe* ATCC 18224] >gi|212530242|ref|XP_002145278.1|:3-95 chaperonin, putative [*Talaromyces marneffe* ATCC 18224] >gi|213404320|ref|XP_002172932.1|:1-102 mitochondrial heat shock protein Hsp10 [*Schizosaccharomyces japonicus* yFS275] >gi|225557301|gb|EEH05587.1|:381-478 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* G186AR] >gi|225684092|gb|EEH22376.1|:3-100 heat shock protein [*Paracoccidioides brasiliensis* Pb03]

>gi|238490330|ref|XP_002370502.1|:2-104 chaperonin, putative [*Aspergillus flavus* NRRL3357
 >gi|238878220|gb|EEQ41858.1|:1-106 10 kDa heat shock protein, mitochondrial [*Candida albicans* WO-1] >gi|240280207|gb|EER43711.1|:426-523 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* H143] >gi|241950445|ref|XP_002417945.1|:1-103 10 kDa chaperonin, putative; 10 kDa heat shock protein mitochondrial (hsp10), putative [*Candida dubliniensis* CD36] >gi|242819222|ref|XP_002487273.1|:90-182 chaperonin, putative [*Talaromyces stipitatus* ATC] >gi|254566327|ref|XP_002490274.1|:1-102 Putative protein of unknown function [*Komagataella pastoris* GS115] >gi|254577241|ref|XP_002494607.1|:1-103 ZYRO0A05434p [*Zygosaccharomyces rouxii*] >gi|255717999|ref|XP_002555280.1|:1-103 KLTH0G05588p [*Lachancea thermotolerans*] >gi|255956581|ref|XP_002569043.1|:2-101 Pc21g20560 [*Penicillium chrysogenum* Wisconsin 54-1255] >gi|258572664|ref|XP_002545094.1|:16-108 chaperonin GroS [*Uncinocarpus reesii* 1704] >gi|261190594|ref|XP_002621706.1|:3-100 chaperonin [*Ajellomyces dermatitidis* SLH14081] >gi|295664909|ref|XP_002793006.1|:3-100 10 kDa heat shock protein, mitochondrial [*Paracoccidioides* sp. 'lutzii' Pb01] >gi|296412657|ref|XP_002836039.1|:76-177 hypothetical protein [*Tuber melanosporum* Mel28] >gi|302307854|ref|NP_984626.2|:2-102 AEL235Wp [*Ashbya gossypii* ATCC 10895] >gi|302894117|ref|XP_003045939.1|:1-102 predicted protein [*Nectria haematococca* mpVI 77- 13-4] >gi|303318351|ref|XP_003069175.1|:3-100 10 kDa heat shock protein, mitochondrial , putative [*Coccidioides posadasii* C735 delta SOWgp] >gi|310795300|gb|EFQ30761.1|:1-102 chaperonin 10 kDa subunit [*Glomerella graminicola* M1.001] >gi|315053085|ref|XP_003175916.1|:12-109 chaperonin GroS [*Arthroderma gypseum* CBS 118893] >gi|317032114|ref|XP_001394060.2|:334-433 heat shock protein [*Aspergillus niger* CBS 513.88] >gi|317032116|ref|XP_001394059.2|:2-101 heat shock protein [*Aspergillus niger* CBS 513.88] >gi|320583288|gb|EFW97503.1|:6-106 chaperonin, putative heat shock protein, putative [*Ogataea parapolymorpha* DL-1] >gi|320591507|gb|EFX03946.1|:1-102 heat shock protein [*Grosmannia clavigera* kw1407] >gi|322700925|gb|EFY92677.1|:1-102 chaperonin [*Metarhizium acridum* CQMa 102] >gi|325096696|gb|EGC50006.1|:409-506 pre-mRNA polyadenylation factor fip1 [*Ajellomyces capsulatus* H88] >gi|326471604|gb|EGD95613.1|:14-111 chaperonin 10 Kd subunit [*Trichophyton tonsurans* CBS112818] >gi|327293056|ref|XP_003231225.1|:3-100 chaperonin [*Trichophyton rubrum* CBS 118892] >gi|330942654|ref|XP_003306155.1|:37-136 hypothetical protein PTT_19211 [*Pyrenophora teres* f. *teres* 0-1] >gi|336268042|ref|XP_003348786.1|:47-147 hypothetical protein SMAC_01809 [*Sordaria macrospora* khell] >gi|340519582|gb|EGR49820.1|:1-109 predicted protein [*Trichoderma reesei* QM6a] >gi|340960105|gb|EGS21286.1|:3-103 putative mitochondrial 10 kDa heat shock protein [*Chaetomium thermophilum* var. *thermophilum* DSM 1495] >gi|342883802|gb|EGU84224.1|: 1-102 hypothetical protein FOXB_05181 [*Fusarium oxysporum* Fo5176] >gi|344302342|gb|EGW32647.1|:2-102 hypothetical protein SPAPADRAFT_61712 [*Spathaspora passalidarum* NRRL Y-27907] >gi|345570750|gb|EGX53571.1|:1-102 hypothetical protein AOL_s00006g437 [*Arthrobotrys oligospora* ATCC 24927] >gi|346321154|gb|EGX90754.1|:1-102 chaperonin [*Cordyceps militaris* CM01] >gi|346970393|gb|EGY13845.1|:1-102 heat shock protein [*Verticillium dahliae* VdLs. 17] >gi|354548296|emb|CCE45032.1|:1-106 hypothetical protein CPAR2_700360 [*Candida parapsilosis*] >gi|358385052|gb|EHK22649.1|:1-102 hypothetical protein TRIVIDRAFT_230640 [*Trichoderma virens* Gv 29-8] >gi|358393422|gb|EHK42823.1|:1-101 hypothetical protein TRIATDRAFT_258186 [*Trichoderma atroviride* IMI 206040] >gi|361126733|gb|EHK98722.1|:1-97 putative 10 kDa heat shock protein, mitochondrial [*Glare lozoyensis* 74030] >gi|363753862|ref|XP_003647147.1|:2-102 hypothetical protein Ecym_5593 [*Eremothecium cymbalariae* DBVPG#7215] >gi|365758401|gb|EHN00244.1|:1-106 Hsp10p [*Saccharomyces cerevisiae* x *Saccharomyces kudriavzevii* VIN7] >gi|365987664|ref|XP_003670663.1|:1-103 hypothetical protein NDAI_0F01010 [*Naumovozyma dairenensis* CBS 421] >gi|366995125|ref|XP_003677326.1|:1-103 hypothetical protein NCAS_0G00860 [*Naumovozyma*

castellii CBS 4309] >gi|366999797|ref|XP_003684634.1|:1-103 hypothetical protein
TPHA_0C00430 [*Tetrapisispora phaffii* CBS 4417] >gi|367009030|ref|XP_003679016.1|:1-103
hypothetical protein TDEL_0A04730 [*Torulaspora delbruekii*]
>gi|367023138|ref|XP_003660854.1|:1-104 hypothetical protein MYCTH_59302 [*Myceliophthora
thermophila* ATCC 42464] >gi|367046344|ref|XP_003653552.1|:1-102 hypothetical protein
THITE_2116070 [*Thielavia terrestris* NRRL8126] >gi|378726440|gb|EHY52899.1|:9-109
chaperonin GroES [*Exophiala dermatitidis* NIH/UT8656] >gi|380493977|emb|CCF33483.1|:1-102
chaperonin 10 kDa subunit [*Colletotrichum higginsianu*] >gi|385305728|gb|EIF49680.1|:1-102 10
kda heat shock mitochondrial [*Dekkera bruxellensis* AWRI1499]
>gi|389628546|ref|XP_003711926.1|:1-102 hsp10-like protein [*Magnaporthe oryzae* 70-15]
>gi|396462608|ref|XP_003835915.1|:1-101 similar to 10 kDa heat shock protein [*Leptosphaeria
maculans* JN3] >gi|398392541|ref|XP_003849730.1|:1-102 hypothetical protein
MYCGRDRAFT_105721 [*Zymoseptoria tritici* IPO323] >gi|400597723|gb|EJP65453.1|:24-124
chaperonin 10 kDa subunit [*Beauveria bassiana* ARSEF 2860] >gi|401623646|gb|EJS41738.1|:1-
106 hsp10p [*Saccharomyces arboricola* H-6] >gi|401842164|gb|EJT44422.1|:1-92 HSP10-like
protein [*Saccharomyces kudriavzevii* IFO 1802] >gi|402084027|gb|EJT79045.1|:1-102 hsp10-like
protein [*Gaeumannomyces graminis* var. *tritici*] >gi|403215209|emb|CCK69709.1|:1-104 hypothetical
protein KNAG_0C06130 [*Kazachstania naganishii* CBS 8797]
>gi|406604629|emb|CCH43969.1|:4-100 hypothetical protein BN7_3524 [*Wickerhamomyces
ciferrii*] >gi|406867021|gb|EKD20060.1|:56-156 hypothetical protein MBM_02012 [*Marssonina
brunnea* f.sp. '*multigermtubi*' MB_m1] >gi|407926227|gb|EKG19196.1|:74-174 GroES-like protein
[*Macrophomina phaseolina* MS6] >gi|408398157|gb|EKJ77291.1|:11-111 hypothetical protein
FPSE_02566 [*Fusarium pseudograminearum* CS3096] >gi|410082063|ref|XP_003958610.1|:1-103
hypothetical protein KAFR_0H00660 [*Kazachstania africana* CBS2517]
>gi|425777664|gb|EKV15823.1|:58-157 Chaperonin, putative [*Penicillium digitatum* Pd1]
>gi|440639680|gb|ELR09599.1|:1-102 chaperonin GroES [*Geomyces destructans* 20631-21]
>gi|444323906|ref|XP_004182593.1|:1-105 hypothetical protein TBLA_OJ00760
[*Tetrapisisporablattae* CBS 6284] >gi|448083208|ref|XP_004195335.1|:2-101 Piso0_005888
[*Millerozyma farinosa* CBS 7064] >gi|448087837|ref|XP_004196425.1|:2-102 Piso0_005888
[*Millerozyma farinosa* CBS 7064] >gi|448534948|ref|XP_003870866.1|:1-106 Hsp10 protein
[*Candida orthopsilosis* Co 90-125] >gi|449295977|gb|EMC91998.1|:1-102 hypothetical protein
BAUCODRAFT_39148 [*Baudoinia compn*] >gi|46123659|ref|XP_386383.1|:3-103 hypothetical
protein FG06207.1 [*Gibberella zeae* PH-1] >gi|50289455|ref|XP_447159.1|:1-103 hypothetical
protein [*Candida glabrata* CBS 138] >gi|50308731|ref|XP_454370.1|:1-103 hypothetical protein
[*Kluyveromyces lactis* NRRL Y-1140] >gi|50411066|ref|XP_457014.1|:1-106 DEHA2B01122p
[*Debaryomyces hansenii* CBS767] >gi|50545998|ref|XP_500536.1|:1-102 YALI0B05610p
[*Yarrowia lipolytica*] >gi|51013895|gb|AAT93241.1|:1-106 YOR020C [*Saccharomyces cerevisiae*]
>gi|6324594|ref|NP_014663.1|:1-106 Hsp10p [*Saccharomyces cerevisiae* S288c]
>gi|67523953|ref|XP_660036.1|:2-101 hypothetical protein AN2432.2 [*Aspergillus nidulans* FGSC
A4] >gi|70992219|ref|XP_750958.1|:12-106 chaperonin [*Aspergillus fumigatus* Af293]
>gi|85079266|ref|XP_956315.1|:1-104 hypothetical protein NCU04334 [*Neurospora crassa*
OR74A]

[0261] As an alternative to GroEL a functional homologue of GroEL may be present, in particular a
functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at
least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least
85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid
sequence of GroEL, respectively the amino sequence of SEQ ID NO: 29.

[0262] SEQ ID NO:29 provides a preferred translated protein sequence, based on GroEL of
Escherichia coli. SEQ ID NO: 30 provides a synthetic nucleic acid sequence, based on GroEL from
Escherichia coli, codon optimized for expression in *Saccharomyces cerevisiae*.

[0263] Suitable natural chaperones polypeptides homologous to GroEL are given in Table 9.

TABLE-US-00009 TABLE 9 Natural chaperones homologous to GroEL polypeptides suitable for expression

>gi|115443330|ref|XP_001218472.1| heat shock protein 60, mitochondrial precursor [*Aspergillus terreus* NIH2624] >gi|114188341|gb|EAU30041.1| heat shock protein 60, mitochondrial precursor [*Aspergillus terreus* NIH2624] >gi|119480793|ref|XP_001260425.1| antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181] >gi|119408579|gb|EAW18528.1| antigenic mitochondrial protein HSP60, putative [*Neosartorya fischeri* NRRL 181] >gi|126138730|ref|XP_001385888.1| hypothetical protein PICST_90190 [*Scheffersomyces stipitis* CBS 6054] >gi|126093166|gb|ABN67859.1| mitochondrial groEL-type heat shock protein [*Scheffersomyces stipitis* CBS 6054] >gi|145246630|ref|XP_001395564.1| heat shock protein 60 [*Aspergillus niger* CBS 513.88] >gi|134080285|emb|CAK46207.1| unnamed protein product [*Aspergillus niger*] >gi|350636909|gb|EHA25267.1| hypothetical protein ASPNIDRAFT_54001 [*Aspergillus niger* ATCC 1015] >gi|146413148|ref|XP_001482545.1| heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260] >gi|154277022|ref|XP_001539356.1| heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAm1] >gi|150414429|gb|EDN09794.1| heat shock protein 60, mitochondrial precursor [*Ajellomyces capsulatus* NAm1] >gi|154303540|ref|XP_001552177.1| heat shock protein 60 [*Botryotinia fuckeliana* B05.10] >gi|347840915|emb|CCD55487.1| similar to heat shock protein 60 [*Botryotinia fuckeliana*] >gi|156063938|ref|XP_001597891.1| heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980] >gi|154697421|gb|EDN97159.1| heat shock protein 60, mitochondrial precursor [*Sclerotinia sclerotiorum* 1980 UF-70] >gi|156844469|ref|XP_001645297.1| hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi|156115957|gb|EDO17439.1| hypothetical protein Kpol_1037p35 [*Vanderwaltozyma polyspora* DSM 70294] >gi|16416029|emb|CAB91379.2| probable heat-shock protein hsp60 [*Neurospora crassa*] >gi|350289516|gb|EGZ70741.1| putative heat-shock protein hsp60 [*Neurospora tetrasperma* FGSC 2509] >gi|169626377|ref|XP_001806589.1| hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15] >gi|111055053|gb|EAT76173.1| hypothetical protein SNOG_16475 [*Phaeosphaeria nodorum* SN15] >gi|169783766|ref|XP_001826345.1| heat shock protein 60 [*Aspergillus oryzae* RIB40] >gi|238493601|ref|XP_002378037.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi|83775089|dbj|BAE65212.1| unnamed protein product [*Aspergillus oryzae* RIB40] >gi|220696531|gb|EED52873.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus flavus* NRRL3357] >gi|391869413|gb|EIT78611.1| chaperonin, Cpn60/Hsp60p [*Aspergillus oryzae* 3.042] >gi|189190432|ref|XP_001931555.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi|187973161|gb|EDU40660.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP] >gi|190348913|gb|EDK41467.2| heat shock protein 60, mitochondrial precursor [*Meyerozyma guilliermondii* ATCC 6260] >gi|225554633|gb|EEH02929.1| hsp60-like protein [*Ajellomyces capsulatus* G186AR] >gi|238880068|gb|EEQ43706.1| heat shock protein 60, mitochondrial precursor [*Candida albicans* WO-1] >gi|239613490|gb|EEQ90477.1| chaperonin GroL [*Ajellomyces dermatitidis* ER-3] >gi|240276977|gb|EER40487.1| hsp60-like protein [*Ajellomyces capsulatus* H143] >gi|241958890|ref|XP_002422164.1| heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi|223645509|emb|CAX40168.1| heat shock protein 60, mitochondrial precursor, putative [*Candida dubliniensis* CD36] >gi|254572906|ref|XP_002493562.1| Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi|238033361|emb|CAY71383.1| Tetradecameric mitochondrial chaperonin [*Komagataella pastoris* GS115] >gi|254579947|ref|XP_002495959.1| ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi|238938850|emb|CAR27026.1| ZYRO0C07106p [*Zygosaccharomyces rouxii*] >gi|255712781|ref|XP_002552673.1| KLTH0C10428p [*Lachancea thermotolerans*] >gi|238934052|emb|CAR22235.1| KLTH0C10428p [*Lachancea thermotolerans*]

CBS 6340] >gi|255721795|ref|XP_002545832.1| heat shock protein 60, mitochondrial precursor
 [*Candida tropicalis* MYA-3404] >gi|240136321|gb|EER35874.1| heat shock protein 60,
 mitochondrial precursor [*Candida tropicalis* MYA-3404] >gi|255941288|ref|XP_002561413.1|
 Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54- 1255] >gi|211586036|emb|CAP93777.1|
 Pc16g11070 [*Penicillium chrysogenum* Wisconsin 54- 1255] >gi|259148241|emb|CAY81488.1|
 Hsp60p [*Saccharomyces cerevisiae* EC1118] >gi|260950325|ref|XP_002619459.1| heat shock
 protein 60, mitochondrial precursor [*Clavispora lusitaniae* ATCC 42720]
 >gi|238847031|gb|EEQ36495.1| heat shock protein 60, mitochondrial precursor [*Clavispora*
lusitaniae ATCC 42720] >gi|261194577|ref|XP_002623693.1| chaperonin GroL [*Ajellomyces*
dermatitidis SLH14081] >gi|239588231|gb|EEQ70874.1| chaperonin GroL [*Ajellomyces*
dermatitidis SLH14081] >gi|327355067|gb|EGE83924.1| chaperonin GroL [*Ajellomyces*
dermatitidis ATCC 18188] >gi|296422271|ref|XP_002840685.1| hypothetical protein [*Tuber*
melanosporum Mel28] >gi|295636906|emb|CAZ84876.1| unnamed protein product [*Tuber*
melanosporum] >gi|296809035|ref|XP_002844856.1| heat shock protein 60 [*Arthroderma otae*
 CBS 113480] >gi|238844339|gb|EEQ34001.1| heat shock protein 60 [*Arthroderma otae* CBS
 113480] >gi|302308696|ref|NP_985702.2| AFR155Wp [*Ashbya gossypii* ATCC 10895]
 >gi|299790751|gb|AAS53526.2| AFR155Wp [*Ashbya gossypii* ATCC 10895]
 >gi|374108933|gb|AEY97839.1| FAFR155Wp [*Ashbya gossypii* FDAG1]
 >gi|302412525|ref|XP_003004095.1| heat shock protein [*Verticillium albo-atrum* VaMs. 102]
 >gi|261356671|gb|EEY19099.1| heat shock protein [*Verticillium albo-atrum* VaMs. 102]
 >gi|302505585|ref|XP_003014499.1| hypothetical protein ARB_07061 [*Arthroderma benhamiae*
 CBS 112371] >gi|291178320|gb|EFE34110.1| hypothetical protein ARB_07061 [*Arthroderma*
benhamiae CBS 112371] >gi|302656385|ref|XP_003019946.1| hypothetical protein TRV_05992
 [*Trichophyton verrucosum* HKI 0517] >gi|291183723|gb|EFE39322.1| hypothetical protein
 TRV_05992 [*Trichophyton verrucosum* HKI 0517] >gi|302915513|ref|XP_003051567.1| predicted
 protein [*Nectria haematococca* mpVI 77-13-4] >gi|256732506|gb|EEU45854.1| predicted protein
 [*Nectria haematococca* mpVI 77-13-4] >gi|310794550|gb|EFQ30011.1| chaperonin GroL
 [*Glomerella graminicola* M1.001] >gi|315048491|ref|XP_003173620.1| chaperonin GroL
 [*Arthroderma gypseum* CBS 118893] >gi|311341587|gb|EFR00790.1| chaperonin GroL
 [*Arthroderma gypseum* CBS 118893] >gi|320580028|gb|EFW94251.1| Tetradecameric
 mitochondrial chaperonin [*Ogataea parapolymorpha* DL-1] >gi|320586014|gb|EFW98693.1| heat
 shock protein mitochondrial precursor [*Grosmannia clavigera* kw1407]
 >gi|322692465|gb|EFY84374.1| Heat shock protein 60 precursor (Antigen HIS-62) [*Metarhizium*
acridum CQMa 102] >gi|322705285|gb|EFY96872.1| Heat shock protein 60 (Antigen HIS-62)
 [*Metarhizium anisopliae* ARSEF 23] >gi|323303806|gb|EGA57589.1| Hsp60p [*Saccharomyces*
cerevisiae FostersB] >gi|323307999|gb|EGA61254.1| Hsp60p [*Saccharomyces cerevisiae*
 FostersO] >gi|323332364|gb|EGA73773.1| Hsp60p [*Saccharomyces cerevisiae* AWRI796]
 >gi|326468648|gb|EGD92657.1| heat shock protein 60 [*Trichophyton tonsurans* CBS 112818]
 >gi|326479866|gb|EGE03876.1| chaperonin GroL [*Trichophyton equinum* CBS 127.97]
 >gi|330915493|ref|XP_003297052.1| hypothetical protein PTT_07333 [*Pyrenophora teres* f. *teres*
 0-1] >gi|311330479|gb|EFQ94847.1| hypothetical protein PTT_07333 [*Pyrenophora teres* f. *teres*
 0-1] >gi|336271815|ref|XP_003350665.1| hypothetical protein SMAC_02337 [*Sordaria*
macrospora k-hell] >gi|380094827|emb|CCC07329.1| unnamed protein product [*Sordaria*
macrospora k- hell] >gi|336468236|gb|EGO56399.1| hypothetical protein
 NEUTE1DRAFT_122948 [*Neurospora tetrasperma* FGSC 2508] >gi|340522598|gb|EGR52831.1|
 hsp60 mitochondrial precursor-like protein [*Trichoderma reesei* QM6a]
 >gi|341038907|gb|EGS23899.1| mitochondrial heat shock protein 60-like protein [*Chaetomium*
thermophilum var. *thermophilum* DSM 1495] >gi|342886297|gb|EGU86166.1| hypothetical protein
 FOXB_03302 [*Fusarium oxysporum* Fo5176] >gi|344230084|gb|EGV61969.1| chaperonin GroL
 [*Candida tenuis* ATCC 10573] >gi|344303739|gb|EGW33988.1| hypothetical protein

SPADRAFT_59397[*Spathaspora passalidarum* NRRL Y-27907]

>gi|345560428|gb|EGX43553.1| hypothetical protein AOL_s00215g289 [*Arthrobotrys oligospora* ATCC 24927] >gi|346323592|gb|EGX93190.1| heat shock protein 60 (Antigen HIS-62) [*Cordyceps militaris* CM01] >gi|346975286|gb|EGY18738.1| heat shock protein [*Verticillium dahliae* VdLs.17]

>gi|354545932|emb|CCE42661.1| hypothetical protein CPAR2_203040 [*Candida parapsilosis*] >gi|358369894|dbj|GAA86507.1| heat shock protein 60, mitochondrial precursor [*Aspergillus kawachii* IFO 4308] >gi|358386867|gb|EHK24462.1| hypothetical protein TRIVIDRAFT_79041 [*Trichoderma virens* Gv29-8] >gi|358399658|gb|EHK48995.1| hypothetical protein TRIATDRAFT_297734 [*Trichoderma atroviride* IMI 206040]

>gi|363750488|ref|XP_003645461.1| hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215] >gi|356889095|gb|AET38644.1| Hypothetical protein Ecym_3140 [*Eremothecium cymbalariae* DBVPG#7215] >gi|365759369|gb|EHN01160.1| Hsp60p [*Saccharomyces cerevisiae* x *Saccharomyces kudriavzevii* VIN7] >gi|365764091|gb|EHN05616.1| Hsp60p [*Saccharomyces cerevisiae* x *Saccharomyces kudriavzevii* VIN7] >gi|365985626|ref|XP_003669645.1| hypothetical protein NDAI_0D00880 [*Naumovozya dairenensis* CBS 421] >gi|343768414|emb|CCD24402.1| hypothetical protein NDAI_0D00880 [*Naumovozya dairenensis* CBS 421]

>gi|366995970|ref|XP_003677748.1| hypothetical protein NCAS_0H00890 [*Naumovozya castellii* CBS 4309] >gi|342303618|emb|CCC71399.1| hypothetical protein NCAS_0H00890 [*Naumovozya castellii* CBS 4309] >gi|367005154|ref|XP_003687309.1| hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417] >gi|357525613|emb|CCE64875.1| hypothetical protein TPHA_0J00520 [*Tetrapisispora phaffii* CBS 4417] >gi|367017005|ref|XP_003683001.1| hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*] >gi|359750664|emb|CCE93790.1| hypothetical protein TDEL_0G04230 [*Torulaspora delbrueckii*]

>gi|367035486|ref|XP_003667025.1| hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464] >gi|347014298|gb|AEO61780.1| hypothetical protein MYCTH_2097570 [*Myceliophthora thermophila* ATCC 42464]

>gi|367055018|ref|XP_003657887.1| hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126] >gi|347005153|gb|AEO71551.1| hypothetical protein THITE_127923 [*Thielavia terrestris* NRRL 8126] >gi|378728414|gb|EHY54873.1| heat shock protein 60 [*Exophiala dermatitidis* NIH/UT8656] >gi|380494593|emb|CCF33032.1| heat shock protein 60 [*Colletotrichum higginsianum*] >gi|385305893|gb|EIF49836.1| heat shock protein 60 [*Dekkera bruxellensis* AWRI1499] >gi|389638386|ref|XP_003716826.1| heat shock protein 60 [*Magnaporthe oryzae* 70-15] >gi|351642645|gb|EHA50507.1| heat shock protein 60 [*Magnaporthe oryzae* 70-15]

>gi|440474658|gb|ELQ43388.1| heat shock protein 60 [*Magnaporthe oryzae* Y34] >gi|440480475|gb|ELQ61135.1| heat shock protein 60 [*Magnaporthe oryzae* P131] >gi|393243142|gb|EJD50658.1| chaperonin GroL [*Auricularia delicata* TFB-10046 SS5]

>gi|396494741|ref|XP_003844378.1| similar to heat shock protein 60 [*Leptosphaeria maculans* JN3] >gi|312220958|emb|CBY00899.1| similar to heat shock protein 60 [*Leptosphaeria maculans* JN3] >gi|398393428|ref|XP_003850173.1| chaperone ATPase HSP60 [*Zymoseptoria tritici* IPO323] >gi|339470051|gb|EGP85149.1| hypothetical protein MYCGRDRAFT_75170 [*Zymoseptoria tritici* IPO323] >gi|401624479|gb|EJS42535.1| hsp60p [*Saccharomyces arboricola* H-6]

>gi|401842294|gb|EJT44530.1| HSP60-like protein [*Saccharomyces kudriavzevii* IFO 1802] >gi|402076594|gb|EJT72017.1| heat shock protein 60 [*Gaeumannomyces graminis* var. *tritici* R3-111a-1] >gi|403213867|emb|CCK68369.1| hypothetical protein KNAG_0A07160 [*Kazachstania naganishii* CBS 8797] >gi|406606041|emb|CCH42514.1| Heat shock protein 60, mitochondrial [*Wickerhamomyces ciferrii*] >gi|406863285|gb|EKD16333.1| heat shock protein 60 [*Marssonina brunnea* f. sp. '*multigermutubi*' MB_m1] >gi|407922985|gb|EKG16075.1| Chaperonin Cpn60 [*Macrophomina phaseolina* MS6] >gi|408399723|gb|EKJ78816.1| hypothetical protein FPSE_00959 [*Fusarium pseudograminearum* CS3096] >gi|410083028|ref|XP_003959092.1| hypothetical protein KAFR_0I01760 [*Kazachstania africana* CBS 2517]

>gi|37246582|emb|CCF59957.1| hypothetical protein KAFR_0101760 [*Kazachstania africana* CBS 2517] >gi|444315528|ref|XP_004178421.1| hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi|387511461|emb|CCH58902.1| hypothetical protein TBLA_0B00580 [*Tetrapisispora blattae* CBS 6284] >gi|448090588|ref|XP_004197110.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|448095015|ref|XP_004198141.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|359378532|emb|CCE84791.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|359379563|emb|CCE83760.1| Piso0_004347 [*Millerozyma farinosa* CBS 7064] >gi|448526196|ref|XP_003869293.1| Hsp60 heat shock protein [*Candida orthopsilosis* Co 90- 125] >gi|380353646|emb|CCG23157.1| Hsp60 heat shock protein [*Candida orthopsilosis*] >gi|46123737|ref|XP_386422.1| HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [*Gibberella zeae* PH-1] >gi|50292099|ref|XP_448482.1| hypothetical protein [*Candida glabrata* CBS 138] >gi|49527794|emb|CAG61443.1| unnamed protein product [*Candida glabrata*] >gi|50310975|ref|XP_455510.1| hypothetical protein [*Kluyveromyces lactis* NRRL Y-1140] >gi|49644646|emb|CAG98218.1| KLLA0F09449p [*Kluyveromyces lactis*] >gi|50422027|ref|XP_459575.1| DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi|49655243|emb|CAG87802.1| DEHA2E05808p [*Debaryomyces hansenii* CBS767] >gi|50555023|ref|XP_504920.1| YALI0F02805p [*Yarrowia lipolytica*] >gi|49650790|emb|CAG77725.1| YALI0F02805p [*Yarrowia lipolytica* CLIB122] >gi|6323288|ref|NP_013360.1| Hsp60p [*Saccharomyces cerevisiae* S288c] >gi|123579|sp|P19882.1|HSP60_YEAST RecName: Full=Heat shock protein 60, mitochondrial; AltName: Full=CPN60; AltName: Full=P66; AltName: Full=Stimulator factor | 66 kDa component; Flags: Precursor >gi|171720|gb|AAA34690.1| heat shock protein 60 (HSP60) [*Saccharomyces cerevisiae*] >gi|577181|gb|AAB67380.1| Hsp60p: Heat shock protein 60 [*Saccharomyces cerevisiae*] >gi|151941093|gb|EDN59473.1| chaperonin [*Saccharomyces cerevisiae* YJM789] >gi|190405319|gb|EDV08586.1| chaperonin [*Saccharomyces cerevisiae* RM11-1a] >gi|207342889|gb|EDZ70518.1| YLR259Cp-like protein [*Saccharomyces cerevisiae* AWRI1631] >gi|256271752|gb|EEU06789.1| Hsp60p [*Saccharomyces cerevisiae* JAY291] >gi|285813676|tpg|DAA09572.1| TPA: chaperone ATPase HSP60 [*Saccharomyces cerevisiae* S288c] >gi|323353818|gb|EGA85673.1| Hsp60p [*Saccharomyces cerevisiae* VL3] >gi|349579966|dbj|GAA25127.1| K7_Hsp60p [*Saccharomyces cerevisiae* Kyokai no. 7] >gi|392297765|gb|EIW08864.1| Hsp60p [*Saccharomyces cerevisiae* CEN.PK113-7D] >gi|226279|prf|1504305A mitochondrial assembly factor >gi|68485963|ref|XP_713100.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|68486010|ref|XP_713077.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|6016258|sp|O74261.1|HSP60_CANAL RecName: Full=Heat shock protein 60, mitochondrial; AltName: Full=60 kDa chaperonin; AltName: Full=Protein Cpn60; Flags: Precursor >gi|3552009|gb|AAC34885.1| heat shock protein 60 [*Candida albicans*] >gi|46434552|gb|EAK93958.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|46434577|gb|EAK93982.1| heat shock protein 60 [*Candida albicans* SC5314] >gi|71001164|ref|XP_755263.1| antigenic mitochondrial protein HSP60 [*Aspergillus fumigatus* Af293] >gi|66852901|gb|EAL93225.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* Af293] >gi|159129345|gb|EDP54459.1| antigenic mitochondrial protein HSP60, putative [*Aspergillus fumigatus* A1163] >gi|90970323|gb|ABE02805.1| heat shock protein 60 [*Rhizophagus intraradices*]

[0264] The recombinant yeast cell preferably comprises, respectively functionally expresses, a GroES chaperone and a GroEL chaperone. Preferably a 10 kDa chaperone (“GroES”) from Table 8 is combined with a matching 60 kDa chaperone (“GroEL”) from Table 9 of the same organism genus or species for expression in the recombinant yeast cell.

[0265] For instance: >gi|189189366|ref|XP_001931022.1|: 71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis*] expressed together with matching

>gi|189190432|ref|XP_001931555.1|heat shock protein 60, mitochondrial precursor [*Pyrenophora*

tritici-repentis Pt-1C-BFP]. All other combinations from Table 8 and 9 similarly made with same organism source are also available to the skilled person for expression. Furthermore, one may combine a chaperone from Table 8 from one organism with a chaperone from Table 9 from another organism, or one may combine GroES with a chaperone from Table 9, or one may combine GroEL with a chaperone from Table 8.

[0266] Preferably the molecular chaperone(s) thus comprise or consist of: [0267] an amino acid sequence of SEQ ID NO: 29 and/or SEQ ID NO: 31; or [0268] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31; or [0269] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31, more preferably one or more functional homologue(s) that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31.

[0270] Preferable the nucleic acid sequence(s) encoding the molecular chaperones comprise or consist of: [0271] a nucleic acid sequence of SEQ ID NO: 30 and/or SEQ ID NO: 32; or [0272] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32; or [0273] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32, more preferably one or more functional homologue(s) of that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32.

[0274] The nucleic acid sequence(s) encoding for the molecular chaperones may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

Phosphoketolase

[0275] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0276] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphoketolase activity. Such a protein having phosphoketolase activity is herein also referred to as “phosphoketolase protein”, “phosphoketoase enzyme” or simply as “phosphoketolase”. Phosphoketolase is further herein abbreviated as “PKL” or “XFP”.

[0277] As used herein, a phosphoketolase catalyzes at least the conversion of D-xylulose 5-phosphate to D-glyceraldehyde 3-phosphate and acetyl phosphate. The phosphoketolase is involved in at least one of the following the reactions:

EC 4.1.2.9:

[0278]

D-xylulose-5-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-glyceraldehyde 3-phosphate+H.sub.2O (IV)

D-ribulose-5-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-glyceraldehyde 3-phosphate+H.sub.2O (V)

EC 4.1.2.22:

[0279]

D-fructose 6-phosphate+phosphate $\rightleftharpoons$ acetyl phosphate+D-erythrose 4-phosphate+H.sub.2O (VI)

[0280] A suitable enzymatic assay to measure phosphoketolase activity is described e.g. in Sonderegger et al., “*Metabolic Engineering of a Phosphoketolase Pathway for Pentose Catabolism in Saccharomyces cerevisiae*”, (2004), Applied & Environmental Microbiology, vol. 70(5), pages 2892-2897, incorporated herein by reference.

[0281] Preferably the protein having phosphoketolase (PKL) activity comprises or consists of:

[0282] an amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or [0283] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or [0284] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36.

[0285] Suitable nucleic acid sequences coding for an phosphoketolase protein may in be found in an organism selected from the group of *Aspergillus niger*, *Neurospora crassa*, *L. casei*, *L. plantarum*, *L. plantarum*, *B. adolescentis*, *B. bifidum*, *B. gallicum*, *B. animalis*, *B. lactis*, *L. pentosum*, *L. acidophilus*, *P. chrysogenum*, *A. nidulans*, *A. clavatus*, *L. mesenteroides*, and *O. oenii*.

[0286] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoketolase (PKL) activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0287] The recombinant cell may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity.

Phosphotransacetylase

[0288] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0289] As used herein, a phosphotransacetylase catalyzes at least the conversion of acetyl phosphate to acetyl-CoA.

[0290] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphotransacetylase activity. Such a protein having phosphotransacetylase activity is herein also referred to as “phosphotransacetylase protein”, “phosphotransacetylase enzyme” or simply as “phosphotransacetylase”. phosphotransacetylase is further herein abbreviated as “PTA”.

[0291] Preferably the protein having phosphotransacetylase (PTA) activity comprises or consists

of: [0292] an amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or [0293] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or [0294] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40.

[0295] Suitable nucleic acid sequences coding for an enzyme having phosphotransacetylase may in be found in an organism selected from the group of *B. adolescentis*, *B. subtilis*, *C. cellulolyticum*, *C. phytofermentans*, *B. bifidum*, *B. animalis*, *L. mesenteroides*, *Lactobacillus plantarum*, *M. thermophila*, and *O. oeniis*.

[0296] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphotransacetylase (PTA) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetate Kinase

[0297] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0298] As used herein, an acetate kinase catalyzes at least the conversion of acetate to acetyl phosphate.

[0299] The recombinant cell may comprise one or more, preferably heterologous, genes coding for a protein having acetate kinase activity (EC 2.7.2.12). Such a protein having acetate kinase activity is herein also referred to as “acetate kinase protein”, “acetate kinase enzyme” or simply as “acetate kinase”. Acetate kinase is further herein abbreviated as “ACK”.

[0300] Preferably the protein having acetate kinase (ACK) activity comprises or consists of: [0301] an amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or [0302] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or [0303] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42.

[0304] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetate kinase (ACK) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetylating Acetaldehyde Dehydrogenase

[0305] As indicated above, the recombinant yeast cell can advantageously comprise and functionally express a, preferably heterologous, nucleic acid sequence encoding a protein

comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10). [0306] If an acetylating acetaldehyde dehydrogenase is present, more preferably, the recombinant yeast cell functionally expresses: [0307] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and [0308] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having NAD^{sup.}+-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2); and [0309] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0310] Acetylating acetaldehyde dehydrogenase is an enzyme that catalyzes the conversion of acetyl-Coenzyme A to acetaldehyde (EC 1.2.1.10). This conversion can be represented by the equilibrium reaction formula:

acetyl-Coenzyme A + NADH + H^{sup.} + <-> acetaldehyde + NAD^{sup.} + Coenzyme A

[0311] A protein having acetylating acetaldehyde dehydrogenase activity is herein also referred to as “acetylating acetaldehyde dehydrogenase protein”, “acetylating acetaldehyde dehydrogenase enzyme” or simply “acetylating acetaldehyde dehydrogenase”. Preferences for a acetylating acetaldehyde dehydrogenase and the nucleic sequences encoding for such are as described in WO2011/010923 and WO2019/063507, incorporated herein by reference.

[0312] The nucleic acid sequence encoding a protein having NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10) is preferably a heterologous nucleic acid sequence. The encoded NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase may therefore preferably be a heterologous NAD^{sup.}+-dependent acetylating acetaldehyde dehydrogenase.

[0313] It is possible for the protein having acetylating acetaldehyde dehydrogenase activity to be monofunctional or bifunctional.

[0314] The nucleic acid sequence encoding the NAD^{sup.}+ dependent acetylating acetaldehyde dehydrogenase may in principle originate from any organism comprising a nucleic acid sequence encoding said dehydrogenase. Known acetylating acetaldehyde dehydrogenases that can catalyse the NADH-dependent reduction of acetyl-Coenzyme A to acetaldehyde may in general be divided in three types of NAD^{sup.}+ dependent acetylating acetaldehyde dehydrogenase functional homologues: [0315] 1) Bifunctional proteins that catalyse the reversible conversion of acetyl-CoA to acetaldehyde, and the subsequent reversible conversion of acetaldehyde to ethanol. These type of proteins advantageously have both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. An example of this type of proteins is the AdhE protein in *E. coli* (Gen Bank No: NP_415757). AdhE appears to be the evolutionary product of a gene fusion. The NH₂-terminal region of the AdhE protein is highly homologous to aldehyde:NAD⁺ oxidoreductases, whereas the COOH-terminal region is homologous to a family of Fe^{sup.2+} dependent ethanol:NAD⁺ oxidoreductases (see Membrillo-Hernandez et al., “*Evolution of the adhE Gene Product of Escherichia coli from a Functional Reductase to a Dehydrogenase*”, (2000) J. Biol. Chem. 275: pages 33869-33875, herein incorporated by reference). The *E. coli* AdhE is subject to metal-catalyzed oxidation and therefore oxygen-sensitive (see Tamarit et al.

“*Identification of the Major Oxidatively Damaged Proteins in Escherichia coli Cells Exposed to Oxidative Stress*” (1998) J. Biol. Chem. 273: pages 3027-3032, herein incorporated by reference).

[0316] 2) Proteins that catalyse the reversible conversion of acetyl-Coenzyme A to acetaldehyde in strictly or facultative anaerobic micro-organisms but do not possess alcohol dehydrogenase activity. An example of this type of proteins has been reported in *Clostridium kluyveri* (see Smith et al. “*Purification, Properties, and Kinetic Mechanism of Coenzyme A-Linked Aldehyde Dehydrogenase from Clostridium kluyveri*” (1980) Arch. Biochem. Biophys. Vol. 203: pages 663-675, incorporated herein by reference). An acetylating acetaldehyde dehydrogenase has been annotated in the genome of *Clostridium kluyveri* DSM 555 (GenBank No: EDK33116). A homologous protein AcdH is

identified in the genome of *Lactobacillus plantarum* (GenBank No: NP_784141). Another example of this type of proteins is the said gene product in *Clostridium beijerinckii* NRRL B593 (see Toth et al. "The ald Gene, Encoding a Coenzyme A-Acylating Aldehyde Dehydrogenase, Distinguishes *Clostridium beijerinckii* and Two Other Solvent-Producing Clostridia from *Clostridium acetobutylicum*", (1999), Appl. Environ. Microbiol. Vol. 65: pages 4973-4980, GenBank No: AAD31841, incorporated herein by reference). [0317] 3) Proteins that are part of a bifunctional aldolase-dehydrogenase complex involved in 4-hydroxy-2-ketovalerate catabolism. Such bifunctional enzymes catalyze the final two steps of the meta-cleavage pathway for catechol, an intermediate in many bacterial species in the degradation of phenols, toluates, naphthalene, biphenyls and other aromatic compounds (Powlowski and Shingler "Genetics and biochemistry of phenol degradation by *Pseudomonas* sp. CF600" (1994) Biodegradation Vol. 5, pages 219-236, herein incorporated by reference). 4-Hydroxy-2-ketovalerate is first converted by 4-hydroxy-2-ketovalerate aldolase to pyruvate and acetaldehyde, subsequently acetaldehyde is converted by acetylating acetaldehyde dehydrogenase to acetyl-CoA. An example of this type of acetylating acetaldehyde dehydrogenase is the DmpF protein in *Pseudomonas* sp CF600 (GenBank No: CAA43226) (Shingler et al., "Nucleotide Sequence and Functional Analysis of the Complete Phenol/3,4-Dimethylphenol Catabolic Pathway of *Pseudomonas* sp. Strain CF600", (1992), J. Bacteriol., Vol. 174, pages 711-724, incorporated herein by reference). The *E. coli* MphF protein (Ferrandez et al., "Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of *Escherichia coli* K-12" (1997) J. Bacteriol. 179: pages 2573-2581, GenBank No: NP_414885, incorporated herein by reference) is homologous to the DmpF protein in *Pseudomonas* sp. CF600.

[0318] In a preferred embodiment, the protein having acetylating acetaldehyde dehydrogenase activity is bifunctional and comprises both NAD.sup.+ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity and NAD.sup.+ dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2).

[0319] A suitable nucleic acid sequence may in particular be found in an organism selected from the group of *Escherichia*, in particular *E. coli*; *Mycobacterium*, in particular *Mycobacterium marinum*, *Mycobacterium ulcerans*, *Mycobacterium tuberculosis*; *Carboxydotherrnus*, in particular *Carboxydotherrnus hydrogenoformans*; *Entamoeba*, in particular *Entamoeba histolytica*; *Shigella*, in particular *Shigella sonnei*; *Burkholderia*, in particular *Burkholderia pseudo mallei*, *Klebsiella*, in particular *Klebsiella pneumoniae*; *Azotobacter*, in particular *Azotobacter vinelandii*; *Azoarcus* sp; *Cupriavidus*, in particular *Cupriavidus taiwanensis*; *Pseudomonas*, in particular *Pseudomonas* sp. CF600; *Pelomaculum*, in particular *Pelotomaculum thermopropionicum*. Preferably, the nucleic acid sequence encoding the NAD.sup.+ dependent acetylating acetaldehyde dehydrogenase originates from *Escherichia*, more preferably from *E. coli*.

[0320] Particularly suitable is an mhpF gene from *E. coli*, or a functional homologue thereof. This gene is described in Ferrandez et al., "Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of *Escherichia coli* K-12" (1997) J. Bacteriol. 179: pages 2573-2581. Good results have been obtained with *S. cerevisiae*, wherein an mhpF gene from *E. coli* has been incorporated. In a further advantageous embodiment the nucleic acid sequence encoding an (acetylating) acetaldehyde dehydrogenase is from *Pseudomonas*, in particular dmpF, e.g. from *Pseudomonas* sp. CF600.

[0321] Further, an acetylating acetaldehyde dehydrogenase (or nucleic acid sequence encoding such activity) may for instance be selected from the group of *Escherichia coli* adhE, *Entamoeba histolytica* adh2, *Staphylococcus aureus* adhE, *Piromyces* sp. E2 adhE, *Clostridium kluyveri* EDK33116, *Lactobacillus plantarum* acdH, *Escherichia coli* eutE, *Listeria innocua* acdH, and *Pseudomonas putida* YP 001268189.

[0322] Preferably the protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity comprises or consists of: [0323] an amino acid sequence of SEQ ID NO:

43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or [0324] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or [0325] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48.

[0326] Most preferably the acetylating acetaldehyde dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity.

[0327] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetylating acetaldehyde dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0328] Examples of suitable enzymes are further illustrated below in tables 10(a) to 10(e) for BLAST of the listed enzymes, giving suitable alternative alcohol/acetaldehyde dehydrogenases.

TABLE-US-00010 TABLE 10 Description Identity (%) Accession number (a) BLAST Query - adHE from *Escherichia coli* bifunctional acetaldehyde-CoA/alcohol dehydrogenase 100

NP_309768.1 [*Escherichia coli* O157:H7 str. Sakai] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 99 YP_540449.1 [*Escherichia coli* UTI89] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 95 YP_001177024.1 [*Enterobacter* sp. 638] (b) BLAST Query - acdH from

Lactobacillus plantarum acetaldehyde dehydrogenase [*Lactobacillus plantarum* 100 YP_004888365.1 WCFS1] acetaldehyde dehydrogenase [*Lactobacillus pentosus* 95 CCC16763.1 IG1] aldehyde-alcohol dehydrogenase [*Enterococcus* 58 WP_016251441.1 *cecorum*] aldehyde-alcohol dehydrogenase 2 [*Enterococcus* 57 WP_016623694.1 *faecalis*] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 55 WP_010493695.1 [*Lactobacillus zae*] alcohol dehydrogenase [*Bacillus thuringiensis*] 54 WP_003280110.1 bifunctional acetaldehyde-CoA/alcohol

dehydrogenase, 53 WP_009931954.1 partial [*Listeria monocytogenes*] (c) BLAST Query - eutE from *Escherichia coli* aldehyde oxidoreductase, ethanolamine utilization 100 NP_416950.1 protein [*Escherichia coli* str. K-12 substr. MG1655] ethanolamine utilization; acetaldehyde dehydrogenase 99 NP_289007.1 [*Escherichia coli* O157:H7 str. EDL933] aldehyde dehydrogenase [*Escherichia albertii*] 99 WP_001075674.1 (d) BLAST Query - Lin1129 from *Listeria innocua* aldehyde dehydrogenase [*Listeria innocua*] 100 NP_470466.1 >emb|CAC96360.1| lin1129 [*Listeria innocua* Clip11262] ethanolamine utilization protein EutE [*Listeria innocua*] 99 WP_003761764.1 aldehyde dehydrogenase [*Listeria monocytogenes*] 95 AGR09081.1 hypothetical protein [*Enterococcus malodoratus*] 64 WP_010739890.1 aldehyde dehydrogenase [*Yersinia aldovae*] 59

WP_004699364.1 aldehyde dehydrogenase EutE [*Klebsiella pneumoniae*] 58 WP_004205473.1 (e) BLAST Query - adhE from *Staphylococcus aureus* bifunctional acetaldehyde-CoA/alcohol dehydrogenase 100 NP_370672.1 [*Staphylococcus aureus* subsp. *aureus* Mu50] aldehyde dehydrogenase family protein [*Staphylococcus* 99 YP_008127042.1 *aureus* CA-347] bifunctional acetaldehyde-CoA/alcohol dehydrogenase 85 WP_002495347.1 [*Staphylococcus epidermidis*] aldehyde-alcohol dehydrogenase 2 [*Enterococcus* 75 WP_016623694.1 *faecalis*]

Acetyl-Coenzyme A Synthetase

[0329] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing: [0330] a nucleic acid sequence encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or or EC1.1.1.2); and/or [0331] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0332] A protein having acetyl-Coenzyme A synthetase activity can herein also be referred to as “acetyl-Coenzyme A synthetase protein”, “acetyl-Coenzyme A synthetase enzyme” or simply “acetyl-Coenzyme A synthetase” or even “acetyl CoA synthetase”. The protein is further abbreviated herein as “ACS”.

[0333] The acetyl-Coenzyme A synthetase, also known as acetate-CoA ligase or acetyl-activating enzyme, catalyses the formation of acetyl-CoA from acetate, coenzyme A (CoA) and ATP as shown below:

[00002] $\text{ATP} + \text{acetate} + \text{CoA} = \text{AMP} + \text{diphosphate} + \text{acetyl-CoA}$

[0334] It is understood that the recombinant yeast cell may naturally comprise an endogenous gene encoding an acetyl-Coenzyme A synthetase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0335] For example, the recombinant yeast cell according to the invention may comprise an acetyl-Coenzyme A synthetase, which may be present in the wild-type cell, as is for instance the case with *S. cerevisiae* which contains two acetyl-Coenzyme A synthetase isoenzymes encoded by the ACS1 (amino acid sequence illustrated as SEQ ID NO: 49) and ACS2 (amino acid sequence illustrated as SEQ ID NO: 50) genes (van den Berg et al (1996) J. Biol. Chem. 271: pages 28953-28959, incorporated herein by reference), or a host cell may be provided with one or more heterologous gene(s) encoding this activity, e.g. the ACS1 and/or ACS2 gene of *S. cerevisiae* or a functional homologue thereof may be incorporated into a cell lacking acetyl-Coenzyme A synthetase isoenzyme activity.

[0336] Preferably the protein having NAD.sup.+ -dependent acetyl-Coenzyme A synthetase activity comprises or consists of: [0337] an amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or [0338] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or [0339] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50.

[0340] Preferably the recombinant yeast cell is a recombinant yeast cell wherein the, endogenous or heterologous, acetyl-Coenzyme A synthetase protein, is overexpressed, most preferably by using a suitable promoter as described for example in WO2011/010923, incorporated herein by reference. Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having acetyl-Coenzyme A synthetase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0341] Examples of suitable proteins having acetyl-Coenzyme A synthetase activity are listed in table 11. At the top of table 11 the ACS2 used in the examples and that is BLASTED is mentioned. TABLE-US-00011 TABLE 11 BLAST Query - ACS2 from *Saccharomyces cerevisiae* Description Identity (%) Accession number acetate-CoA ligase ACS2 [*Saccharomyces cerevisiae* 100 NP_013254.1 S288c] acetyl CoA synthetase [*Saccharomyces cerevisiae* 99 EDN59693.1 YJM789]

acetate-CoA ligase [*Kluyveromyces lactis* NRRL Y- 85 XP_453827.1 1140] acetate-CoA ligase [*Candida glabrata* CBS 138] 83 XP_445089.1 acetate-CoA ligase [*Scheffersomyces stipitis* CBS 6054] 68 XP_001385819.1 acetyl-coenzyme A synthetase FacA [*Aspergillus* 63 EDP50475.1 *fumigatus* A1163] acetate-CoA ligase facA-*Penicillium chrysogenum* 62 XP_002564696.1 [*Penicillium chrysogenum* Wisconsin 54-1255]

Alcohol Dehydrogenase

[0342] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing: [0343] a nucleic acid sequence encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or or EC1.1.1.2); and/or [0344] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0345] A protein having alcohol dehydrogenase activity is herein also referred to as “alcohol dehydrogenase protein”, “alcohol dehydrogenase enzyme” or simply “alcohol dehydrogenase”. The protein is further abbreviated herein as “ADH”.

[0346] The alcohol dehydrogenase enzyme catalyses the conversion of acetaldehyde into ethanol.

[0347] It is understood that the recombinant yeast cell may naturally comprise an endogenous nucleic acid sequence encoding an alcohol dehydrogenase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having alcohol dehydrogenase activity

[0348] For example, the recombinant yeast cell may naturally comprise a gene encoding alcohol dehydrogenase, as is the case with *S. cerevisiae* (Amino acid sequences of the native *S. cerevisiae* alcohol dehydrogenases ADH1, ADH2, ADH3, ADH4 and ADH5 are illustrated respectively as SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 and SEQ ID NO: 55), see Lutstorf and Megnet, “*Multiple Forms of Alcohol Dehydrogenase in Saccharomyces cerevisiae*”, (1968), Arch. Biochem. Biophys., vol. 126, pages 933-944, incorporated herein by reference, or Ciriacy, “*Genetics of Alcohol Dehydrogenase in Saccharomyces cerevisiae I. Isolation and genetic analysis of adh mutants*”, (1975), Mutat. Res. 29, pages 315-326, incorporated herein by reference).

[0349] Preferably, however, the recombinant yeast cell comprises alcohol dehydrogenase activity within a, suitably heterologous, bifunctional enzyme having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity as described herein above. That is, most preferably the alcohol dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. When the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity, any native nucleic acid sequences encoding for any native protein encoding alcohol dehydrogenase activity may or may not be disrupted and/or deleted.

[0350] The recombinant yeast cell may therefore advantageously be a recombinant yeast cell functionally expressing: [0351] one or more heterologous nucleic acid sequence(s) encoding a bifunctional protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and [0352] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1), wherein optionally one or more native nucleic acid sequence(s) encoding a protein having NAD.sup.+ -dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2) are disrupted or deleted.

[0353] Alternatively the recombinant yeast cell may advantageously be a recombinant yeast cell functionally expressing: [0354] one or more, native or heterologous, nucleic acid sequence(s) encoding a monofunctional protein having NAD.sup.+ -dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and [0355] one or more, native or heterologous, nucleic acid

sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1); and [0356] one or more, native or heterologous, nucleic acid sequences(s) encoding a protein having NAD.sup.+/-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2).

[0357] Preferences for the bifunctional protein are provided above and are as listed for the acetylating acetaldehyde dehydrogenase protein. If the protein is not bifunctional, the NAD.sup.+/-dependent alcohol dehydrogenase protein is preferably a protein having NAD.sup.+/-dependent alcohol dehydrogenase activity that comprises or consists of: [0358] an amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or [0359] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or [0360] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55.

[0361] Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having NAD.sup.+/-dependent alcohol dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Deletion or Disruption of Glycerol 3-Phosphate Phosphohydrolase and/or Glycerol 3-Phosphate Dehydrogenase

[0362] The recombinant yeast cell further may or may not comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase gene and/or encoding a glycerol 3-phosphate dehydrogenase gene.

[0363] Preferably enzymatic activity needed for the NADH-dependent glycerol synthesis in the yeast cell is reduced or deleted. The reduction or deletion of the enzymatic activity of glycerol 3-phosphate phosphohydrolase and/or glycerol 3-phosphate dehydrogenase can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase (GPD) and/or one or more genes encoding a glycerol phosphate phosphatase (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encodes a polypeptide with reduced activity. Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO2011010923, and are disclosed in SEQ ID NO: 24-27 of that application.

[0364] Preferably the recombinant yeast is a recombinant yeast that further comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase (GPD) gene. The one or more of the glycerol phosphate phosphatase (GPP) genes may or may not be deleted or disrupted.

[0365] More preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene. The glycerol-3-phosphate dehydrogenase 2 (GPD2) gene may or may not be deleted or disrupted.

[0366] Most preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene, whilst the glycerol-3-

phosphate dehydrogenase 2 (GPD2) gene remains active and/or intact. Preferably therefore, only one of the *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes is disrupted and deleted, whereas most preferably only GPD1 is chosen from the group consisting of GPD1, GPD2, GPP1 and GPP2 genes to be disrupted or deleted.

[0367] Without wishing to be bound to any kind of theory it is believed that a recombinant yeast according to the invention wherein the GPD1 gene, but not the GPD2 gene, is deleted or disrupted, can be advantageous when applied in a fermentation process where the glucose at the start of or during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0368] Preferably at least one gene encoding a GPD and/or at least one gene encoding a GPP is entirely deleted, or at least a part of the gene is deleted that encodes a part of the enzyme that is essential for its activity. Good results can be achieved with a *S. cerevisiae* cell, wherein the open reading frames of the GPD1 gene and/or of the GPD2 gene have been inactivated. Inactivation of a structural gene (target gene) can be accomplished by a person skilled in the art by synthetically synthesizing or otherwise constructing a DNA fragment consisting of a selectable marker gene flanked by DNA sequences that are identical to sequences that flank the region of the host cell's genome that is to be deleted. Suitably, good results can be obtained with the inactivation of the GPD1 and GPD2 genes in *Saccharomyces cerevisiae* by integration of the marker genes kanMX and hphMX4. Subsequently this DNA fragment is transformed into a host cell.

Transformed cells that express the dominant marker gene are checked for correct replacement of the region that was designed to be deleted, for example by a diagnostic polymerase chain reaction or Southern hybridization.

[0369] Thus, in the recombinant yeast cells of the invention, glycerol 3-phosphate phosphohydrolase activity in the cell and/or glycerol 3-phosphate dehydrogenase activity in the cell can be advantageously reduced.

Glucoamylase

[0370] Preferably, the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding for a glucoamylase (EC 3.2.1.20 or 3.2.1.3).

[0371] A protein having glucoamylase activity is herein also referred to as “glucoamylase enzyme”, “glucoamylase protein” or simply “glucoamylase”. Glucoamylase has herein been abbreviated as “GA”.

[0372] Glucoamylase, also referred to as amyloglucosidase, alpha-glucosidase, glucan 1,4-alpha glucosidase, maltase glucoamylase, and maltase-glucoamylase, catalyses at least the hydrolysis of terminal 1,4-linked alpha-D-glucose residues from non-reducing ends of amylose chains to release free D-glucose. A glucoamylase may be further defined by its amino acid sequence. Likewise a glucoamylase may be further defined by a nucleotide sequence encoding the glucoamylase. As explained in detail above under definitions, a certain glucoamylase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glucoamylase.

[0373] Preferably the protein having glucoamylase activity comprises or consists of: [0374] an amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or [0375] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or [0376] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence

of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58.

[0377] The polypeptide of SEQ ID NO: 56 encodes a “mature glucoamylase”, referring to the enzyme in its final form after translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc.

[0378] In an embodiment the nucleotide sequence encodes a polypeptide having an amino acid sequence of SEQ ID NO: 57 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 57. Amino acids 1-17 of the SEQ ID NO: 57 may encode for a native signal sequence.

[0379] In another embodiment the nucleotide sequence allowing the expression of a glucoamylase encodes a polypeptide having an amino acid sequence of SEQ ID NO: 58 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 58. Amino acids 1-19 of the SEQ ID NO: 58 may encode for a signal sequence.

[0380] A signal sequence (also referred to as signal peptide, targeting signal, localization signal, localization sequence, transit peptide, leader sequence or leader peptide) can be present at the N-terminus of a polypeptide (here, the glucoamylase) where it signals that the polypeptide is to be excreted, for example outside the cell and into the media.

[0381] The nucleic acid sequence (e.g. the gene) encoding for the protein having glucoamylase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Nitrate Reductase

[0382] The recombinant yeast cell may also advantageously comprise, respectively functionally express, a nucleic acid sequences encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequences encoding an enzyme having NADH-dependent nitrite reductase activity. Details for the expression of such an alternative redox sink have been described in non-pre-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

[0383] Nitrate reductase (NR) catalyzes the reduction of nitrate (NO.sub.3.sup.-) to nitrite (NO.sub.2.sup.-). Nitrite reductase catalyzes the reduction of nitrite to ammonia (NH.sub.3). Nitrate reductase and/or nitrite reductase can be part of a so-called nitrogen assimilation pathway in certain cells. Cells comprising nitrate reductase activity and/or nitrite reductase activity include certain plant cells and bacterial cells and a few yeast cells. As indicated by Linder, the ability to assimilate inorganic nitrogen sources other than ammonia is thought to be rare among budding yeasts. Among the few fungi that are naturally capable to assimilate nitrate or nitrite are *Blastobotrys adeninivorans* (family Trichomonascaceae) *Candida boidinii* (family Pichiaceae), *Cyberlindnera jadinii* (family Phaffomycetaceae), and *Ogataea polymorpha* (family Pichiaceae).

[0384] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrate reductase.

[0385] By a NADH-dependent nitrate reductase is herein understood a nitrate reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrate reductase has a ratio of catalytic efficiency for NADPH/NADP⁺ as a cofactor (k.sub.cat/K.sub.m).sup.NADP⁺ to NADH/NAD⁺ as cofactor (k.sub.cat/K.sub.m).sup.NAD⁺ i.e. a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺: (k.sub.cat/K.sub.m).sup.NAD⁺, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than

10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrate reductase may have a catalytic efficiency ratio $(k_{\text{sub.cat}}/K_{\text{sub.m}})_{\text{sup.NADP}^+}/(k_{\text{sub.cat}}/K_{\text{sub.m}})_{\text{sup.NAD}^+}$ of equal to or less than 1,000,000,000:1 (i.e. $1.\text{Math.}10.\text{sup.}9$). Most preferably the NADH-dependent nitrate reductase is exclusively depended on NADH/NAD⁺ as a co-factor. That is, most preferably the NADH-dependent nitrate reductase has an absolute requirement for NADH/NAD⁺ as a cofactor instead of NADPH/NADP⁺ as a cofactor.

[0386] Preferably the NADH-dependent nitrate reductase is a NADH-dependent nitrate reductase with enzyme classification EC 1.7.1.1. (i.e. with EC number EC 1.7.1.1) or enzyme classification EC.1.6.6.1 (i.e. with EC number 1.6.6.1). Suitably the NADH-dependent nitrate reductase, also referred to as NADH-dependent nitrate oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:

nitrate+NADH+H^{sup.}+.fwdarw.nitrite+NAD^{sup.}++H^{sub.2O}

[0387] Suitable NADH-dependent nitrate reductases may include one or more NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp *durum*, *Ulva* sp. and/or *Zea mays*; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and/or functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0388] Preferred NADH-dependent nitrate reductases include the NADH-dependent nitrate reductases as obtained or derived from *Candida boidinii* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Candida utilis* (a nitrate reductase capable of utilizing both NADH and NADPH as electron donors), *Fusarium oxysporum* (as described by Fujii et al, in their article titled "Denitrification by the Fungus *Fusarium oxysporum* Involves NADH-Nitrate Reductase" published in Biosci. Biotechnol. Biochem., 72 (2), pages 412-420, 2008, incorporated herein by reference), *Spinacia oleracea* and *Zea mays*.

[0389] Preferred NADH-dependent nitrate reductases hence include: NADH-dependent nitrate reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:74 and/or SEQ ID NO:75, as described herein; and/or functional homologues of SEQ ID NO:74 and/or SEQ ID NO:75 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:74 and/or SEQ ID NO:75 respectively; and/or functional homologues of SEQ ID NO:74 and/or SEQ ID NO:75 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:74 and/or SEQ ID NO:75 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:74 and/or SEQ ID NO:75 respectively.

[0390] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity selected from the group consisting of NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticribra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*, *Oryza sativa*, *Phaeocystis antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia Testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp *durum*, *Ulva* sp. and *Zea mays*, and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0391] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:74 and/or SEQ ID NO:75 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:74 and/or SEQ ID NO:75. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:74 and/or SEQ ID NO:75 respectively.

[0392] The recombinant yeast cell may combine one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding an NADPH-dependent nitrite

reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding a NADH-dependent nitrite reductase.

[0393] Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:74, are listed in Table 12 below.

[0394] Table 12: Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:74, are listed in Table 12 below.

TABLE-US-00012 TABLE 12 Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 74, are listed in Table 12 below. UNIPROT DATABASE ACCESSION IDENTITY TO NUMBER DESCRIPTION ORGANISM SEQ ID NO: 74 [%]

K3YG56 Cluster: Nitrate PACMAD clade 94.06 reductase P16081 Cluster: Nitrate cellular organisms 90.54 reductase [NADH] 1 UPI0004DEBCF9 Cluster: nitrate *Zea mays* 88.57 reductase [NADH]-like A0A6G1CIJ1 Cluster: *Oryza meyeriana* 85.7 Uncharacterized var. *granulata* protein P27967 Cluster: Nitrate Pooideae 85.37 reductase [NADH] A0A3B6T8X1 Cluster: Nitrate Triticinae 86.11 reductase A0A4S8IHP6 Cluster: Nitrate Musaceae 78.55 reductase A0A2T7D896 Cluster: Nitrate *Panicum* 78.22 reductase A0A199UYK8 Cluster: Nitrate *Ananas comosus* 77.45 reductase A0A4S8J849 Cluster: Nitrate Mesangiospermae 77.67 reductase A0A2H3Y7X8 Cluster: Nitrate Arecaceae 76.68 reductase A0A4S8JVI1 Cluster: Nitrate *Musa balbisiana* 75.58 reductase A0A200QET5 Cluster: Nitrate *Macleaya cordata* 72.39 reductase F6H0M5 Cluster: Nitrate *Vitis* 72.61 reductase A0A4Y7J3A8 Cluster: Nitrate *Papaver* 71.95 reductase *somniferum* A0A218XU44 Cluster: Nitrate *Punica granatum* 71.73 reductase A0A2C9PIN3 Cluster: Nitrate *Phellodendron* 72.06 reductase *amurense* UPI000B8CEBA2 Cluster: nitrate *Carica papaya* 71.51 reductase [NADH]-like A0A2R6RP67 Cluster: Nitrate *Actinidia chinensis* 70.63 reductase var. *chinensis* A0A5J5BUB1 Cluster: Nitrate *Nyssa sinensis* 71.29 reductase V4TCH9 Cluster: Nitrate Citrus 71.51 reductase P36859 Cluster: Nitrate *Petunia hybrida* 71.62 reductase [NADH] A0A0E0GDD5 Cluster: Nitrate Oryzinae 74.76 reductase UPI001263D0A7 Cluster: nitrate *Pistacia vera* 71.4 reductase [NAD(P)H]- like A0A061FFV7 Cluster: Nitrate Byttnerioideae 70.85 reductase K3YPT2 Cluster: Nitrate Panicoideae 75.21 reductase A0A1U7ZDA9 Cluster: Nitrate *Nelumbo nucifera* 72.17 reductase A0A0P0VPX4 Cluster: Nitrate *Oryza* 74.17 reductase A0A6A3CP56 Cluster: Nitrate *Hibiscus syriacus* 69.97 reductase A0A067KLT8 Cluster: Nitrate *Jatropha curcas* 70.08 reductase A0A2I4H0D3 Cluster: Nitrate *Juglans regia* 69.86 reductase A0A5J9UYH6 Cluster: PACMAD clade 74.26 Uncharacterized protein (Fragment) A0A5N6RHU6 Cluster: Nitrate fabids 70.63 reductase A0A1U8KZC0 Cluster: Nitrate Malvaceae 70.74 reductase UPI000C0393D8 Cluster: nitrate *Durio zibethinus* 73.33 reductase [NADH]-like B9H7A1 Cluster: Nitrate Saliceae 70.41 reductase P17569 Cluster: Nitrate Cucurbitaceae 70.74 reductase [NADH] UPI000581598E Cluster: nitrate *Sesamum indicum* 70.41 reductase [NADH] 2- like UPI000F7C3BDF Cluster: nitrate *Abrus precatorius* 74.12 reductase [NADH] 2 isoform X1 P08509 Cluster: Nitrate Solanaceae 70.08 reductase [NADH] 2 P39867 Cluster: Nitrate Brassiceae 71.65 reductase [NADH], clone PBNBR1405 UPI000C1D548F Cluster: nitrate *Olea europaea* var. 71.22 reductase [NADH] 2- *sylvestris* like P39866 Cluster: Nitrate Phaseoleae 73.88 reductase [NADH] 2 P27968 Cluster: Nitrate Triticeae 72.82 reductase [NAD(P)H] A8IJ84 Cluster: Nitrate *Beta vulgaris* 71.14 reductase A0A2U1KIT7 Cluster: Nitrate *Artemisia annua* 69.64 reductase A0A165XNP4 Cluster: Nitrate *Daucus carota* 69.31 reductase subsp. *sativus* B9RJB1 Cluster: Nitrate Mesangiospermae 74 reductase P43101 Cluster: Nitrate Asteraceae 69.86 reductase [NADH] I1LW34 Cluster: Nitrate

Glycine subgen. 74.49 reductase *Soja* UPI000980E182 Cluster: nitrate *Ananas comosus* 75.54 reductase [NADH]-like A0A4P1R4X1 Cluster: Nitrate *Lupinus* 71.22 reductase *angustifolius* Q8H1T7 Cluster: Nitrate *Tilia platyphyllos* 69.42 reductase I1IER4 Cluster: Nitrate *Brachypodium* 73.47 reductase *distachyon* A0A1U7Z2Y3 Cluster: Nitrate *Nelumbo nucifera* 72.45 reductase A0A068VGB5 Cluster: Nitrate *Coffea* 68.87 reductase A0A2T7F896 Cluster: Nitrate *Panicum* 73.27 reductase A0A2Z7C1Z2 Cluster: Nitrate *Dorcoceras* 69.31 reductase *hygrometricum* UPI00053C7726 Cluster: nitrate *Tarenaya* 69.64 reductase [NADH] 2 *hassleriana* A0A445BT68 Cluster: Nitrate *Arachis* 72.55 reductase (NADH) P39869 Cluster: Nitrate *Lotus japonicus* 73.76 reductase [NADH] UPI0010FB6302 Cluster: nitrate *Cajanus cajan* 73.64 reductase [NADH] 2 Q93XS1 Cluster: Nitrate *Prunus* 73.76 reductase A0A022Q6Q6 Cluster: Nitrate *Erythranthe guttata* 69.2 reductase A0A1U8NV83 Cluster: Nitrate *Gossypium* 73.04 reductase P27783 Cluster: Nitrate fabids 69.53 reductase [NAD(P)H] A0A2T8KWM5 Cluster: Nitrate *Panicum hallii* 72.93 reductase A0A2I4GBD5 Cluster: Nitrate *Juglans regia* 69.42 reductase A0A5N6PSR2 Cluster: Nitrate *Mikania micrantha* 69.2 reductase A0A2P6PI42 Cluster: Nitrate Mesangiospermae 73.88 reductase E7ELN7 Cluster: Nitrate IRL clade 70.41 reductase A0A4U5QSR8 Cluster: Nitrate *Populus* 73.64 reductase A0A2K2BFU1 Cluster: Nitrate Saliceae 74.12 reductase A0A1Z5RNB1 Cluster: Nitrate *Sorghum bicolor* 76.67 reductase A0A0M4P4G3 Cluster: Nitrate rosids 73.28 reductase M4CVT6 Cluster: Nitrate *Brassica* 70.47 reductase *campestris* A0A251SAE9 Cluster: Nitrate *Helianthus annuus* 72.55 reductase A0A2G5DUU4 Cluster: Nitrate *Aquilegia coerulea* 71.24 reductase P23312 Cluster: Nitrate Chenopodioideae 72.79 reductase [NADH] P11832 Cluster: Nitrate Brassicaceae 71.22 reductase [NADH] 1 P11035 Cluster: Nitrate rosids 68.76 reductase [NADH] 2 W1NJD0 Cluster: Nitrate Magnoliopsida 69.31 reductase UPI000C048389 Cluster: nitrate *Durio zibethinus* 71.36 reductase [NADH]-like A0A6A3C902 Cluster: Nitrate *Hibiscus syriacus* 73.16 reductase P39870 Cluster: Inducible *Glycine* subgen. 73.52 nitrate reductase *Soja* [NADH] 2 UPI000F7C9CC8 Cluster: inducible *Abrus precatorius* 72.79 nitrate reductase [NADH] 2 A0A6A5MI86 Cluster: Nitrate *Lupinus* 70.31 reductase

Nitrite Reductase

[0395] As indicated above, nitrite reductase catalyzes the reduction of nitrite to ammonia (NH.sub.3).

[0396] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrite reductase.

[0397] By a NADH-dependent nitrite reductase is herein understood a nitrite reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrite reductase has a ratio of catalytic efficiency for NADPH/NADP⁺ as a cofactor (k.sub.cat/K.sub.m).sup.NADP⁺ to NADH/NAD⁺ as cofactor (k.sub.cat/K.sub.m).sup.NAD⁺, i.e. a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺: (k.sub.cat/K.sub.m).sup.NAD⁺, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrite reductase may have a catalytic efficiency ratio (k.sub.cat/K.sub.m).sup.NADP⁺:(k.sub.cat/K.sub.m).sup.NAD⁺ of equal to or less than 1,000,000,000:1 (i.e. 1.Math.10.sup.9). Most preferably the NADH-dependent nitrite reductase is exclusively depended on NADH/NAD⁺.sup.+ as a co-factor. That is, most preferably the NADH-dependent nitrite reductase has an absolute requirement for NADH/NAD⁺ as a cofactor instead of NADPH/NADP⁺ as a cofactor.

[0398] Preferably the NADH-dependent nitrite reductase is a NADH-dependent nitrite reductase with enzyme classification EC 1.7.1.15 (i.e. with EC number EC 1.7.1.15). Suitably the NADH-dependent nitrite reductase, also referred to as NADH-dependent nitrite oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:

nitrite+3NADH+5H^{sup.}+fwdarw.ammonia+3NAD^{sup.}++2H^{sub.}2O

The person skilled in the art will understand that the ammonia may also be present and/or referred to as so-called ammonium hydroxide NH₄OH

[0399] Suitable NADH-dependent nitrite reductases may include one or more NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus*, *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0400] *Escherichia coli* utilizes several distinct enzymes in its nitrite assimilation pathway. The nirD gene encodes a NADH-dependent nitrite reductase (NADH) small subunit, whilst the nirB gene encodes a NADH-dependent nitrite reductase (NADH) large subunit.

[0401] Preferred NADH-dependent nitrite reductases include the NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), a nitrite reductase capable of utilizing both NADH and NADPH as electron donors, and/or *Escherichia coli*. At high nitrate and/or nitrite concentrations, the nitrite reductase encoded by the nirB gene of *Escherichia coli* is especially preferred.

[0402] Preferred NADH-dependent nitrite reductases hence include: NADH-dependent nitrite reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:76 (*E. coli* nitrite reductase small subunit encoded by nirD) and/or SEQ ID NO:77 (*E. coli* nitrite reductase large subunit encoded by nirB) and/or SEQ ID NO:78 (*Emericella nidulans* nitrate reductase encoded by niiA), as described herein; and/or functional homologues of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively; and/or functional homologues of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively.

[0403] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity selected from the group consisting of NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus*, *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90,

95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0404] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:76 (*E. coli* nitrate reductase small subunit encoded by nirD) and/or SEQ ID NO:77 (*E. coli* nitrate reductase large subunit encoded by nirB) and/or SEQ ID NO:78 (*Emericella nidulans* nitrate reductase encoded by niiA), or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:76 and/or SEQ ID NO:77 and/or SEQ ID NO:78 respectively.

[0405] The recombinant yeast cell may combine one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding an NADPH-dependent nitrate reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding a NADH-dependent nitrate reductase.

[0406] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:76 (small subunit encoded by nirD), are listed in Table 13 below.

[0407] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:77 (large subunit encoded by nirB), are listed in Table 14 below.

TABLE-US-00013 TABLE 13 Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 76 (small subunit encoded by nirD). IDENTITY TO UNIPROT DATABASE SEQ ID NO: 76 ACCESSION NUMBER DESCRIPTION ORGANISM [%] E2QFM2 Cluster: Nitrite Gammaproteobacteria 99.07 reductase A0A078LHD3 Cluster: Nitrite Enterobacteriaceae 94.44 reductase small subunit P0A230 Cluster: Nitrite Bacteria 93.52 reductase (NADH) small subunit A0A170JME2 Cluster: Nitrite Enterobacteriaceae 91.67 reductase [NAD(P)H] small subunit A0A0F6TX22 Cluster: Nitrite Gammaproteobacteria 91.67 reductase A0A6D0EPI7 Cluster: Nitrite Enterobacteriaceae 87.04 reductase small subunit NirD A0A198H1N5 Cluster: Nitrite Bacteria 86.11 reductase [NAD(P)H] small subunit A0A447MQ21 Cluster: Nitrite Enterobacteriaceae 84.26 reductase [NAD(P)H] small subunit UPI00077B73C7 Cluster: nitrite *Enterobacter* 83.33 reductase small subunit *timonensis* NirD A0A094XNF5 Cluster: Nitrite Enterobacteriales 82.41 reductase (NADH) small subunit UPI00105F841D Cluster: nitrite *Enterobacter* sp. 81.48 reductase small subunit OV724 NirD A0A1X0XB32 Cluster: Nitrite Enterobacteriales 81.48 reductase small subunit A0A1A9FFN3 Cluster: Nitrite *Lelliottia amnigena* 81.48 reductase small subunit UPI0002D5E0A7 Cluster: nitrite *Metakosakonia* 80.56 reductase small subunit *massiliensis* NirD UPI0004E24438 Cluster: nitrite *Siccibacter colletis* 79.63 reductase small subunit NirD A0A514EKN7 Cluster: Nitrite Enterobacteriaceae 77.78 reductase [NAD(P)H] small subunit A0A2P8VHS5 Cluster: Nitrite Enterobacteriaceae 80.56 reductase small

subunit NirD A0A4V3BP91 Cluster: Nitrite cellular organisms 80.56 reductase (NADH) small subunit A0A3R9F800 Cluster: Nitrite Enterobacteriaceae 79.63 reductase small subunit NirD R9VMQ4 Cluster: Nitrite Enterobacteriaceae 78.7 reductase small subunit A0A285KH74 Cluster: Nitrite *Enterobacter* sp. 80.37 reductase (NADH) CC120223-11 small subunit A0A658U6S1 Cluster: Assimilatory *Enterobacter* 77.78 nitrite reductase (NAD(P)H) small subunit A0A085A429 Cluster: Nitrite Enterobacteriaceae 78.7 reductase [NAD(P)H] small subunit A0A1V0L823 Cluster: Nitrite *Kosakonia* 76.85 reductase small subunit UPI001431C56A Cluster: nitrite *Cedecea* colo 75 reductase small subunit NirD A0A085G1A1 Cluster: Nitrite Enterobacteriaceae 75 reductase [NAD(P)H] small subunit A0A089PL75 Cluster: Nitrite *Pluralibacter gergoviae* 76.85 reductase A0A3C0H3I1 Cluster: Nitrite Enterobacteriaceae 77.78 reductase small subunit A0A3N2DXC8 Cluster: Assimilatory unclassified 75.93 nitrite reductase *Enterobacter* (NAD(P)H) small subunit A0A0L0ATR1 Cluster: Nitrite Enterobacteriaceae 78.5 reductase K8AGI3 Cluster: Nitrite *Cronobacter* 75 reductase [NAD(P)H] *condimenti* 1330 small subunit A0A1I6EAU0 Cluster: Nitrite *Enterobacter* sp. kpr-6 75 reductase (NADH) small subunit A0A0D5X0Q1 Cluster: Nitrite Enterobacteriaceae 75 reductase C9Y259 Cluster: Nitrite Enterobacteriaceae 74.07 reductase [NAD(P)H] small subunit A0A4R0GWC0 Cluster: Nitrite *Kosakonia* 75 reductase small subunit quasisacchari NirD UPI00057BF926 Cluster: nitrite *Enterobacter* sp. 73.15 reductase small subunit Bisph1 NirD A0A495AHM0 Cluster: Nitrite *Enterobacter* sp. 72.22 reductase small subunit R1(2018) NirD A0A447QKU3 Cluster: Nitrite Yersiniaceae 75.47 reductase [NAD(P)H] small subunit I2B4C9 Cluster: Nitrite *Shimwellia blattae* 73.83 reductase (NAD(P)H) (strain ATCC 29907/ DSM 4481/JCM 1650/ NBRC 105725/CDC 9005-74) UPI0012B7CAE2 Cluster: nitrite *Erwinia* sp. CPCC 72.22 reductase small subunit 100877 NirD UPI00073DB139 Cluster: nitrite [*Erwinia*] *teleogrylli* 73.83 reductase small subunit NirD UPI000237D150 Cluster: nitrite *Enterobacter mori* 80.56 reductase small subunit NirD A0A4P8YE33 Cluster: Nitrite Enterobacteriaceae 72.22 reductase small subunit NirD UPI0005DC1E97 Cluster: nitrite *Yersinia rohdei* 71.03 reductase small subunit NirD A0A0U1HQI4 Cluster: Nitrite *Yersinia* 71.03 reductase small subunit A0A0K0HM15 Cluster: Nitrite Enterobacterales 72.9 reductase [NAD(P)H] small subunit H8NP95 Cluster: Nitrite Yersiniaceae 69.16 reductase small subunit A0A2S4QQ85 Cluster: Nitrite Enterobacteriaceae 69.44 reductase small subunit A0A0A7RXK3 Cluster: Assimilatory *Frischella perrara* 71.3 nitrite reductase (NAD(P)H) small subunit D4DYL5 Cluster: Nitrite *Serratia odorifera* DSM 73.58 reductase [NAD(P)H], 4582 small subunit A0A370QTS9 Cluster: Assimilatory *Enterobacillus tribolii* 71.7 nitrite reductase (NAD(P)H) small subunit A0A1B9JUB0 Cluster: Nitrite *Gilliamella apicola* 69.16 reductase small subunit A0A085U6Z8 Cluster: Nitrite *Yersinia ruckeri* 72.22 reductase [NAD(P)H] small subunit UPI00156B8D43 Cluster: unknown 70.75 UPI00156B8D43 related cluster A0A0T9KRE5 Cluster: Nitrite *Yersinia* 70.37 reductase small subunit A0A066TFS8 Cluster: Nitrite *Gilliamella* 69.16 reductase [NAD(P)H] small subunit A0A1B9M9X4 Cluster: Nitrite *Gilliamella* 68.22 reductase small subunit A0A0E1NM95 Cluster: Nitrite Yersiniaceae 68.52 reductase (NAD(P)H) A0A080K0G0 Cluster: Ferredoxin *Snodgrassella alvi* 69.16 subunits of nitrite SCGC AB-598-O02 reductase and ring- hydroxylating dioxygenase A0A0Q4N870 Cluster: Nitrite Enterobacterales 67.29 reductase small subunit UPI0004A391D4 Cluster: nitrite *Tatumella saanichensis* 69.81 reductase small subunit NirD A0A1B9ZVE6 Cluster: Nitrite *Gilliamella apicola* 67.29 reductase small subunit A0A0T9JI92 Cluster: Nitrite *Yersinia* 68.52 reductase small subunit A0A2N5EJM2 Cluster: Nitrite *Chimaeribacter* 67.59 reductase small subunit NirD A0A2V4E983 Cluster: Nitrite *Gilliamella* 65.74 reductase small subunit A0A419N5U0 Cluster: Nitrite Yersiniaceae 67.29 reductase small subunit NirD A0A0F7H970 Cluster: Nitrite *Serratia* 70.75 reductase A0A318NZT5 Cluster: Nitrite Yersiniaceae 69.81 reductase small subunit NirD A0A250B1W9 Cluster: Nitrite *Gibbsiella* 69.81 reductase small subunit *quercinecans* A0A6G9I849 Cluster: Nitrite *Orbus* sp. IPMB12 65.74 reductase small subunit NirD A0A419M923 Cluster: Nitrite *Rahnella inusitata* 67.92 reductase small subunit NirD A0A389M9P6 Cluster: Nitrite Enterobacterales 67.29 reductase (NADH) small

subunit A0A0N8GUB4 Cluster: Nitrite Vibrionales 67.96 reductase (NADH) small subunit A0A443IEH9 Cluster: Nitrite *Pantoea* sp. LMG 68.87 reductase 27579 A0A1X1MLP8 Cluster: Nitrite *Vibrio* sp. qd031 69.9 reductase A0A542BGK2 Cluster: Assimilatory *Serratia fonticola* 68.87 nitrite reductase (NAD(P)H) small subunit A0A2U3B6U3 Cluster: Nitrite *Vibrio* sp. E4404 67.92 reductase small subunit A0A240CDV1 Cluster: Nitrite Gammaproteobacteria 67.92 reductase [NAD(P)H] small subunit A0A2S9I8J3 Cluster: Nitrite Bacteria 67.29 reductase small subunit A0A495RE76 Cluster: Assimilatory *Orbus hercynius* 67.59 nitrite reductase (NAD(P)H) small subunit A0A178KL05 Cluster: Nitrite *Photobacterium* 66.04 reductase small subunit A0A198GJ99 Cluster: Nitrite *Cosenzaea* 63.55 reductase [NAD(P)H] *myxofaciens* ATCC small subunit 19692 A0A2N0W0U3 Cluster: Nitrite Enterobacterales 66.36 reductase small subunit A0A1B9JKC9 Cluster: Nitrite *Gilliamella apicola* 66.67 reductase small subunit UPI0011BEE383 Cluster: nitrite *Pantoea* sp. CCBC3-3-1 65.09 reductase small subunit NirD A0A085G554 Cluster: Nitrite *Ewingella americana* 67.29 reductase [NAD(P)H] small subunit A0A1B1NU77 Cluster: Nitrite *Vibrio* 67.96 reductase (NADH) A0A2N7G6C9 Cluster: Nitrite *Vibrio* sp. 67.96 reductase small subunit 10N.261.55.A7 A0A349FFE1 Cluster: Nitrite *Vibrio* sp. 69.52 reductase small subunit A0A1B9JS56 Cluster: Nitrite *Gilliamella apicola* 64.49 reductase small subunit A0A366XH21 Cluster: Nitrite Vibrionales 66.67 reductase small subunit UPI00062A46AC Cluster: nitrite *Tatumella morbirosei* 67.59 reductase small subunit NirD UPI00046A4834 Cluster: nitrite *Pantoea* sp. IMH 68.87 reductase small subunit NirD

TABLE-US-00014 TABLE 14 Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 77 (large subunit encoded by nirB). IDENTITY TO UNIPROT DATABASE SEQ ID NO: 77 ACCESSION NUMBER DESCRIPTION ORGANISM [%]

Cluster	Protein	Organism	Identity [%]
W9B2S1	NirB protein	Bacteria	93.27
A0A486K457	Nitrite reductase [NAD(P)H]	Enterobacterales	93.15
A0A1B7KIV8	Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	85.41
A0A087L3N6	Nitrite reductase	<i>Serratia</i>	84.6
A0A654DEI2	Nitrite reductase, large subunit, nucleotide- and Fe/S-cluster binding	Enterobacterales	84.36
UPI00156BAB07	unknown related cluster		81.3
A0A2I0FWV2	Nitrite reductase (NAD(P)H) bacterium	CwR94	81.52
I2B4D0	Nitrite reductase (NAD(P)H) (strain ATCC 29907/ DSM 4481/JCM 1650/NBRC 105725/ CDC 9005-74)	<i>Shimwellia blattae</i>	85.24
A0A370QTZ7	Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Enterobacillus tribolii</i>	83.39
UPI000DEFBD5D	Nitrite reductase large subunit	<i>Edaphovirga cremea</i>	83.53
A0A380ASL5	Nitric oxide cellular organisms		84.36
FIRd-NAD(+)	Nitrite reductase	unclassified <i>Pantoea</i>	81.42
A0A221T4L0	Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	80.45
A0A2N5EJN0	Nitrite reductase (NAD(P)H)	Enterobacterales	82.68
A0A377N6S4	Benzene 1,2- dioxxygenase system ferredoxin--NAD(+) reductase subunit	Enterobacterales	82.82
A0A4P8SFP8	Nitrite reductase (NAD(P)H)	<i>Nissabacter</i> sp.	82.11
SGAir0207	Nitrite reductase large subunit	<i>Erwinia iniecta</i>	79.62
A0A0L7TEE8	Nitrite reductase large subunit	Erwiniaceae	79.88
A0A506V9E3	Nitrite reductase large subunit	<i>Mixta</i>	79.41
UPI000468AD1D	Nitrite reductase large subunit	Erwiniaceae	81.02
A0A0N0Z921	Nitrite reductase [NAD(P)H] large subunit	<i>Moellerella wisconsensis</i>	79.34
B4EXR2	Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	78.34
A0A198GKA3	Nitrite reductase [NAD(P)H]	<i>Cosenzaea myxofaciens</i> ATCC	78.11
19692	Nitrite reductase	Enterobacterales	78.46
E1SGS9	Nitrite reductase (NAD(P)H) subunit	Erwiniaceae	77.78
A0A1X1EZA9	Nitrite reductase large subunit	<i>Pantoea cypripedii</i>	78.25
A0A3A9I4K8	Nitrite reductase large subunit	Erwiniaceae	77.9
A0A2S9I8J4	Nitrite reductase (NAD(P)H)	Erwiniaceae	79.48
A0A0R4FQI6	Nitrite reductase, large subunit, nucleotide- binding	<i>Xenorhabdus nematophila</i>	76.42

UPI001265FD68 Cluster: nitrite *Erwinia endophytica* 77.49 reductase large subunit
UPI00146D663B Cluster: nitrite *Rahnella* sp. SAP-1 78.32 reductase large subunit A0A3S4CWL7
Cluster: Rubredoxin- Enterobacterales 78.65 NAD(+) reductase A0A1Y2S8S4 Cluster: NADH
oxidase *Xenorhabdus* 76.78 *vietnamensis* A0A0A3YW59 Cluster: Nitrite *Erwinia typographi*
79.31 reductase A0A506PS78 Cluster: Nitrite *Pantoea* 76.12 reductase large subunit D8MY05
Cluster: Nitrite *Erwinia* 79.12 reductase [NAD(P)H] large subunit E6WGY1 Cluster: Nitrite
unclassified *Pantoea* 75.62 reductase (NAD(P)H), large subunit A0A3R8ZPT1 Cluster: Nitrite
unclassified *Erwinia* 75.3 reductase large subunit A0A5M7NEQ9 Cluster: Nitrite Enterobacterales
75.3 reductase large subunit A0A2V2BEC9 Cluster: Assimilatory *Pantoea* 76.36 nitrite reductase
(NAD(P)H) large subunit A0A1X0W567 Cluster: Nitrite *Rouxiiella* 77.88 reductase large subunit
A0A6G9I851 Cluster: Nitrite *Orbus* sp. IPMB12 77.66 reductase large subunit UPI0011BF5221
Cluster: nitrite *Pantoea* sp. CCBC3- 76.66 reductase large subunit 3-1 A0A556RIT3 Cluster:
Nitrite *Gilliamella* 74.91 reductase large subunit A0A1R4J5I6 Cluster: Nitrite *Vibrio* 72.82
reductase [NAD(P)H] large subunit A0A1B9MVL8 Cluster: Nitrite *Gilliamella* 75.03 reductase
large subunit UPI0013596BEF Cluster: nitrite unclassified 72.59 reductase large subunit
Psychromonas A0A1B9L8J9 Cluster: Nitrite *Gilliamella apicola* 74.79 reductase large subunit
A0A095TED5 Cluster: Nitrite *Tatumella* 74.67 reductase A0A2S7VXZ8 Cluster: Nitrite
Photobacterium 73.6 reductase large subunit A0A495REZ8 Cluster: Assimilatory *Orbus hercynius*
75.15 nitrite reductase (NAD(P)H) large subunit A0A2T3QB92 Cluster: Nitric oxide
Photobacterium 75.18 reductase FIRd-NAD(+) *damselae* reductase A0A1X0WF55 Cluster: Nitrite
Gammaproteobacteria 76.51 reductase large subunit A0A2I8TVB9 Cluster: Nitrite *Vibrio* 75.33
reductase large subunit A0A2T3NPM2 Cluster: Nitrite *Photobacterium* 74.85 reductase large
subunit *sanctipauli* R9PFS3 Cluster: Nitrite *Agarivorans* 73.18 reductase A0A0B7J736 Cluster:
Nitrite *Photobacterium* 73.6 reductase, large subunit, NAD(P)H- binding U4KF90 Cluster: Nitrite
Vibrio nigripulchritudo 73.78 reductase [NAD(P)H] large subunit A0A1Y6IN13 Cluster: Nitrite
Vibrio mangrovi 73.54 reductase [NAD(P)H] A0A1L9L1Z8 Cluster: Nitrite Vibrionaceae 73.78
reductase [NAD(P)H] A0A0A7RZE6 Cluster: Assimilatory *Frischella perrara* 73.24 nitrite
reductase (NAD(P)H) large subunit A0A1Q9GSK0 Cluster: Nitrite *Photobacterium* 74.97
reductase large subunit UPI000200DAF9 Cluster: nitrite *Vibrio furnissii* 74.97 reductase large
subunit A0A378PW16 Cluster: Nitric oxide Vibrionaceae 72.82 reductase FIRd-NAD(+) reductase
A0A1B9JKB8 Cluster: Nitrite *Gilliamella apicola* 72.65 reductase large subunit D4ZBU5 Cluster:
Nitrite *Shewanella violacea* 72.4 reductase [NAD(P)H], (strain JCM 10179/ large subunit CIP
106290/LMG 19151/DSS12) UPI00117F52C3 Cluster: nitrite *Vibrio furnissii* 74.97 reductase large
subunit UPI0001B93571 Cluster: nitrite *Vibrio furnissii* 74.97 reductase large subunit
A0A1B1NUU8 Cluster: Nitrite Vibrionaceae 72.94 reductase (NADH) A0A178KL92 Cluster:
Nitrite *Photobacterium* 73.39 reductase large subunit A0A1T4RZM3 Cluster: Nitrite *Vibrio*
cincinnatiensis 74.37 reductase (NADH) large subunit UPI0012ADB8F8 Cluster: nitrite *Vibrio*
furnissii 74.97 reductase large subunit A0A418YIX2 Cluster: Nitrite *Motilimonas* sp. 73 reductase
large subunit PLHSC7-2 Q6LS87 Cluster: Putative nitrite *Photobacterium* 73.67 reductase
(NAD(P)H), large subunit A0A4R3I7X5 Cluster: Nitrite *Reinekea* 73.72 reductase (NADH)
marinisedimentorum large subunit UPI0003F87F52 Cluster: nitrite *Psychromonas* 72.02 reductase
large subunit UPI00037B4F3D Cluster: nitrite *Uliginosibacterium* 70.2 reductase large subunit
gangwonense A0A2N7DBN5 Cluster: Nitrite *Vibrio* sp. 73.06 reductase large subunit
10N.286.49.B3 A0A484GE91 Cluster: Nitrite *Candidatus* 72.16 reductase large subunit
Schmidhempelia bombi str. Bimp A0A2M8H0G2 Cluster: Nitrite unclassified *Vibrio* 73.84
reductase (NAD(P)H) U3CNX4 Cluster: Nitrite *Vibrio* 73.78 reductase large subunit
UPI0010A63CB5 Cluster: nitrite *Vibrio* sp. H11 73.12 reductase large subunit A0A090IJG5
Cluster: Nitrite *Moritella* 71.92 reductase (NAD(P)H) large subunit A0A2N0WYN5 Cluster:
Nitrite *Psychromonas* 72.87 reductase (NAD(P)H) UPI000B356675 Cluster: nitrite
Thaumasiobacter 73.12 reductase large subunit *subtropicus* A0A1R4LRM2 Cluster: Nitrite *Vibrio*

69.41 reductase [NAD(P)H] UPI00082A845F Cluster: nitrite *Erwinia* sp. ErVv1 72.27 reductase large subunit B8KAR6 Cluster: Nitrite *Vibrio* 73.3 reductase (NAD(P)H), large subunit A0A1G8E1S6 Cluster: Assimilatory *Vibrio xiamenensis* 72.59 nitrite reductase (NAD(P)H) large subunit A0A0C5WZ76 Cluster: Nitrite *Photobacterium* 72.94 reductase (NAD(P)H), large subunit A0A2G4AWG1 Cluster: Nitrite Vibrionaceae 73.18 reductase [NAD(P)H]

Nitrate/Nitrite Transporter

[0408] Preferably, the recombinant yeast cell further comprises one or more genetic modifications that result in an increased transport of oxidized nitrogen source, such as nitrate or nitrite, into the yeast cell. More preferably the recombinant yeast cell further comprising one or more genes encoding a nitrate and/or nitrite transporter.

[0409] Suitable transporters may include the sulphite transporters Ssu1 and SSu2 (as described by Cabrera et al in their article titled “Molecular Components of Nitrate and Nitrite Efflux in Yeast”, published February 2014 Volume 13 Number 2 Eukaryotic Cell p. 267-278, herein incorporated by reference); and the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* (also referred to as *Hansenula polymorpha*) and/or a functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of the aforementioned nitrate/nitrite transporters; and/or functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporters, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0410] Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* and/or a functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with nitrate/nitrite transporter YNT1; and/or functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporter YNT1, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0411] Preferred nitrate/nitrite transporter hence include: nitrate/nitrite transporters comprising a polypeptide having an amino acid sequence of SEQ ID NO:79, as described herein; and/or functional homologues of SEQ ID NO:79 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with SEQ ID NO:79; and/or functional homologues of SEQ ID NO:79 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO:79. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:79.

[0412] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of SEQ ID NO:79 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:79. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:79 respectively.

[0413] Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:79 are listed in Table 15 below.

TABLE-US-00015 TABLE 15 Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 79. IDENTITY TO UNIPROT DATABASE SEQ ID NO: 79
 ACCESSION NUMBER DESCRIPTION ORGANISM [%] W1QLT0 Cluster: Nitrate/nitrite
Ogataea 95.66 transporter W6MFQ5 Cluster: Nitrate/nitrite *Kuraishia capsulata* 56.89 transporter CBS 1993 A0A1E4TX12 Cluster: Nitrate/nitrite *Pachysolen tannophilus* 56.55 transporter NRRL Y-2460 A0A1E3NV96 Cluster: Nitrate/nitrite *Wickerhamomyces* 56.52 transporter *anomalus* (strain ATCC 58044/CBS 1984/ NCYC 433/NRRL Y- 366-8) K0KNB9 Cluster: Nitrate/nitrite *Wickerhamomyces* 56.46 transporter *ciferrii* (strain F-60-10/ ATCC 14091/CBS 111/JCM 3599/ NBRC 0793/NRRL Y- 1031) A0A0H5C0H7 Cluster: Nitrate/nitrite *Cyberlindnera jadinii* 54.44 transporter (strain ATCC 18201/ CBS 1600/CCRC 20928/JCM 3617/ NBRC 0987/NRRL Y- 1542) A0A5P8N9I7 Cluster: Nitrate/nitrite *Cyberlindnera* 54.58 transporter *americana* A0A061BB65 Cluster: *Cyberlindnera fabianii* 52.72 CYFA0S24e00188g1_1 A0A060T893 Cluster: Nitrate/nitrite *Blastobotrys* 47.18 transporter *adeninivorans* B3WFS7 Cluster: Nitrate/nitrite *Blastobotrys* 47.79 transporter *adeninivorans* A0A261XZN8 Cluster: Nitrate/nitrite *Bifiguratus adelaidae* 47.49 transporter A0A0E9N9H8 Cluster: Nitrate/nitrite *Saitoella complicata* 46.17 transporter (strain BCRC 22490/ CBS 7301/JCM 7358/ NBRC 10748/NRRL Y-17804) A0A060T786 Cluster: Nitrate/nitrite *Blastobotrys* 43.91 transporter *adeninivorans* A0A0A1T938 Cluster: Nitrate/nitrite *Torribiella* 42.57 transporter *hemipterigena* A0A3D8R1D7 Cluster: Nitrate/nitrite *Coleophoma* 42.66 transporter A0A370TMI7 Cluster: Nitrate/nitrite *Venustampulla* 41.72 transporter *echinocandica* A0A3N4HFN0 Cluster: Nitrate/nitrite *Ascobolus immersus* 42.16 transporter RN42 A0A167RLM4 Cluster: Nitrate/nitrite *Cordyceps* 40.04 transporter *fumosorosea* ARSEF 2679 A0A179I4H5 Cluster: Nitrate/nitrite *Akanthomyces lecanii* 40.94 transporter A0A0G2E479 Cluster: Nitrate/nitrite *Phaeomoniella* 42.33 transporter *chlamydospora* A0A6A6CV94 Cluster: Nitrate/nitrite *Zasmidium cellare* 42.91 transporter ATCC 36951 S3CRP8 Cluster: Nitrate/nitrite *Glarea lozoyensis* 40.4 transporter A0A1B8DXT4 Cluster: Nitrate/nitrite unclassified 42.74 transporter *Pseudogymnoascus* A0A094B0S5 Cluster: Nitrate/nitrite *Pseudogymnoascus* 42.94 transporter A0A6A6WTZ3 Cluster: Nitrate/nitrite *Melanomma pulvis-* 42.51 transporter *pyrius* CBS 109.77 G3JS45 Cluster: Nitrate/nitrite *Cordyceps militaris* 40.65 transporter A0A0E9NCH7 Cluster: Nitrate/nitrite *Saitoella complicata* 44.27 transporter (strain BCRC 22490/ CBS 7301/JCM 7358/ NBRC 10748/NRRL Y-17804) A0A0C3CTX3 Cluster: Nitrate/nitrite *Oidiodendron maius* Zn 43.15 transporter A0A6G1J7V7 Cluster: Nitrate *Lentithecium fluviatile* 42.63 transporter-like protein CBS 122367 A0A4U0XSZ7 Cluster: Nitrate/nitrite *Cryomyces minteri* 41.8 transporter UPI000CE1C7E3 Cluster: nitrate *Quercus suber* 41.33 transporter-like A0A6A5XQJ5 Cluster: Nitrate/nitrite *Aaosphaeria arxii* CBS 42.24 transporter 175.79 A0A1W5CWS1 Cluster: Nitrate/nitrite *Lasallia pustulata* 43.11 transporter A0A6A6R7R9 Cluster: Nitrate/nitrite Mytiliniaceae 43.99 transporter U1GKS4 Cluster: Nitrate/nitrite *Endocarpon pusillum* 42.43 transporter (strain Z07020/HMAS- L-300199) D5GLI8 Cluster: Nitrate/nitrite *Tuber melanosporum* 41.75 transporter (strain Mel28) A0A5J5EIC4 Cluster: Nitrate/nitrite *Sphaerospora* 41.7 transporter *brunnea* UPI00144AC6B7 Cluster: nitrate *Lindgomyces* 42.6 transporter *ingoldianus* A0A6A6RHP6 Cluster: Nitrate/nitrite *Massarina eburnea* 42.6 transporter CBS 473.64 A0A6A6JM23 Cluster: Nitrate/nitrite *Westerdykella ornata* 42.63 transporter A0A317SYP8 Cluster: Nitrate/nitrite *Tuber* 41.34 transporter A0A232LMW8 Cluster: Nitrate/nitrite *Elaphomyces* 43.09 transporter *granulatus* A0A1L7WDP7 Cluster: Nitrate/nitrite *Phialocephala* 42.04 transporter *subalpina* A0A6A5UCT1 Cluster: Nitrate/nitrite *Byssothecium*

circinans 43.38 transporter A0A6A6BJV5 Cluster: Nitrate/nitrite *Aplosporella prunicola* 44.08 transporter CBS 121167 A0A6G1LEQ8 Cluster: Nitrate *Teratosphaeria* 41.68 transporter *nubilosa* S3DBU1 Cluster: Nitrate/nitrite *Glarea lozoyensis* 43.51 transporter A0A4U0TQA4 Cluster: Nitrate/nitrite *Hortaea thailandica* 40.68 transporter A0A4U0X4U1 Cluster: Nitrate/nitrite *Friedmanniomyces* 40.28 transporter A0A507R2R1 Cluster: Nitrate/nitrite *Monascus purpureus* 41.84 transporter A0A2V1DKP2 Cluster: Nitrate/nitrite *Periconia* 41.46 transporter *macrospinos* A0A1L9RJV0 Cluster: Nitrate/nitrite *Aspergillus wentii* DTO 42.54 transporter 134E9 A0A167P4D7 Cluster: Nitrate/nitrite *Sporothrix insectorum* 41.12 transporter RCEF 264 UPI000CE20498 Cluster: nitrate *Quercus suber* 42.71 transporter-like A0A4Z1P351 Cluster: Nitrate/nitrite *Venturia* 41.63 transporter A0A6G1K4M7 Cluster: Nitrate *Pleomassaria siparia* 42.16 transporter-like protein CBS 279.74 M7U767 Cluster: Nitrate/nitrite Sclerotiniaceae 42.65 transporter A0A0D2AUR8 Cluster: Nitrate/nitrite *Verruconis gallopava* 41.41 transporter A0A6G1H5R5 Cluster: Nitrate *Aulographum hederarum* 41.73 transporter CBS 113979 A0A1Y2TEF0 Cluster: Nitrate/nitrite unclassified *Hypoxylon* 41.77 transporter A0A177BV64 Cluster: Nitrate/nitrite *Paraphaeosphaeria* 41.75 transporter *sporulosa* A0A6A5WR45 Cluster: Nitrate/nitrite *Amniculicola lignicola* 42.63 transporter CBS 123094 A0A6G1KNX3 Cluster: Nitrate *Pleomassaria siparia* 41.41 transporter-like protein CBS 279.74 W9WHT3 Cluster: Nitrate/nitrite Herpotrichiellaceae 42.14 transporter A0A2T2P8J8 Cluster: Nitrate/nitrite *Corynespora cassiicola* 43.06 transporter Philippines A0A6A6T246 Cluster: Nitrate/nitrite *Lophiostoma* 41.41 transporter *macrostomum* CBS 122681 A0A2P8ABJ3 Cluster: Nitrate/nitrite *Elsinoe australis* 42.25 transporter A0A1Y1ZNY9 Cluster: Nitrate/nitrite *Clohesyomyces* 40.77 transporter *aquaticus* W2RSI4 Cluster: Nitrate/nitrite *Cyphellophora* 41.24 transporter *europaea* CBS 101466 A0A2V1E4S3 Cluster: Nitrate/nitrite *Periconia* 41.41 transporter *macrospinos* A0A6A5V026 Cluster: Nitrate/nitrite *Bimuria novae-* 41.15 transporter *zelandiae* CBS 107.79 A0A2K1QG31 Cluster: Nitrate/nitrite Elsinoaceae 42.25 transporter A0A094A2I4 Cluster: Nitrate/nitrite *Pseudogymnoascus* 43.54 transporter sp. VKM F-4281 (FW- 2241) A0A6A6XX03 Cluster: Nitrate/nitrite *Melanomma pulvis-* 42.07 transporter *pyrius* CBS 109.77 A0A2J5HKF9 Cluster: Nitrate/nitrite *Aspergillus* 40.56 transporter A0A6A6IDB6 Cluster: Nitrate/nitrite *Trematosphaeria* 42.45 transporter *pertusa* A0A6A5ST79 Cluster: Nitrate/nitrite *Clathrospora elyanae* 41.85 transporter A0A3N4JV10 Cluster: Nitrate/nitrite *Choiromyces venosus* 42.06 transporter 120613-1 A0A6A5XFW9 Cluster: Nitrate/nitrite *Aaosphaeria arxii* CBS 41.43 transporter 175.79 A0A553IBL5 Cluster: Nitrate/nitrite *Xylaria flabelliformis* 40.24 transporter R7Z451 Cluster: Nitrate/nitrite *Coniosporium apollinis* 40.61 transporter (strain CBS 100218) A0A139I046 Cluster: Nitrate/nitrite *Pseudocercospora* 40.24 transporter *musae* A0A2T2NBP3 Cluster: Nitrate/nitrite *Corynespora cassiicola* 41.84 transporter Philippines A0A177E3H1 Cluster: Nitrate/nitrite *Alternaria* 42.21 transporter B0XPC3 Cluster: Nitrate/nitrite *Neosartorya fumigata* 42.89 transporter A0A150V198 Cluster: Nitrate/nitrite *Acidomyces* 42.08 transporter *richmondensis* BFW A0A3N4KQT8 Cluster: Nitrate/nitrite *Morchella conica* 40.44 transporter CCBAS932 A0A1Y2DSQ4 Cluster: Nitrate/nitrite *Pseudomassariella* 41.05 transporter *vexata* A0A2V1BSJ8 Cluster: Nitrate/nitrite Helotiales *incertae* 42.04 transporter *sedis* A0A1V1TKZ5 Cluster: Nitrate/nitrite Fungi 40.24 transporter K1WJ63 Cluster: Nitrate/nitrite *Marssonina brunnea* f. 42.06 transporter sp. *multigermmtubi* (strain MB_m1) A0A6A6E950 Cluster: Nitrate/nitrite *Zopfia rhizophila* CBS 41.92 transporter 207.26 A0A2K1R263 Cluster: Nitrate/nitrite *Sphaceloma murrayae* 41.83 transporter A0A2S7Q2X6 Cluster: Nitrate/nitrite unclassified 42.32 transporter *Rutstroemia*

Co-Factors

[0414] Preferably the recombinant yeast cell further comprises suitable co-factors to enhance the activity of the above mentioned NADH-dependent nitrate reductase and/or NADH-dependent nitrite reductase. Preferred cofactors include flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). Preferably the recombinant yeast cell may therefore

further comprise one or more genes encoding enzymes for the synthesis of one or more of flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). For example, the recombinant yeast cell may comprise one or more genes encoding for an enzyme having FAD synthase activity. Preferred co-factors are as exemplified in non-pre-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

Recombinant Expression

[0415] The recombinant yeast cell is a recombinant cell. That is to say, a recombinant yeast cell comprises, or is transformed with or is genetically modified with a nucleotide sequence that does not naturally occur in the cell in question. Techniques for the recombinant expression of enzymes in a cell, as well as for the additional genetic modifications of a recombinant yeast cell are well known to those skilled in the art. Typically such techniques involve transformation of a cell with nucleic acid construct comprising the relevant sequence. Such methods are, for example, known from standard handbooks, such as Sambrook and Russel (2001) “*Molecular Cloning: A Laboratory Manual*”, (3rd edition), published by Cold Spring Harbor Laboratory Press, or F. Ausubel et al., eds., “*Current protocols in molecular biology*”, Green Publishing and Wiley Interscience, New York (1987). Methods for transformation and genetic modification of fungal host cells are known from e.g. EP-A-0635574, WO98/46772, WO 99/60102, WO00/37671, WO90/14423, EP-A-0481008, EP-A-0635574 and U.S. Pat. No. 6,265,186.

Fermentation Process

[0416] The invention further provides a process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate or another organic carbon source, using a recombinant yeast cell as described in this specification, thereby forming ethanol.

[0417] The feed for this fermentation process suitably comprises one or more fermentable carbon sources. The fermentable carbon source preferably comprises or is consisting of one or more fermentable carbohydrates. More preferably, the fermentable carbon source comprises one or more mono-saccharides, disaccharides and/or polysaccharides. For example, the fermentable carbon source may comprise one or more carbohydrates selected from the group consisting of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose, mannose and trehalose. The fermentable carbon source, preferably comprising or consisting of one or more carbohydrates, may suitably be obtained from starch, cellulose, hemicellulose lignocellulose, and/or pectin. Suitably the fermentable carbon source may be in the form of a, preferably aqueous, slurry, suspension, or a liquid.

[0418] The concentration of fermentable carbohydrate, such as for example glucose, during fermentation is preferably equal to or more than 80 g/L. That is, the initial concentration of glucose at the start of the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L. The start of the fermentation may be the moment when the fermentable carbohydrate is brought into contact with the recombinant cell of the invention.

[0419] The fermentable carbon source may be prepared by contacting starch, lignocellulose, and/or pectin with an enzyme composition, wherein one or more mono-saccharides, disaccharides and/or polysaccharides are produced, and wherein the produced mono-saccharides, disaccharides and/or polysaccharides are subsequently fermented to give a fermentation product.

[0420] Before enzymatic treatment, the lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-

pretreatment, e.g. between 150-220° C. for 1 to 30 minutes. Subsequently the pretreated material can be subjected to enzymatic hydrolysis to release sugars that may be fermented according to the invention. This may be executed with conventional methods, e.g. contacting with cellulases, for instance cellobiohydrolase(s), endoglucanase(s), beta-glucosidase(s) and optionally other enzymes, The conversion with the cellulases may be executed at ambient temperatures or at higher temperatures, at a reaction time to release sufficient amounts of sugar(s). The result of the enzymatic hydrolysis is hydrolysis product comprising C5/C6 sugars, herein designated as the sugar composition.

[0421] In one embodiment the fermentable carbohydrate is, or is comprised by a biomass hydrolysate, such as a corn stover or corn fiber hydrolysate. Such biomass hydrolysate may in its turn comprise, or be derived from corn stover and/or corn fiber.

[0422] By a “hydrolysate” is herein understood a polysaccharide-comprising material (such as corn stover, corn starch, corn fiber, or lignocellulosic material, which polysaccharides have been depolymerized through the addition of water to form mono and oligosaccharide sugars.

Hydrolysates may be produced by enzymatic or acid hydrolysis of the polysaccharide-containing material.

[0423] A biomass hydrolysate may be a lignocellulosic biomass hydrolysate. Lignocellulose herein includes hemicellulose and hemicellulose parts of biomass. Also lignocellulose includes lignocellulosic fractions of biomass. Suitable lignocellulosic materials may be found in the following list: orchard primings, chaparral, mill waste, urban wood waste, municipal waste, logging waste, forest thinnings, short-rotation woody crops, industrial waste, wheat straw, oat straw, rice straw, barley straw, rye straw, flax straw, soy hulls, rice hulls, rice straw, corn gluten feed, oat hulls, sugar cane, corn stover, corn stalks, corn cobs, corn husks, switch grass, miscanthus, sweet sorghum, canola stems, soybean stems, prairie grass, gamagrass, foxtail; sugar beet pulp, citrus fruit pulp, seed hulls, cellulosic animal wastes, lawn clippings, cotton, seaweed, algae (including macroalgae and microalgae), trees, softwood, hardwood, poplar, pine, shrubs, grasses, wheat, wheat straw, sugar cane bagasse, corn, corn husks, corn hobs, corn kernel, fiber from kernels, products and by-products from wet or dry milling of grains, municipal solid waste, waste paper, yard waste, herbaceous material, agricultural residues, forestry residues, municipal solid waste, waste paper, pulp, paper mill residues, branches, bushes, canes, corn, corn husks, an energy crop, forest, a fruit, a flower, a grain, a grass, a herbaceous crop, a leaf, bark, a needle, a log, a root, a sapling, a shrub, switch grass, a tree, a vegetable, fruit peel, a vine, sugar beet pulp, wheat midlings, oat hulls, hard or soft wood, organic waste material generated from an agricultural process, forestry wood waste, or a combination of any two or more thereof. Algae, such as macroalgae and microalgae have the advantage that they may comprise considerable amounts of sugar alcohols such as sorbitol and/or mannitol. Lignocellulose, which may be considered as a potential renewable feedstock, generally comprises the polysaccharides cellulose (glucans) and hemicelluloses (xylans, heteroxylans and xyloglucans). In addition, some hemicellulose may be present as glucomannans, for example in wood-derived feedstocks. The enzymatic hydrolysis of these polysaccharides to soluble sugars, including both monomers and multimers, for example glucose, cellobiose, xylose, arabinose, galactose, fructose, mannose, rhamnose, ribose, galacturonic acid, glucuronic acid and other hexoses and pentoses occurs under the action of different enzymes acting in concert. In addition, pectins and other pectic substances such as arabinans may make up considerably proportion of the dry mass of typically cell walls from non-woody plant tissues (about a quarter to half of dry mass may be pectins). Lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes.

[0424] The process for the production of ethanol may comprise an aerobic propagation step and an

anaerobic fermentation step. More preferably the process according to the invention is a process comprising an aerobic propagation step wherein the population of the recombinant yeast cell is increased; and an anaerobic fermentation step wherein the carbon source is converted to ethanol by using the recombinant yeast cell population.

[0425] By propagation is herein understood a process of recombinant yeast cell growth that leads to increase of an initial recombinant yeast cell population. Main purpose of propagation is to increase the population of the recombinant yeast cell using the recombinant yeast cell's natural reproduction capabilities as living organisms. That is, propagation is directed to the production of biomass and is not directed to the production of ethanol. The conditions of propagation may include adequate carbon source, aeration, temperature and nutrient additions. Propagation is an aerobic process, thus the propagation tank must be properly aerated to maintain a certain level of dissolved oxygen. Adequate aeration is commonly achieved by air inductors installed on the piping going into the propagation tank that pull air into the propagation mix as the tank fills and during recirculation. The capacity for the propagation mix to retain dissolved oxygen is a function of the amount of air added and the consistency of the mix, which is why water is often added at a ratio of between 50:50 to 90:10 mash to water. "Thick" propagation mixes (80:20 mash-to-water ratio and higher) often require the addition of compressed air to make up for the lowered capacity for retaining dissolved oxygen. The amount of dissolved oxygen in the propagation mix is also a function of bubble size, so some ethanol plants add air through spargers that produce smaller bubbles compared to air inductors. Along with lower glucose, adequate aeration is important to promote aerobic respiration during propagation, making the environment during propagation different from the anaerobic environment during fermentation.

[0426] By an anaerobic fermentation process is herein understood a fermentation step run under anaerobic conditions.

[0427] The anaerobic fermentation is preferably run at a temperature that is optimal for the cell. Thus, for most recombinant yeast cells, the fermentation process is performed at a temperature which is less than about 50° C., less than about 42° C., or less than about 38° C. For recombinant yeast cell or filamentous fungal host cells, the fermentation process is preferably performed at a temperature which is lower than about 35, about 33, about 30 or about 28° C. and at a temperature which is higher than about 20, about 22, or about 25° C.

[0428] The ethanol yield, based on xylose and/or glucose, in the process according to the invention is preferably at least about 50, about 60, about 70, about 80, about 90, about 95 or about 98%. The ethanol yield is herein defined as a percentage of the theoretical maximum yield.

[0429] The process according to the invention, and the propagation step and/or fermentation step suitably comprised therein can be carried out in batch, fed-batch or continuous mode. A separate hydrolysis and fermentation (SHF) process or a simultaneous saccharification and fermentation (SSF) process may also be applied.

[0430] The recombinant yeast and process according to the invention advantageously allow for a more robust process. Advantageously the process, or any anaerobic fermentation during the process can be carried out in the presence of high concentrations of carbon source. The process, respectively any anaerobic fermentation step therein, is therefore preferably carried out in the presence of a glucose concentration of 25 g/L or more, 30 g/L or more, 35 g/L or more, 40 g/L or more, 45 g/L or more, 50 g/L or more, 55 g/L or more, 60 g/L or more, 65 g/L or more, 70 g/L or more, 75 g/L or more, 80 g/L or more, 85 g/L or more, 90 g/L or more, 95 g/L or more, 100 g/L or more, 110 g/L or more, 120 g/L or more or may for example be in the range of 25 g/L-250 g/L, 30 g/L-200 g/L, 40 g/L-200 g/L, 50 g/L-200 g/L, 60 g/L-200 g/L, 70 g/L-200 g/L, 80 g/L-200 g/L, or 90 g/L-200 g/L.

[0431] For the recovery of the fermentation product existing technologies are used. For different fermentation products different recovery processes are appropriate. Existing methods of recovering ethanol from aqueous mixtures commonly use fractionation and adsorption techniques. For

example, a beer still can be used to process a fermented product, which contains ethanol in an aqueous mixture, to produce an enriched ethanol-containing mixture that is then subjected to fractionation (e.g., fractional distillation or other like techniques). Next, the fractions containing the highest concentrations of ethanol can be passed through an adsorber to remove most, if not all, of the remaining water from the ethanol. In an embodiment in addition to the recovery of fermentation product, the yeast may be recycled.

[0432] All patent and literature references cited in the present specification are hereby incorporated by reference in their entirety.

[0433] The following examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way

EXAMPLES

General Molecular Biology Techniques

[0434] Unless indicated otherwise, the methods used are standard biochemical techniques.

Examples of suitable general methodology textbooks include Sambrook et al., *Molecular Cloning, a Laboratory Manual* (1989) and Ausubel et al., *Current Protocols in Molecular Biology* (1995), John Wiley & Sons, Inc.

HPLC Analysis

[0435] HPLC analysis is typically conducted as described in “*Determination of sugars, byproducts and degradation products in liquid fraction in process sample*”; Laboratory Analytical Procedure (LAP, Issue date: Dec. 8, 2006; by A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, and D. Templeton; Technical Report (NREL/TP-51042623); January 2008; National Renewable Energy Laboratory.

[0436] After fermentation, samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 µm pore size filter.

Strains and DNA Sequences Used in the Examples

[0437] Table 16 provides an overview of the genotypes of the strain

[0438] Table 17 provides an overview of the nucleic acid sequences referred to in these examples.

TABLE-US-00016 TABLE 16 *S. cerevisiae* strains used in the examples

Strain name	Genotype
Source Parental strain	Wildtype
Commercial <i>Saccharomyces cerevisiae</i> strain, available from Lesaffre	Intermediate strain IX15
PRK, cbbM, GroES, GroEL	Following strain construction described in WO2013/144257A1 and Reference strain RX16
PRK, cbbM, GroES, GroEL	Example 2
DAK1, gldA, ACT1.pro-T5	Example strain NX17
PRK, cbbM, GroES, GroEL	Example 3
DAK1, gldA, ANB1.pro-T5	Example strain NX18
PRK, cbbM, GroES, GroEL	Example 4
DAK1, gldA, HEM13.pro-T5	

TABLE-US-00017 TABLE 17 DNA sequences used in the examples

SEQ ID NO:	Short description
59	25-Sc_EFT2.pro-Sc_DAK1.orf-Sc_ENO1.ter-2A 2 (“fragment A”).
60	2A-Sc_HHF2.pro-Ec_gldA.orf-Sc CTC1.ter-2B 2 (“fragment B”).
61	2B-Sc_ACT1.pro_0001-Zrou_T5.orf-Sc_TEF2.ter_0001-2C 2 (“fragment C”).
62	25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A 3, 4 (“fragment D”).
63	2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B 3, 4 (“fragment E”).
64	2B-Sc_ANB1.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C 3 (“fragment F”).
65	2B-Sc_HEM13.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C 4 (“fragment G”).
66	Connector 25: 2, 3, 4 ATCCGTACACGTTTGTGCCGAACCCGAAGTATTGCAACA GGTCTGAAGTTC
67	Connector 2A: 2, 3, 4 ACCCGATCCCGAAGAAAGTCCAACCTCCAGTTCGATTCCC TGCTTTCGAGT
68	Connector 2B: 2, 3, 4 TCCTCGGATCGGCCAGTTGGGAGCAAGGTGGTACTTCT GTCGTCGTATTT
69	Connector 2C: 2, 3, 4

ATTGGACAGCAGGAGGTATTGGTCGAGAGAAGAGGGTGTTTCCGAG SEQ ID
NO: 70 Sc_INT7.03_FLANK5 2 SEQ ID NO: 71 Sc_INT7.03_FLANK3 2 SEQ ID
NO: 72 INT28_FLANK5 3, 4 SEQ ID NO: 73 INT28_FLANK3 3, 4

Starter Strains

[0439] Strains were prepared using Ethanol Red® as starting strain. Ethanol Red® is a commercial *Saccharomyces cerevisiae* strain, available from Lesaffre.

[0440] A strain construction approach that can be followed is described in WO2013/144257A1 and WO2015/028582, incorporated herein by reference.

[0441] Expression cassettes from various genes of interest can be recombined in vivo into a pathway at a specific locus upon transformation of this yeast (U.S. Pat. No. 9,738,890 B2). The promoter, ORF and terminator sequences are assembled into expression cassettes with Golden Gate technology, as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference, and ligated into BsaI-digested backbone vectors that decorated the expression cassettes with the connectors for the in vivo recombination step. The expression cassettes including connectors are amplified by PCR. In addition, a 5'- and a 3'-DNA fragment of the up- and downstream part of the integration locus was amplified using PCR and decorated by a connector sequence. Upon transformation of yeast cells with these DNA fragments, in vivo recombination and integration into the genome takes place at the desired location. CRISPR-Cas9 technology is used to make a unique double stranded break at the integration locus to target the pathway to this specific locus (see DiCarlo et al., “*Genome engineering in Saccharomyces cerevisiae using CRISPR-Cas systems*”, (2013), *Nucleic Acids Res* Vol. 41, pages 4336-4343, incorporated herein by reference) and WO16110512 and US2019309268. The gRNA was expressed from a multi-copy yeast shuttling vector that contains a natMX marker which confers resistance to the yeast cells against the antibiotic substance nourseothricin (NTC). The backbone of this plasmid is based on pRS305 (see Sikorski and Hieter, “*A System of Shuttle Vectors and Yeast Host Strains Designed for Efficient Manipulation of DNA in Saccharomyces cerevisiae*”, (1989), *Genetics*, vol. 122, pages 19-27, incorporated herein by reference), including a functional 2 micron ORI sequence. The *Streptococcus pyogenes* CRISPR-associated protein 9 (Cas9) was expressed from a pRS414 plasmid (see Sikorski and Hieter, 1989, as indicated above) with kanMX marker which confers resistance to the yeast cells against the antibiotic substance geneticin (G418). The guide RNA and protospacer sequences were designed with a gRNA designer tool (known by a person skilled in the art and for example described in <https://www.atum.bio/eCommerce/cas9/input>).

Example 1: Construction of “Rubisco” Strain (Intermediate Strain IX15)

[0442] In the current example the starter strain was transformed with the cbbM gene encoding the single subunit of ribulose-1,5-biphosphate-carboxylase (RuBisCO) from *Thiobacillus denitrificans*, genes encoding chaperonins GroEL and GroES from *E. coli* to aid in the proper folding of the RuBisCO protein in the cytosol of *S. cerevisiae*, a gene encoding phosphoribulokinase (prk) from *S. oleacea* as described by Guadalupe-Medina et al., “*Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast*”, published in *Biotechnol, Biofuels*, 2013, vol. 6, page 125 onwards, incorporated herein by reference. This resulted in reference strain IX15 which contained cbbM, prk, groEL and groES (see Table 16 for detailed genotypes).

Example 2: Construction of Reference Strain RX16

[0443] A reference strain RX16 was construed, comprising a glycerol transporter derived from *Z. rouxii* “Zrou_T5”, preceded by a constitutive promoter according to the prior art “Sc_ACT1.pro_0001”.

[0444] Reference strain RX16 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes; [0445] Expression cassette “fragment A”:

25-EFT2p.Sc_DAK1.ENO1t-2A; [0446] Expression cassette “fragment B”: 2A-HHF2p.Ec_gldA.CTC1t-2B; and [0447] Expression cassette “fragment C”: 2B-Sc_ACT1.pro_0001-Zrou_T5.orf-Sc_TEF2.ter_0001-2C.

[0448] Expression cassette “fragment A”: The first cassette contained a DNA fragment named “fragment A” was compiled using Golden Gate Cloning and comprised the *S. cerevisiae* EFT2 promoter (Sc_EFT2.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* ENO1 terminator (Sc_ENO1.ter). The cassette was decorated with 50 bp connectors 25 and 2A. Connector 25 had a nucleic acid sequence as illustrated in: SEQ ID NO: 66. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. The nucleic acid sequence of the DNA fragment “fragment A” is illustrated in SEQ ID NO: 59.

[0449] Expression cassette “fragment B”: The second cassette contained a DNA fragment named “fragment B”, and comprised the *S. cerevisiae* HHF2 promoter (Sc_HHF2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* CTC1 terminator (Sc_CTC1.ter). The cassette was decorated with 50 bp connectors 2A and 2B. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. The nucleic acid sequence of the DNA fragment “fragment B” is illustrated in SEQ ID NO: 60.

[0450] Expression cassette “fragment C”: The third cassette contained a DNA fragment named “fragment C”, and comprised the *S. cerevisiae* ACT1 promoter (Sc_ACT1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* TEF2 terminator (Sc_TEF2.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment C” is illustrated in SEQ ID NO: 61.

[0451] The above three cassettes were integrated in intermediate strain IX15 in the locus INT7.03 located on a non-coding region on chromosome VII between coding sequences PUP2 (YGR253C) and ENO1 (YGR254W) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration: [0452] Sc_INT7.03_FLANK5 (illustrated by SEQ ID NO: 70); and [0453] Sc_INT7.03_FLANK3 (illustrated by SEQ ID NO: 71).

[0454] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT7.03 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in new strain RX16 (see Table 16 for detailed genotypes).

Example 3: Construction of New NX17

[0455] New strain NX17 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes: [0456] Expression cassette “fragment D”: 25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A; [0457] Expression cassette “fragment E”: 2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B; and [0458] Expression cassette “fragment F”: 2B-Sc_ANB1.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C.

[0459] Expression cassette “fragment D”: The first cassette named “fragment D” was compiled using Golden Gate Cloning and comprised the *S. cerevisiae* MYO4 promoter (Sc_MYO4.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* GPM1 terminator (Sc_GPM1.ter). The cassette was decorated with 50 bp connectors 25 and 2A. Connector 25 had a nucleic acid sequence as illustrated in: SEQ ID NO: 66. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. The nucleic acid sequence of the DNA fragment “fragment D” is illustrated in SEQ ID NO: 62.

[0460] Expression cassette “fragment E”: The second cassette named “fragment E” comprised *S. cerevisiae* HHF2 promoter (Sc_HHF2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The cassette was decorated with 50 bp connectors 2A and 2B. Connector 2A had a nucleic acid sequence as illustrated in: SEQ ID NO: 67. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. The nucleic acid sequence of the DNA fragment “fragment E” is illustrated in SEQ ID NO: 63.

[0461] Expression cassette “fragment F”: The third cassette named “fragment F”, comprised the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment F” is illustrated in SEQ ID NO: 64.

[0462] The above three cassettes were integrated in intermediate strain IX15 in the INT28 locus using CRISPR-Cas9 using. These three cassettes were integrated in the locus INT28 located on a non-coding region on Chromosome IV between YDR345C (HXT3) and YDRT246C (SVF1) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration: [0463] INT28_FLANK5 (illustrated by SEQ ID NO: 72); and [0464] INT28_FLANK3 (illustrated by SEQ ID NO: 73)

[0465] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT28 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX17 (see Table 16 for detailed genotypes).

Example 4: Construction of New NX18

[0466] New strains NX18 was constructed by transforming the intermediate strain IX15 obtained in example 1 with three expression cassettes: [0467] Expression cassette “fragment D”: 25-Sc_MYO4.pro-Sc_DAK1.orf-Sc_GPM1.ter-2A; [0468] Expression cassette “fragment E”: 2A-Sc_HHF2.pro-Ec_gldA.orf-Sc_EFM1.ter-2B; and [0469] Expression cassette “fragment G”: 2B-Sc_HEM13.pro_0001-Zrou_T5.orf-Sc_TEF1.ter_0001-2C.

Fragment D and fragment E were as described above under example 3.

[0470] The third cassette named “fragment G” comprised the *S. cerevisiae* HEM13 promoter (Sc_HEM13.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The cassette was decorated with 50 bp connectors 2B and 2C. The cassette was decorated with 50 bp connectors 2B and 2C. Connector 2B had a nucleic acid sequence as illustrated in: SEQ ID NO: 68. Connector 2C had a nucleic acid sequence as illustrated in: SEQ ID NO: 69. The nucleic acid sequence of the DNA fragment “fragment G” is illustrated in SEQ ID NO: 65.

[0471] The above three cassettes were integrated in intermediate strain IX15 in the locus INT28 located on a non-coding region on Chromosome IV between YDR345C (HXT3) and YDRT246C (SVF1) of *S. cerevisiae* using CRISPR-Cas9 using the following sequences for homologous integration. INT28_FLANK5 (SEQ ID NO: 72) and INT28_FLANK3 (SEQ ID NO: 73) for homologous integration.

[0472] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT28 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX18 (see Table 16 for detailed genotypes).

Example 5: Fermentation

[0473] Preculture preparation and conditions: Glycerol stocks (−80° C.) were thawed at room temperature and used to inoculate 0.2 L filter-sterilized mineral medium (as described by Luttkik et al, “*The Saccharomyces cerevisiae ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism*”, (2000), JOURNAL OF BACTERIOLOGY, pages 7007-7013, herein incorporated by reference (Luttkik et al., 2000) at pH 6.0 (adjusted with 2M H2SO4/4N KOH) supplemented with 2% (w/v) glucose, in non-baffled 0.5 L shake-flasks. Precultures were incubated for 16 to 20 hours at 32° C., shaking at 200 RPM. After determination of the yeast biomass (CDW) content of the culture (via OD600 vs CDW calibration), a quantity of preculture corresponding to the required 0.5 g CDW/L inoculum concentration for the propagation was centrifuged (3 min, 5300×g), washed once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0474] Propagation: Propagation media consisted of 20 ml diluted corn mash (70% v/v Corn mash: 30% v/v demineralized water), at pH 5.0 (adjusted with 4N KOH/2M H.sub.2SO.sub.4) in 100 ml non-baffled shake flasks. Urea (1.25 g/L) was added as N-source and a standard antibiotic mix (1 ml 100 µg/L PenG & 1 ml 50 µg/L Neomycin stock per liter of corn mash) was added to prevent outgrowth of bacterial contaminants. The Glucoamylase (Spirizyme, Novozymes) dosage for all the strains was 0.1 g/kg. The amount of preculture material used to inoculate the propagation phase (0.5 g CDW/L) was determined by OD & strain specific OD.sub.600/CDW conversion factors. The required quantity of preculture was centrifuged (3 min, 4000 rpm), washed once with one culture volume cold (4° C.) sterile demi-water, centrifuged once more, resuspended in 500 µL sterile demi-water and transferred to the propagation. The propagations ran for 6 hrs at 32° C. shaking at 140 rpm.

[0475] Fermentation: Corn mash was used for all test described here. 1 g/L Urea was added as N-source, while the standard antibiotics mix was applied (100 mg/ml PenG stock+50 mg/ml Neomycin stock). pH was adjusted to 5.0 using 2M H.sub.2SO.sub.4/4N KOH. The Glucoamylase (Spirizyme, Novozymes) dosage applied was 0.24 g/kg. Fermentations were performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. The pressure development was measured in psi units (pound-force per square inch) and the results are illustrated in Table 18 and Table 19. FIG. 1 illustrates the results of Table 18 graphically and FIG. 2 illustrates the results of Table 19 graphically. The pressure listed is the cumulative pressure generated, expressed in psi.

[0476] As illustrated by Table 18 and FIG. 1, over the whole fermentation run, more ethanol and CO₂ was formed by the new strains NX17 and NX18 than by the reference strain RX16, illustrating more conversion of sugars. This is evidenced by the total area below the curve.

[0477] Table 19 and FIG. 2 further illustrate that the strains according to the invention, NX17 and NX18, comprising a promotor as claimed, have a steeper onset in fermentation than reference strain RX16 comprising a standard constitutive promoter, that is, the strains according to the invention are quicker in starting the fermentation.

[0478] pH was not controlled during fermentation. Fermentations were stopped after 66 h.

[0479] Sampling and analysis: All cultivations were sampled at end-of fermentation. Since the fermentation broths contained active GA enzyme, 50 µl of a 10 g/L acarbose stock solution was added to approximately 5 g sample to stop glucoamylase activity. Samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 µm pore size filter. HPLC analysis was conducted as described in (Sluiter, et al., 2008). The total sugar content (g/L) of the samples at end-of-fermentation (EOF) was determined with HPLC and the results are provided in Table 20.

[0480] As illustrated by Table 20, the total sugar content for the wild-type strain was 13.0 g/L and the total sugar content (g/L) for reference strain RX16 was 14.0 g/L. New strains NX17 and NX18 both had a total sugar content (g/L) at EOF of 12.6 g/L. These results illustrate that the strains according to the invention result in an improved consumption of available sugars.

TABLE-US-00018 TABLE 18 ethanol and CO₂ gas production (in psi) during fermentation NX17 (average of NX18 (average of time (h) RX16 duplo) duplo) 0 0 0 0 5 5 7 7 10 41 56 55 15 128 160 158 20 216 247 245 25 280 307 305 30 330 358 354 35 376 403 399 40 415 434 433 45 440 447 448 50 452 454 456 55 458 458 460 60 461 460 463 65 463 461 464

TABLE-US-00019 TABLE 19 ethanol and CO₂ gas production (in psi) during fermentation (first 10 hours) time (h) RX16 NX17 NX18 0 0 0 0 1 1 1 1 2 2 2 2 3 2 3 3 4 3 4 4 5 5 7 7 6 8 11 11 7 13 18 17 8 20 27 27 9 29 40 39 10 41 56 55

TABLE-US-00020 TABLE 20 Total sugar content (g/L) at end of fermentation (66 hours of fermentation) Total sugar content Strain (g/L) Wild-type 13 RX16 14 NX17 12.6 * NX18 12.6 * * Average of duplicate experiment

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Claims

1. A recombinant yeast cell that functionally expresses: a nucleic acid sequence encoding a protein having glycerol dehydrogenase activity; a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity; and a nucleic acid sequence encoding a protein having glycerol transporter activity, wherein the expression of the nucleic acid sequence encoding the protein having glycerol transporter activity is under control of a promoter (the "GT promoter"), which GT promoter has an anaerobic/aerobic expression ratio for the glycerol transporter of 2 or more.
2. The recombinant yeast cell according to claim 1, wherein the GT promoter is the promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C, SML1, TIR2, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.
3. The recombinant yeast strain according to claim 1, wherein the GT promoter is a synthetic oligonucleotide.
4. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is a protein having glycerol-proton symporter activity.
5. The recombinant yeast cell according to claim 1, wherein the protein having glycerol dehydrogenase activity is a protein having NAD⁺ dependent glycerol dehydrogenase activity.
6. The recombinant yeast cell according to claim 1, wherein the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity is a heterologous nucleic acid sequence.
7. The recombinant yeast cell according to claim 1, wherein the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.
8. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is under control of a promoter.
9. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell comprises one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.
10. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses: a nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or a nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or a nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.
11. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell

functionally expresses: a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

12. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

13. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequence encoding an enzyme having NADH-dependent nitrite reductase activity.

14. The recombinant yeast cell according to claim 13, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding an enzyme having nitrate and/or nitrite transporter activity.

15. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further comprises a deletion or disruption of a nucleic acid sequence encoding a protein having glycerol-3-phosphate dehydrogenase (GPD) activity and/or a nucleic acid sequence encoding a protein having glycerol phosphate phosphatase (GPP) activity.

16. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having glucoamylase activity (EC 3.2.1.20 or 3.2.1.3).

17. A process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate, using a recombinant yeast cell according to claim 1.

18. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is an STL1 protein.

19. The recombinant yeast cell according to claim 1, wherein the protein having glycerol transporter activity is an STL1 protein derived from *Zygosaccharomyces rouxii*.
